

What Happens When Inventors Collaborate? Evidence from 19th-Century French Patents*

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Abstract

Collaboration was rare in nineteenth-century France, yet when inventors moved from solo work to co-patenting it reshaped their inventive activity in ways distinct from modern teamwork. Using the full universe of French patents from 1791 to 1900, text-based measures of novelty and influence, and within-inventor event studies around each inventor's first collaboration, I show that collaboration was not associated with higher productivity or originality and produced only a short-lived rise in influence. Its durable effect was a large and persistent expansion in technological scope: first collaboration shifted roughly 35 percentage points of output outside an inventor's technology home class, from a baseline of about 23 percent. This diversification was driven by partnerships that crossed technology-class boundaries and by repeated ties with partners who had already worked in a different field. Women, who did not patent as often as men, entered collaboration at lower rates and faced a clear access barrier. Within the same technology class, those who overcame it built networks more cross-class than men's, despite entering disproportionately through within technology-class family ties, and diversified accordingly. The results suggest that in this historical setting, collaboration functioned primarily as a channel for knowledge access across technology classes rather than as a way to produce more novel or more productive work.

1 Introduction

Modern innovation is overwhelmingly collaborative. A central explanation is the burden of knowledge: as the stock of knowledge grows, inventors specialize and rely on teams to combine expertise (Jones, 2009). In science and patenting, teams have also come to dominate the upper tail of impact (Wuchty et al., 2007). Teams can insert unusual combinations into familiar domains (Uzzi et al., 2013), but size matters: smaller teams tend to disrupt science and technology while larger teams develop existing lines of work (Wu et al., 2019). Yet we know far less about how collaboration operated in earlier historical periods, when technical knowledge was less codified and not yet concentrated inside the firms, research universities, and industrial laboratories that now organize innovation.

Nineteenth-century France is useful because it combines broad formal access to patenting with rare collaboration. The patent system was grounded in natural-rights ideology and appears more accessible and inclusive than older elite-centered accounts suggest (Galvez-Behar, 2019; Merouani and Perrin, 2026b). Khan (2020) offers the strongest contrary reading, emphasizing high fees, secrecy, and institutional discretion as limits on French openness. Collaboration, however, remained uncommon, as most inventors worked alone (Merouani and Perrin, 2026b). If collaboration is as central to innovation as modern evidence suggests, why was it so limited? And when it did occur, what did it change?

A further question concerns gender. Women patented in France at higher rates than in Britain or the United States, yet formal access to patenting did not necessarily imply equal access to collaboration (Merouani and Perrin, 2026b). If collaboration mattered, did women enter collaborative ties at the same rate as men, and did those ties change their inventive outcomes in the same way?

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This paper uses the full universe of French patents from 1791 to 1900 to examine these questions. I focus on inventors who filed at least one solo patent before their first collaboration. The first part of the analysis uses within-inventor event studies to ask whether collaboration changed the outcomes most emphasized in modern work: annual patent output, novelty, and influence. Because historical patents lack citations, I develop text-based measures of novelty and influence from patent titles. The second part asks whether collaboration also changed the direction of inventive activity and uses the reconstructed co-inventor network to test which network features predict this redirection, drawing on the structural holes and cohesion literatures (Burt, 2004; Fleming et al., 2007b; Reagans and McEvily, 2003). Throughout, I ask whether these changes differed between women and men.

Three findings emerge. First, collaboration was not associated with higher productivity or novelty. The only output spike occurs in the year of first collaboration, and it disappears when the first collaborative patent itself is excluded. Novelty remains flat, while influence rises modestly before fading. These patterns differ from the large citation and novelty gains documented in modern teams. Second, the durable change appears when I look beyond these standard outcomes. Inventors redirected roughly 35 percentage points of their output outside their historical home class at first collaboration. Third, this diversification was driven by cross-class partnerships.

I cannot detect a gender difference in the diversification outcome, but the small number of women in the collaboration network prevents me from ruling out moderate differences. The path into collaboration differed sharply by gender. Women, scarce in the patent system and often reliant on family ties for initial access, faced an access barrier to collaboration. Conditional on entering the network, women had as many collaborators as men and belonged to components of similar size, yet their direct collaborator networks had a higher share of cross-class ties, even within the same technology class, despite heavier reliance on within-class family ties for entry. Their diversification outcomes matched what those networks would predict. The men who collaborated with women were not typical male collaborators; they had more patents, more collaborators, and broader technology portfolios than men who collaborated only with men. This pattern fits modern evidence that women may depend more on direct, high-trust collaboration than on looser professional ties (Meng, 2016), and echoes evidence from corporate networks that women may benefit from access to a well-connected contact rather than from occupying brokerage positions themselves (Burt, 1998).

These results suggest that collaboration in nineteenth-century France functioned primarily as a channel for knowledge access and movement across technology classes, not as a means of producing more novel ideas or becoming more productive. The dispersed network structure, with few brokers, no dominant hubs, and many small components, fits this interpretation. Collaboration was rare not because patenting was formally closed, but because useful cross-class matches were hard to form in a fragmented network. Entering the patent record was not the same as entering the collaborative relationships through which some knowledge moved. The paper contributes methodologically by introducing text-based measures of novelty and influence for a historical patent setting where citations are unavailable, building on modern evidence that text-based measures identify expert-selected breakthrough inventions more reliably than citation-based disruption indices (Yang, 2025). Substantively, it shows that collaborative ties shifted the technological direction, rather than the volume or novelty, of inventive work in nineteenth-century France. On gender, it shows that the largest observed inequality was access to collaboration, not the returns to collaboration once women entered the network.

The rest of the paper is organized as follows. Section 2 reviews the literature on collaboration, networks, and knowledge production and sets out working hypotheses that guide the empirical analysis. Section 3 describes the patent data, the record-linking procedure, and the construction of the text-based novelty and influence measures. Section 4 presents the event-study results on productivity, novelty, influence, and diversification around first collaboration. Section 5 reconstructs the inventor network, examines gender differences in access and network composition, and tests which network features predict diversification. Section 6 discusses the results and their implications.

2 Previous literature

2.1 Why collaborate? Complexity in knowledge production and collaboration

As knowledge accumulates and individual scope narrows, collaboration rises to cover missing expertise. Jones (2009), using U.S. patents (1975–1999), finds that age at first invention rises by about 0.6 years per decade, team size grows roughly 17 percent per decade, and specialization deepens, measured by a ~6 percent per decade decline in switching across technological classes between consecutive patents. Across science, social science, arts and humanities, and patents, teams increasingly occupy the top of the citation distribution by the late 1990s, displacing solos at the upper tail (Wuchty et al., 2007).

Complexity is uneven across fields, and collaboration does not automatically raise individual productivity. Teams in patenting are larger within fields where knowledge is deeper (Jones, 2009). At the macro level, sustaining semiconductor progress requires far more researchers: keeping Moore’s Law on pace involves roughly eighteen times the research effort relative to the early 1970s, with semiconductor research productivity declining about seven percent per year (Bloom et al., 2020). In research output, collaboration correlates with higher total publication counts but not with higher per-author productivity (Lee and Bozeman, 2005), echoing the pattern in Bloom et al. (2020) that the volume of collective invention grows while output per inventor falls. At the firm level, collaborative ties nonetheless predict subsequent innovation; at the individual level, brokerage across structural holes generates more valuable ideas (Burt, 2004; Kreutzer and Taalbi, 2026). These patterns document persistent pressure toward teamwork and specialization.

The nature of recombination also differs between solo and team work. Uzzi et al. (2013) show that the most influential scientific papers mix a small share of atypical, cross-domain combinations into otherwise conventional work, and that teams are markedly more likely than solo authors to achieve this balance. Their data show that solo authors match teams in baseline conventionality, but teams are markedly more likely to insert the atypical combinations that distinguish the highest-impact work. Chan et al. (2021) show that collaboration generally increases the likelihood of breakthrough inventions for functional or technological patents, but this advantage disappears for design patents, whose interdependent elements cannot easily be divided among team members. Wu et al. (2019) differentiate between the sizes of teams across the production of scientific papers, patents, and software and find that smaller teams tend to disrupt science and technology with new ideas and opportunities, while larger teams tend to develop existing ones. Patent data also show that collaboration reshapes both tails of the impact distribution: compared with unaffiliated solo inventors, patents produced by teams or within firms are more likely to become highly cited and less likely to be ignored. The combination of teamwork and firm affiliation roughly doubles the chance of producing a breakthrough, with the advantage linked partly to teams’ diverse experience and external networks (Singh and Fleming, 2010).

Rising knowledge complexity, specialization, and the division of problem solving go hand in hand. In knowledge hierarchies, routine problems are solved at lower levels and exceptional problems escalate to specialists (Garicano, 2000). In invention, collaborators supply non-overlapping capabilities, such as conceptual design, methodological execution, implementation, and commercialization, which are costly for one inventor to accumulate as knowledge becomes more complex. Solo breakthroughs persist when problems are simple or when finding the right collaborator is costly, so collaboration is conditional on complexity. Entry into collaboration should coincide with exposure to non-redundant knowledge and a larger feasible problem set; whether such gains persist beyond the first collaboration is an open empirical question, distinct from teams’ higher citation impact.

These patterns are documented almost exclusively in modern data, where corporate labs, universities, and formal intellectual property systems structure the conditions for teamwork. The logic itself is older: wherever knowledge grows more complex than one person can master, collaboration fills the gap.

2.2 Networks, ideas, and knowledge production

2.2.1 Network position, brokerage, and boundary-spanning

Positions in a network shape both where new ideas come from and whether those ideas spread. Brokers who span “structural holes” enjoy a vision advantage. By bridging otherwise disconnected groups, they encounter different perspectives and can combine approaches that those within a single group would miss (Burt, 2004). Classic weak-tie logic explains part of this advantage: weak ties connect dissimilar circles and supply non-redundant information (Burt, 1992) that is especially useful for mobility and diffusion, while strong ties provide trust and motivation but rarely bridge (Granovetter, 1973). Yet novelty creation and knowledge use rely on different structural ingredients. Cohesion (dense, overlapping ties) fosters cooperative norms and eases knowledge transfer, while range (ties into diverse pools) equips actors to reframe ideas for different groups; both cohesion and range independently facilitate knowledge transfer beyond the effect of dyadic tie strength (Reagans and McEvily, 2003). Range retains its effect across structural holes. Individuals with diverse contacts can transfer knowledge even where third-party ties are absent (Reagans and McEvily, 2003). Whether structural holes help or hinder depends on the objective. Among competing firms in the chemicals industry, Ahuja (2000) finds that holes reduce innovation because joint resource sharing demands the trust that closed networks provide. At the individual level, weaker ties are generally creativity-enhancing, whereas strong ties are neutral on average; moreover, being central helps creativity only when external ties are limited—pairing high centrality with many outside ties is not optimal (Perry-Smith, 2006).

Evidence from patenting sharpens these mechanisms. In career-level data on U.S. collaborative inventors (1975–2002), collaborating through brokered structures is linked to more “generative creativity”, the novel combinations of patent subclasses, but brokered ideas are less likely to be used later by others; a diffusion penalty that cohesion can mitigate (Fleming et al., 2007b). Brokerage requires multiple collaborators by definition; for inventors in a single partnership, Fleming et al. (2007b) argue that the partner’s diverse experience can partially substitute, bringing non-redundant information into a cohesive structure that would otherwise recycle familiar knowledge. In contemporary French co-inventor networks, Cassi and Plunket (2015) confirm that bridging ties form precisely when inventors seek technological skills unavailable in their immediate network. At the regional scale, the widely assumed benefits of “small-world” structure do not materialize once measured directly: clustering by itself shows no robust association with innovative output, whereas shorter average path lengths and larger connected components are positively associated with subsequent patenting (Fleming et al., 2007a). Two things matter: (i) access to collaborators who bring knowledge the inventor does not already have, and (ii) sufficient connectedness to spread ideas once generated. Generic clustering delivers neither directly.

These findings rest on modern data: twentieth-century US patents, organizational studies, and corporate supply chains. Burt (2004) draws on historical anecdotes, from Cosimo de’ Medici to the early American cotton and electricity industries, to place the brokerage advantage in wider perspective. Direct evidence on historical collaboration networks is sparse. Berger and Prawitz (2024) show that railroad expansion increased collaborative patenting in nineteenth-century Sweden, but track tie formation rather than downstream consequences for individual inventors. Andersson et al. (2019) map the structure of co-inventor networks in Sweden and Spain during 1878–1914, finding that both were highly fragmented, with roughly ninety percent of patents filed by single applicants and no giant component in either country. Earlier historical work describes collective invention as concentrated in dense, localized communities where competing producers freely exchanged technical information (MacLeod and Nuvolari, 2016).

2.2.2 Gender and networks in knowledge production

Network structure matters for gender gaps in knowledge production. Women’s underrepresentation in science and patenting persists to this day despite decades and centuries of rising participation (Hanson, 1996; Jung and Ejermo, 2014; Khan, 2024; Merouani and Perrin, 2026b; Rossiter, 1995). Structural holes typically confer advantages by providing access to diverse, non-redundant information (Burt, 1992, 2004). Yet networks rich in structural holes can disadvantage women, particularly when they are not recognized as established members of the field (Burt, 1998; Ling et al., 2025). Burt (1998) argues that legitimate insiders build social capital by spanning structural holes directly, while outsiders must borrow it through a strong tie to an established sponsor who

already holds a bridging position. Yang et al. (2019) complicate the sponsorship model, showing that women in a modern graduate program achieve high placement not through a single hierarchical sponsor but through a small inner circle of female peers whose non-overlapping external ties provide both information and access that no single peer could supply alone. The core mechanism is trust. Gatekeepers require stronger signals of competence from outsiders before granting access to resources (Lin, 2001; Meng, 2016). Among contemporary university inventors, men access commercialization knowledge through weak ties like former students and acquaintances; women access the same knowledge through channels requiring higher trust, particularly those they directly collaborate with (Ding et al., 2006; Meng, 2016) or for instance family members. In nineteenth-century France specifically, Khan (2016) argues that family firms served as the primary conduit through which women accessed innovation, compensating for coverture laws, exclusion from formal schooling, and restricted access to capital markets. Once women secure trusted ties, the deficit does not merely disappear: the knowledge and opportunities gained through close collaboration produce outcomes that equal or exceed men's, because for men the same close tie adds little over what they already obtain through looser connections (Meng, 2016). From a network perspective, the problem is not what collaboration provides but who gets access to it.

Context determines whether network advantages translate into outcomes. Tokenism theory predicts that raising women's proportional representation reduces bias (Kanter, 1993), but intrusiveness theories predict the opposite: as minority representation grows, perceived competition triggers backlash (Blalock, 1967; Yoder, 1991). Empirical evidence confirms that the effects of structural holes are contingent on women's proportional representation in the field. Structural holes predict citations positively for women in low-representation settings but become negative once women's field representation exceeds a certain threshold, a pattern that points to backlash rather than inclusion (Ling et al., 2025). The returns to having ties that cross group boundaries vary with the broader institutional environment.

2.3 Patenting and France's economic context in the nineteenth century

For decades, historians saw French industrialization as a failure. Clapham (1921) and Landes (1949; 1969) blamed conservative social structures and stubborn attachment to small-scale craft production. Roehl (1976) and later O'Brien and Keyder (1978) flipped the narrative: France did not lag; it specialized. Where Britain mass-produced, France made quality goods, especially in textiles and luxury items (Walton, 1992). The numbers support specialization. Textiles employed 40 percent of France's industrial workers in 1860. Textiles and food together generated 62 percent of industrial output (Merouani and Perrin, 2026b). This was an economy built on craft skill and niche production, not mass manufacturing. Collaboration networks, if they existed, had to span fragmented workshops and dispersed knowledge.

France made patents a natural right on January 7, 1791 (Marchal, 2009). Enlightenment principles drove the choice. Inventors owned their ideas, and the state could not censor them by judging novelty in advance (Galvez-Behar, 2019). Five-year patents cost 300 livres plus a 50-livre fee; ten years cost 800; fifteen cost 1,500 (Baudry, 2019; Nuvolari et al., 2023). Those sums excluded most artisans and favored inventors with money or backers (Khan, 2020). The law added another burden: use your invention within two years or lose your claim (*Loi relative aux découvertes utiles et aux moyens d'en assurer la propriété à ceux qui seront reconnus en être les auteurs, 1791*). That clause raised the stakes beyond filing cost.

The absence of formal examination did not eliminate scrutiny. The *Comité consultatif des Arts et des Manufactures* reviewed applications informally, offering opinions that could shape outcomes and adding a degree of discretion to what was intended as an open system (Baudry, 2015, 2019; Hirsch and Minard, 1998). Patent details stayed locked in manuscript at local offices, rarely printed, which choked off knowledge spillovers (Galvez-Behar, 2019), in sharp contrast with the American system's requirement of immediate public disclosure (Khan and Sokoloff, 2008). Worse, the law granted "import patents" to anyone bringing foreign inventions to France first, rewarding middlemen over creators (Emptoz and Marchal, 2002; Galvez-Behar, 2019). Import patents moved technology across borders (Nuvolari et al., 2023) but corrupted the link between invention and reward.

The patent reform of July 5, 1844, overhauled the system, but not without contest. In the *Chambre des pairs*, Vicomte Dubouché defended the 1791 framing of invention as a natural right. A patent,

he argued, was “véritable propriété,” true property born of science, labor, and persistent effort, sometimes at considerable expense, by men “peu favorisés de la fortune,” little favored by fortune. Strip the word property from the law, he warned, and what remains is the language of privilege and monopoly, words that would prejudice courts against the very inventors the law was meant to protect ([Gazette des tribunaux, 1843](#)). The new law clarified property rights, patents now explicitly gave inventors exclusive commercial control (Article 1), and killed import patents to favor original work over borrowed ideas ([Galvez-Behar, 2019](#)). Foreigners could file on equal terms, and annual installment payments of 100 francs cut upfront costs ([Galvez-Behar, 2019](#)). Courts, not bureaucrats, judged validity. Patents still carried no government guarantee (Article 33), which preserved the natural-rights principle but left inventors vulnerable to litigation ([Galvez-Behar, 2019](#)).

The reform went beyond lowering costs. Because the French system was a registration regime that did not investigate civil capacity, it had been formally open to married women and minors since 1791 ([Galvez-Behar, 2019](#)). The 1844 law reinforced this openness: Article 5 declared the formalities applicable to “whoever wishes to take out a patent” ([Loi du 5 juillet 1844 sur les brevets d’invention, 1844](#)), and an accompanying administrative instruction specified that the administration was not authorized to inquire into civil capacity, so patents must be issued even when the applicant was a married woman, a minor, or a legally incapacitated person ([Merouani and Perrin, 2024](#)). In Britain, by contrast, married women’s property rights remained restricted by coverture until 1870 and 1882 ([Khan, 2020](#); [Merouani and Perrin, 2026b](#)). Women’s likelihood of filing a patent rose 40 percent after the reform ([Merouani and Perrin, 2026b](#)). Annual patents jumped from 1,200 in the early 1840s to 5,800 by the late 1850s; a contemporary statistician attributed the surge directly to the new fee structure, noting that the law’s “fractionnement en annuités de 100 fr.” had transformed access ([Loua, 1891](#)).

Khan ([2005, 2020](#)) calls the French system oligarchic: high fees, secrecy, and rent-seeking locked out common inventors and protected elites. Galvez-Behar ([2019](#)) counters that natural-rights ideology made French patents more democratic than Khan admits. Merouani and Perrin ([2026b](#)) agree with the latter view. They find that among women patentees with known occupations, 31 percent held medium-skill jobs and 25 percent came from the bottom: seamstresses, tailors, unskilled laborers. France issued more patents per capita than the United States during 1840–1860 ([Merouani and Perrin, 2026b](#)) and matched American rates of one-time patenting, a key accessibility marker ([Nuvolari et al., 2023](#)). Measured in working days of skilled mechanics, French patent costs were 30 to 45 percent lower than British ones, though still well above American fees ([Nuvolari et al., 2023](#)). Over the full nineteenth century, women in France filed 1.8 percent of all patents, compared to 1 percent in Britain and less than 0.5 percent in the United States ([Khan, 2024](#); [Merouani and Perrin, 2026b](#)). High fees mattered, but they did not close the door. The gender gap in patenting was smaller in France than in either Britain or the United States.

2.4 Collaboration in the patent record

Because this unusually accessible but still uneven system is observed through patent records, co-patenting is a conservative and imperfect proxy for the knowledge exchange that surrounded nineteenth-century invention. Most sharing happened informally and left no patent trail. Moser ([2005](#)) finds that only 11 percent of British and 14 percent of American inventions exhibited at the 1851 Crystal Palace were patented. Industries at the technological frontier, from Cornish steam engines to Cleveland ironworks, advanced through collective exchange while largely avoiding the patent system ([Bessen and Nuvolari, 2016](#)). When multiple names did appear on a patent, the relationship behind them varied widely. In the nineteenth-century United States, Lamoreaux et al. ([2013](#)) show that financial backers typically organized their stakes through patent assignments rather than co-inventor status. Co-patenting therefore usually reflected inventive collaboration, though backers occasionally appeared as co-inventors when they contributed technical solutions. In the French system, patent agents and other intermediaries appeared on the filing documents but were not recorded as inventors, which further separates the co-inventor listing from financial or legal intermediation. Even with that separation, co-patenting captured a range of relationships. Chanteux ([2023](#)) documents the full range: women co-patented as capital providers, as commercial fronts for male relatives, and as strategic extensions of family monopolies, but more often as genuine co-inventors in joint ventures with husbands, brothers, and business partners.

The variation is real, yet historians have often resolved the ambiguity by defaulting to the assumption

that women were mere proxies. Merouani and Perrin (2024) show that this assumption frequently fails when tested against the original application documents. Jeanne Geneviève Labrosse, long dismissed as a legal proxy for her husband’s parachute patent, explicitly referred to “this invention, of which I am the author” in the manuscript petition, and the application included documents signed by her husband and his lawyers attesting to her right as co-inventor. Madame Redier held over thirty patents in barometric instruments, a domain entirely distinct from her husband’s specialty, yet biographical sources acknowledge her only as his wife. Co-patenting therefore does not measure symmetric collaboration. What it does capture is a documented link between two individuals who shared a stake in a specific invention, and Singh (2005) demonstrates that such co-inventor ties predict subsequent knowledge diffusion, with the probability of a knowledge flow roughly four times that of unconnected pairs, even after controlling for geographic and organizational proximity.

2.5 Working hypotheses

I take four working hypotheses from this literature. First, if collaboration lets inventors combine specialized skills that would be costly to accumulate alone, first collaboration should improve the reach or quality of inventive output, most plausibly through greater influence and possibly through greater novelty, without raising, and possibly lowering, per-inventor output (Jones, 2009; Lee and Bozeman, 2005; Singh and Fleming, 2010; Uzzi et al., 2013). Section 4 tests annual patenting, novelty, and influence around first collaboration.

Second, if collaboration matters because it brings non-redundant knowledge, first collaboration should redirect inventors toward technology classes outside their previous home class. This prediction is strongest when the first collaborator has experience in a different technological class, because bridging ties expose inventors to information that their existing network is unlikely to contain (Cassi and Plunket, 2015; Fleming et al., 2007b; Granovetter, 1973; Reagans and McEvily, 2003). Section 4 tests the timing and class split, and Section 5 then decomposes the tie structure.

Third, later boundary-crossing should depend on the structure of the accumulated network, not only on the first collaborative tie. Range should increase diversification by exposing inventors to different knowledge pools (Burt, 2004; Fleming et al., 2007b; Reagans and McEvily, 2003), while cohesion and repeated ties should help inventors transfer and use knowledge that is difficult to move through arm’s-length contacts (Ahuja, 2000; Fleming et al., 2007b; Reagans and McEvily, 2003). Section 5 evaluates these channels.

Fourth, modern evidence warns that the same network access need not yield the same returns for women and men (Burt, 1998; Meng, 2016), but nineteenth-century France adds a prior access barrier: legal, educational, and capital-market constraints made family firms an important route into innovation (Khan, 2016). I therefore expect access to be the first-order margin, with lower collaboration entry for women and greater reliance on trusted or well-positioned partners, including family firms. Conditional on collaboration, whether women’s redirection follows their ties or still falls short of men’s is the open question that Section 5 tests.

3 Materials and methods

3.1 Data sources and inventor identification

I begin with the French Patent Database, 1791–1900, which records roughly 390,000 patents as patent-level entries (Merouani and Perrin, 2026b). Each entry lists one or more inventors associated with a single patent ID, but the database offers no way to determine which inventors are unique across the full corpus. To address this, I reshape the data into an inventor–observation table in which each inventor–patent pairing constitutes a distinct row, yielding about 448,000 observations. I then classify these inventor-observations into four types of entities, each requiring its own record-linking strategy: 406,868 male individual inventors, 6,916 women¹ individual inventors, 8,205 family-firm inventors, and 26,156 non-family firm inventors.

¹I follow Merouani and Perrin (2026b) approach to gender identification, which uses gender-specific honorifics for women: *madame*, *mademoiselle*, and *veuve* found in the manual transcription of patent notes. I extend this by detecting the same honorifics directly in the patent application scans using the vision-language model pipeline described in Appendix B, which increases the count of women-linked patents by 10.

The difficulty of record linking varies across these groups. Firms are the most straightforward, because the samples are smaller and names tend to be stable. Women inventors pose greater challenges. The small sample size belies the complexity introduced by naming conventions, including surname changes with marital status. Male inventors are the hardest, largely because of scale. With more than 400,000 observations, even minor ambiguities, such as missing or very common given names, compound into a substantial linking problem.

3.1.1 Firm inventor-identification

I rely on assumptions based on close inspection of the underlying records, including application documents, signatures, dates, and places, to determine when two firm-inventor observations refer to the same entity. These assumptions are chosen to match the qualitative structure of the data for as many entities as possible. A family firm is never linked to a non-family firm, and vice versa. I distinguish between the two by using naming conventions. Non-family firms usually include generic multilingual organizational markers such as *Fabrik*, *Société*, or *Company*. Family firms often signal kinship ties in their names, for example “& Fils,” “& Sons,” or “Frères.” A firm is assumed to keep the same name over time. Even though a firm may persist across generations of directors, I also assume that it will not survive more than fifty years without producing a new invention. Two observations more than fifty years apart are therefore never linked to the same entity unless a third observation falls within fifty years of both and connects them. In practice, almost no observations reach this threshold. It is included mainly to prevent linking firms across implausibly long periods of inactivity.

In addition to these core assumptions, I apply three restrictions separately in order to construct alternative sets of links. In one case, I require inventors to share the same HISCO code, the historical international classification of occupations. In another, I require their inventions to overlap in technical vocabulary, which I capture by identifying at least one common keyword extracted from the patent title.² Throughout the paper, “technology class” refers to the twenty broad technological sectors defined by the Institut national de la propriété industrielle (INPI), such as Textile, Machinery, Medicine and Hygiene, and Clothing. I build on the classification in Merouani and Perrin (2026b), extending it with domain-adaptive pre-training and an expanded training set.³

I set up additional rules to avoid implausible links, such as linking within the same patent application, and stabilize linking across patent applications within the same patent family.⁴ Taken together, these restrictions yield about 201,000 firm comparison pairs and 51,000 family-firm pairs. When clustered, these pairs resolve into 12,295 distinct firms and 3,240 distinct family firms.

3.1.2 Individual inventor-identification

For individuals, the first name is the most important complement to the surname in distinguishing inventors across patents. To enrich the French Patent Database, I therefore fill in missing first names by turning directly to the original patent application documents. For female inventors, I manually transcribe all missing first names. For male inventors, I use an ensemble of two vision-based large language models to extract first names from more than 150,000 handwritten applications that otherwise lacked this information.⁵ I then quality-assure and manually clean the transcriptions. When the models do not agree, I resolve the cases myself, focusing especially on the original applications linked to the twenty most frequent surnames. Altogether, I decrease missingness in the first name field from 65 to 33 percent for male inventor-observations.

As with firms, I rely on blocking rules to generate candidate matches: about 13,000 comparison pairs for women inventors and more than 1.5 million for male inventors.⁶ The female comparison pairs are resolved using 51 rules that codify the manual linking I carried out for every inventor in

²See technical keyword section in Appendix A.1.

³I continued the language model’s training on all French and Italian patent titles filed up to 1930 before asking it to classify, so that it recognizes archaic and technical vocabulary from the period. I then expanded the labeled training data from roughly 40,000 to over 140,000 patents using a consensus of three independently trained classifiers, adding a manual label only when automated methods disagreed. The improved classifier reaches an F1 score of 0.95 at the main class level on a held-out test set of 8,015 patents.

⁴See all firm blocking rules in Appendix A.2.1.

⁵See procedure for machine transcription in Appendix B.

⁶See exhaustive list of blocking rules for male inventors in Appendix A.2.3 and female inventors in Appendix A.2.2.

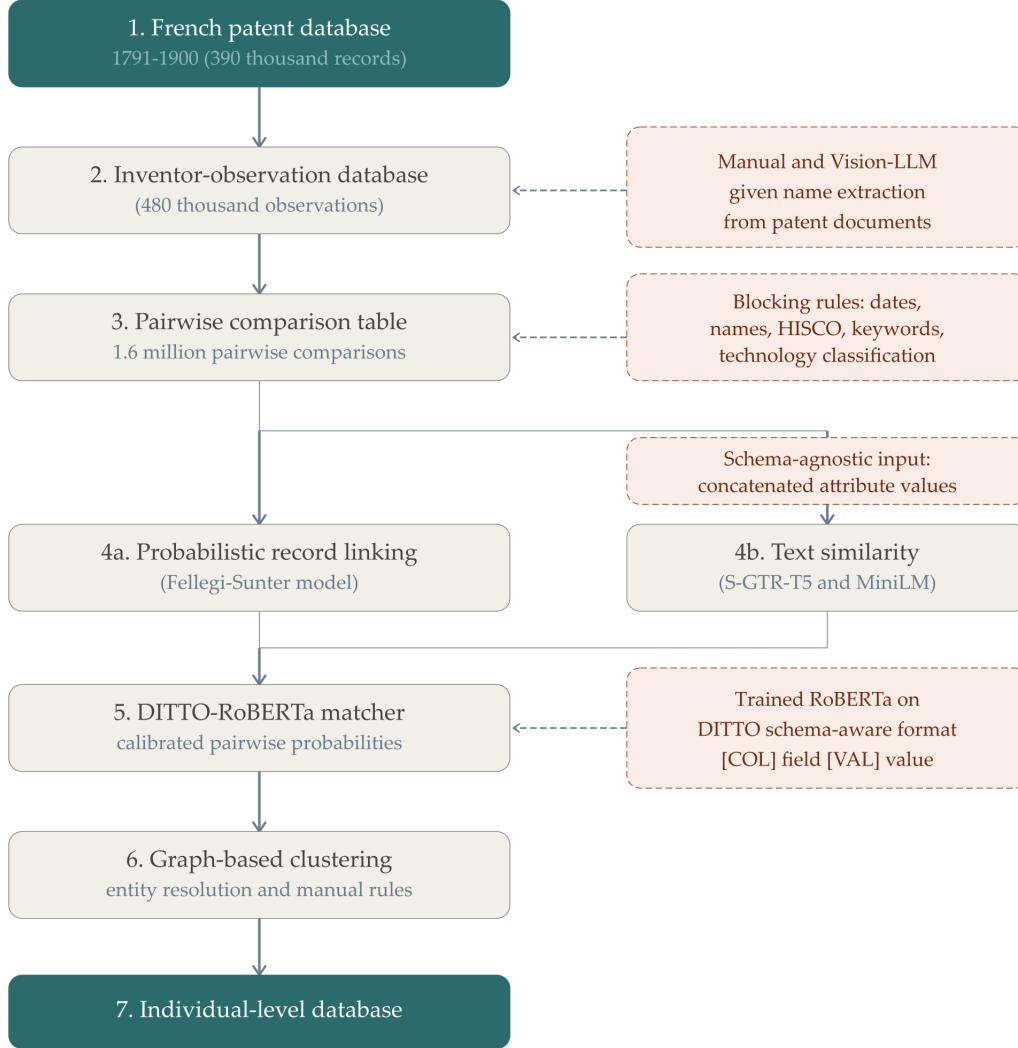


Figure 1: Record linking pipeline for male inventors. The process begins with 390 thousand patent records enriched with Vision-LLM-transcribed given names, generates 1.6 million pairwise comparisons via blocking rules, and resolves them through four stages: probabilistic scoring (Fellegi-Sunter), unsupervised text similarity, supervised matching (DITTO-RoBERTa), and constraint-aware clustering. Solid boxes trace the main pipeline; dashed boxes describe methodological inputs at each stage.

the sample, each tailored to a specific blocking strategy. The rules make the process transparent and replicable as the database grows, but the underlying work was examining each cluster against the original patent applications, biographical details, and co-inventor networks by hand. The rules draw on features such as surname, first name, marital status, maiden name, occupation, location, year of application, patent identifiers, patent titles, and co-inventors.⁷ Clustering the matched pairs yields 4,165 distinct female inventors.

Linking male inventors is the hardest part of the exercise. The pair comparison table holds over 56,000 rows of men with the most frequent surname Martin, and 9,700 of the runner-up “Smith.” A single missing first name, or a slight variation like “Jean” versus “Jean-Baptiste,” can make one person look like several or several people look like one.

I begin with rules, which resolve about 60 percent of the cases. Some inventors appear on every application within a patent family, leaving little doubt about their identity. Others can be ruled out immediately when their first names or important details conflict. These “must-link” and “cannot-link” rules are cheap and reliable, but they run out of power quickly. The remaining 40 percent of pairs are ambiguous, and here I need more sophisticated tools. I build a four-stage process to handle them, sketched in Figure 1.

The first step is to bring in a probabilistic framework. I use the Fellegi–Sunter model (Enamorado et al., 2019; Winkler, 1988), which assigns a probability that two records belong to the same inventor based on their agreements and disagreements. It accounts for the fact that some features are more informative than others. Sharing the surname “Siemens” means more than sharing “Martin.” Yet even here the model has limits. It struggles to link the anglicized name of Carl Wilhelm Siemens, Charles William Siemens, without pulling in many false matches. The model is noisy, but it gives a useful baseline.

The second step is to collapse information into text and let modern language models judge how close two records sound to one another. I combine surname, first name, occupation, technology class, co-inventors, year, and keywords from the patent title into a single string for each record. Models like S-GTR-T5 and MiniLM, which have been shown to excel at this kind of similarity task (Zeakis et al., 2025), produce scores that capture subtle connections: “forge master” and “iron works” may not look alike to a rule-based system, but they do to a model trained on language. These schema-agnostic comparisons widen the lens, picking up cases that fixed fields cannot.

The third step is to refine these signals with supervised learning. I draw a training sample of 25,000 pairs, balanced between matches and non-matches, and I deliberately degrade, dropping first names or occupations, so that the training set mirrors the imperfections of the real data. I then fine-tune a RoBERTa-large model using the DITTO framework (Li et al., 2023; Zeakis et al., 2025). Unlike the schema-agnostic approach, DITTO reads the attributes in a structured format, learning which pieces of information matter most. The result is a calibrated match probability that is sharper than either the rules or the unsupervised similarities alone.

The final stage clusters these predictions into inventor identities. The clustering procedure respects the hard rules from the first step. Matches that must hold are enforced, and conflicts that cannot hold are excluded. Within these bounds, I build graphs of candidate inventors, add edges when probabilities exceed calibrated thresholds, and merge nodes while respecting realistic limits, such as a maximum career span. Where clusters stretch implausibly far, weak edges are cut to restore plausibility.

The logic of this sequence is pragmatic. Rules handle the obvious cases. Probabilistic models add a foundation of structure. Language models broaden the net, capturing similarities that rules miss. Supervised learning sharpens the distinctions. Clustering then imposes discipline to transform noisy fragments into coherent inventor entities. Each step has limits, but together they allow me to move from scattered patent filings to a credible reconstruction of 233,163 distinct male inventors from 406,868 observations. This yields a ratio of 1.7 observations per male inventor entity, closely matching the 1.65 observed for the female inventors, whose entities I linked by hand from the original documents. The two error types have opposite biases for within-inventor analyses. False positive links, merging distinct inventors into one entity, inflate within-inventor variation by combining unrelated career trajectories. False negative links, splitting one inventor into two,

⁷See Appendix A for a list of all features used.

attenuate within-inventor variation by truncating careers. The clustering stage is designed to be conservative. The maximum career-span constraint and the hard cannot-link rules prioritize splitting over merging, which favors false negatives over false positives. A validation exercise on 200 stratified pairs yields a population-weighted error rate below 10 percent, concentrated among high-frequency surnames at borderline confidence. I drop the most frequent surnames in a robustness check to confirm that this does not drive the results.

3.2 Novelty and influence

Economists have long measured novelty by asking how far a new idea sits from the existing stock of knowledge at the moment it arrives. The classic tools use backward citations and technology classes. Hall, Jaffe, and Trajtenberg’s originality index summarizes whether a patent cites prior art concentrated in a narrow set of technology classes or scattered across many (Hall et al., 2001). Jaffe-style proximity treats ideas as points in a high-dimensional space and asks whether a new point lies in a crowded neighborhood or on the fringe (Jaffe, 1986, 1989). The recombinant search tradition, from Fleming through the atypical combinations literature, views novelty as rare pairings of components that are seldom seen together (Fleming, 2001; Uzzi et al., 2013; Weitzman, 1998). The common intuition is simple. An invention is novel when it resembles few of the patents that came before it, especially when it combines ideas from distant technology classes rather than reusing what is already common in its own class. This framing sits squarely in the knowledge production tradition (Griliches, 1990).

Influence or importance is the forward mirror of that logic. The standard measures follow how often later patents cite a focal patent and how widely those citations spread across technology classes. Forward citations are taken as a noisy but informative count of downstream use, and the generality index asks whether those downstream uses stay inside a technology class or spill broadly (Hall et al., 2001). Work that links citations to economic value shows that influential inventions leave many traces in the subsequent record. Forward citations correlate with firms’ market value (Hall et al., 2005), with survey-based value estimates and patent renewal behavior (Harhoff et al., 1999), where renewals themselves proxy for private value (Pakes, 1986; Schankerman and Pakes, 1986), with litigation risk as a proxy for stakes (Lanjouw and Schankerman, 2001), and with the resale and reassignment of patents in secondary markets (Serrano, 2010). The intuition again is straightforward. A patent is influential when it becomes a template that later inventors find it convenient to reuse, and it is broadly influential when that reuse crosses technology class boundaries rather than recycling within one niche.

Nineteenth century France lacks systematic citations and full abstracts, so the usual measures are unavailable. My strategy is to recover the same economic primitives using the language every patent supplies. Using patent titles, I build backward and forward measures that align with the originality and generality intuition. This strategy follows the historical-innovation literature that reconstructs novelty and diffusion when modern citation data are unavailable (MacLeod and Nuvolari, 2016; Nuvolari and Tartari, 2011).

My approach builds on the framework developed by Kelly et al. (2021), who use text-based backward and forward similarity to measure novelty and influence when citation data are not available.⁸ Their core insight is simple. A patent is novel when its language differs from what came before it, and it is influential when later patents pick up its language. La Mela et al. (2024) apply this logic to historical Swedish patents.⁹ Yang (2025) shows, across six million USPTO patents,

⁸Kelly et al. (2021) measure patent quality using the language patents share across time. They represent each patent’s text as a weighted list of words, where unusual technical terms receive more weight than common ones. Critically, they weight terms according to how common they were *up to* the moment the patent was filed, so that early uses of new terminology are recognized as novel rather than penalized because the terms later become widespread. They then measure how similar two patents are by comparing these weighted word lists. For a patent filed in year t , they sum the similarity scores with all patents in the prior five years to get a backward similarity score, and sum similarities with all patents in the subsequent five years to get a forward similarity score. Quality is the ratio of forward to backward similarity, adjusted by removing year-specific averages. They show this measure predicts future citations and correlates with market value. Their approach relies on adding up raw similarity scores, which means a few very similar patents can dominate the measure, and their year-adjustment makes cross-period comparisons difficult. My threshold-based counting approach with explicit normalization for differences in prior-art availability addresses both issues while preserving their core insight.

⁹La Mela et al. (2024) adapt the method of Kelly et al. (2021) to historical Swedish patents (1890–1929). They make one important modification. Instead of summing similarity scores, they average them by dividing by the number

that text-based cosine similarity identifies expert-selected technological breakthroughs more reliably than citation-based disruption indices. The text-based measure ranks these milestones at the 92.8th percentile on average, the citation-based index at the 60.4th, because patent text captures knowledge content directly where citation links often fail to acknowledge foundational prior art.

I follow the same intuition. Where Kelly et al. (2021) compare patents through weighted word-count overlap, I compare the underlying meaning of titles using a domain-adapted language model. This allows me to see that “vulcanizing rubber” and “hardening caoutchouc” describe the same idea even when they share no words. Before laying out the full procedure in the next section, I briefly summarize how my implementation further adapts the approach for historical conditions in four ways.¹⁰ First, some titles use generic language that resembles many others, such as “improved method for manufacturing.” I apply a correction that rewards pairs of patents that are specifically similar rather than generically similar.¹¹ Second, instead of adding up all similarity scores with neighbors, I count how many neighbors are truly close, close enough that the match is unlikely to occur by chance. This yields an interpretable quantity: the number of semantic ancestors or descendants a patent has. Third, I normalize these counts by the amount of prior art available to each patent. A patent filed during a busy period faces more potential matches than one filed in a quiet year, and a patent in a crowded technology class faces more than one in a sparse class. Fourth, I allow patents to belong partly to several technology classes at once, which better reflects multi-purpose inventions. These differences produce novelty and influence measures that remain comparable across time periods and technology classes.

3.2.1 Novelty

I construct a backward-looking measure of a patent’s novelty that asks a simple question: at the moment a patent is filed, how much does its title echo ideas that were already present, and does that resemblance sit within the patent’s own broad technology class or spill across different ones. To answer that, I translate patent titles into a common idea space, look back twenty years, and count the earlier titles that land unusually close to the new patent’s title. I then scale those counts by the amount of prior art available to each patent, since some years and technology classes offer far richer prior art than others, and I finally express the result as a percentile within the patent’s own year and technology class. A value at the 90th percentile means the patent’s title resembles much of its accessible past. I define novelty as one minus this backward percentile, so higher values correspond to greater originality. Formally, I refer to the underlying resemblance process as Backward Semantic Adoption (BSA). I use $BSA_{i,g}$, $BSA_{i,in}$, and $BSA_{i,x}$ for the global, within-class, and cross-class components of patent i .

The core problem is separating genuine resemblance from the mechanical fact that busier periods and crowded technology classes offer more chances for two patents to look alike. I begin with all patents’ titles and their application years. Each patent carries a vector of memberships across twenty broad technology classes, denoted $\mathbf{f}_i = (f_{i1}, \dots, f_{iC})$ with $\sum_c f_{ic} = 1$.¹² I do not treat class membership as all-or-nothing. Instead, a patent can be partly associated with several classes, with each element f_{ic} representing the strength of its connection to class c . When I refer to within-class resemblance, I mean links that fall in the same technology classes where the new patent is most strongly represented, while cross-class resemblance refers to links that occur in other classes. For

of patents in each time window. This adjustment accounts for the fact that patenting activity grew substantially over their period. They validate their measures using patent renewal data and a list of famous Swedish inventors, finding that their importance measure correlates with how long patents were kept in force. However, their averaging does not account for shifts in the technology mix. A patent in a crowded technology class faces more potential connections than one in a sparse class. My denominators adjust for both the total number of available patents and their distribution across technology classes, producing measures that are robust to changes in both patenting intensity and the technological composition of innovation.

¹⁰This is also done more in line with the development of machine learning methods for Out-of-Distribution detection and Novelty Detection. See Barcina-Blanco et al. (2024) for a recent review.

¹¹The correction is known as Cross-domain Similarity Local Scaling (CSLS) in the machine translation literature, where it was developed to match words across languages. In the context of my measure, a patent pair receives a high adjusted score only if they are close to each other but not close to many others. In practice, this correction substantially reduces false positives from generic phrasing. See Section 3.2.1.

¹²These class memberships use the twenty-sector technological classification defined by INPI and documented in Merouani and Perrin (2026b). I train a patent–technology–class classifier on historical patent data to assign memberships to these sectors. The model achieves an F1 score of 0.95, and I interpret each element f_{ic} as the strength of patent i ’s connection to class c .

percentile comparisons, I assign each patent a primary technology class, defined as the class with the highest membership share, purely to identify its peer group.

Patent titles are mapped into a high-dimensional semantic space using a fixed domain-adapted language model.¹³ Each title i is represented by a normalized embedding vector $\hat{\mathbf{e}}_i$, and closeness between two titles is measured by cosine similarity $s_{ij} = \hat{\mathbf{e}}_i^\top \hat{\mathbf{e}}_j$. Two titles are close if they describe the same idea in different words. For example, “process for vulcanizing rubber” and “method to harden caoutchouc” would sit near each other. For a patent filed in year t , I consider all earlier patents filed between $t - 20$ and $t - 1$ and down-weight those that appear further back by $w(\Delta) = 1/(1 + \Delta)$, where $\Delta = t - y_j$. This gives more weight to near-term antecedents and less to distant ones. I use a 20-year window rather than the shorter bins common in the literature to keep threshold calibration stable through the sparse early-century years. The encoder is fixed before scoring. For measurement, I preserve temporal order. For each patent, I compute neighbors, CSLS local averages, thresholds, denominators, and novelty percentiles using only titles filed before that patent.

Nearest-neighbor methods in such spaces have a well-known bias. Some titles act as hubs that look similar to many others simply because they contain common language. To correct this, I adjust each similarity by how generically close both titles are on average. The adjusted measure, known as the Cross-domain Similarity Local Scaling correction (CSLS), is

$$s_{ij}^{\text{CSLS}} = 2s_{ij} - r_i - r_j,$$

where r_i is the new patent’s average similarity to its nearest past neighbors and r_j is the past patent’s average similarity to its nearest seeds in the same rolling window. This adjustment rewards pairs that are specifically similar rather than generally similar to everything. A pair like “process for vulcanizing rubber” and “method to harden caoutchouc” retains a high adjusted similarity, while a generic title such as “method for manufacturing” is penalized because it sits close to many unrelated patents. In practice, I compute r_i within each seed year using the full set of potential predecessors in the twenty-year window. The reverse averages for past patents, which would otherwise require recomputing similarities against all seeds, are estimated using a subsample of seeds within each batch. This approximation keeps the results unchanged in expectation while making the computation manageable.

I define an ancestral link as a case where the adjusted similarity s_{ij}^{CSLS} between a new patent and an earlier patent is unusually high for that time slice. Rather than choosing this cutoff arbitrarily, for each filing year t I take a large random sample of seed–past pairs within the twenty-year window, compute their adjusted similarities, and set the global threshold $\tau_g^B(t)$ at the 99th percentile of that empirical distribution. Because the typical degree of similarity varies across technology classes, I also estimate class-specific thresholds for each seed year. For each class c , I compute a within-class threshold $\tau_{\text{in}}^B(c, t)$ that gives greater weight to past patents more strongly associated with that same class. I further define cross-class thresholds $\tau_x^B(c_s, c_f, t)$ that depend on how strongly the seed patent belongs to one class c_s and the past patent to another c_f , using weights proportional to $f_{ic_s} f_{jc_f}$ with $c_s \neq c_f$. This weighting emphasizes strong cross-links between well-defined classes rather than faint overlaps. All three thresholds are re-estimated for each seed year so that they adjust automatically to gradual shifts in language use and the distribution of technology classes over time.

With the thresholds established, I count earlier patents whose adjusted similarity exceeds them. For each patent i , let $\mathcal{N}_i^B$ denote the set of its 5,000 nearest predecessors in the semantic space among patents filed in the twenty-year backward window, and define the pairwise time-decay weight as $w_{ij} = 1/(1 + (t_i - y_j))$. I use subscripts g , in , and x throughout to denote the global, within-class, and cross-class components, and $\mathbb{1}\{\cdot\}$ for the indicator function, equal to one when its argument is true and zero otherwise. The global count is

$$\text{BSA}_{i,g}^{\text{raw}} = \sum_{j \in \mathcal{N}_i^B} w_{ij} \mathbb{1}\{s_{ij}^{\text{CSLS}} \geq \tau_g^B(t_i)\}.$$

¹³I domain-adapted the InfoXLM-Large language model on all French and Italian patent titles filed up to 1930. This exposes the model to the multilingual technical vocabulary of the era and substantially improves its ability to identify semantic similarity across languages and historical terminology.

To describe how resemblance maps onto technology classes, I construct two soft class-weighted components. The within-class count

$$\text{BSA}_{i,\text{in}}^{\text{raw}} = \sum_{j \in \mathcal{N}_i^B} \sum_c w_{ij} f_{ic} f_{jc} \mathbb{1}\{s_{ij}^{\text{CSLS}} \geq \tau_{\text{in}}^B(c, t_i)\}$$

captures reuse that remains inside the same broad areas. Each passing pair is weighted by the overlap of the seed’s and the past patent’s class profiles. The cross-class count

$$\text{BSA}_{i,x}^{\text{raw}} = \sum_{j \in \mathcal{N}_i^B} \sum_{c_s \neq c_f} w_{ij} f_{ic_s} f_{jc_f} \mathbb{1}\{s_{ij}^{\text{CSLS}} \geq \tau_x^B(c_s, c_f, t_i)\}$$

captures cases where the new patent resembles ideas from different areas, such as a machinery title echoing an earlier chemical process. For each patent, I evaluate the 5,000 nearest predecessors to manage computation. The global, within-class, and cross-class counts use different thresholds and (for the latter two) fractional class weights. They therefore do not sum algebraically; I treat them as three related measures of backward resemblance rather than an exact decomposition.

These counts must be adjusted for differences in the amount of prior art available to each patent. A patent filed in a year that follows many earlier patents simply has more chances to resemble something than one that follows a quieter period. Two patents filed in the same year also face different possibilities if one sits in a crowded technology class and the other in a sparse one. Simply dividing by the total number of prior patents does not account for this. I therefore normalize each count by a denominator that captures how much prior art was available in the relevant window, using the same time-decay weights. For patents filed in year t , the global backward denominator is

$$N_{t,g}^B = \sum_{\Delta=1}^H w(\Delta) n_{t-\Delta}, \quad H = 20,$$

where $n_{t-\Delta}$ is the number of patents filed Δ years earlier. For within- and cross-class rates, I also adjust for the class distribution of prior art. The within-class denominator for patent i is the dot product of the patent’s class-membership vector and a vector of time-weighted class masses:

$$N_{i,\text{in}}^B = \mathbf{f}_i^\top \mathbf{N}_{\text{class}}^B(t_i), \quad \text{where} \quad \mathbf{N}_{\text{class}}^B(t) = \sum_{\Delta=1}^H w(\Delta) \sum_{j: y_j = t-\Delta} \mathbf{f}_j,$$

where $\mathbf{f}_i^\top$ is the transpose of patent i ’s class-membership vector $\mathbf{f}_i$ (defined above), turning the multiplication into a weighted sum across classes. The cross-class denominator is the remainder once the within-class denominator is subtracted:

$$N_{i,x}^B = \left(\sum_c N_{\text{class},c}^B(t_i) \right) - N_{i,\text{in}}^B.$$

These denominators capture how much prior art a patent could have resembled, both within its own technological areas and across others. For consistency with the patent-level rate formula below, I index the global denominator by patent through $N_{i,g}^B \equiv N_{t_i,g}^B$.

I then compute backward rates as

$$\text{BSA}_{i,\ell}^{\text{rate}} = \frac{\text{BSA}_{i,\ell}^{\text{raw}}}{N_{i,\ell}^B}, \quad \ell \in \{g, \text{in}, x\}.$$

This divides each close-neighbor count by the corresponding time-weighted patent mass in the twenty-year window, producing a size-adjusted index of backward resemblance.

Finally, I express the global backward rate as a percentile within each patent’s own year and primary technology class and define novelty as one minus that percentile, so that higher values correspond to greater originality. A novelty score of 0.90 indicates that the patent ranks among the top ten percent of its contemporaries in linguistic originality. The resulting score lies on a common zero-to-one scale and is directly comparable across filing years and technology classes.

3.2.2 Influence

I measure a patent’s influence by asking the forward question of the novelty measure. Instead of asking how much a patent draws on ideas that already existed, I ask how much later inventions draw on it. The goal is to quantify how often a patent’s language is taken up by its successors, and whether that take-up remains within the patent’s broad technology class or moves across others. As before, I represent titles in a common idea space and look forward twenty years to identify later titles that land unusually close to the seed patent’s title. These forward matches, weighted by their temporal proximity, form the basic indicator of how much a patent’s phrasing reappears in the subsequent record. I refer to this process as Forward Semantic Adoption (FSA). The global component provides the forward signal used in my influence measure.

The computation follows the same steps as in Novelty, except that time now runs forward. For a patent i filed in year t_i , I compare its embedding to those of patents j filed between $t_i + 1$ and $t_i + 20$.¹⁴ Similarities are adjusted with the same CSLS correction so that generic phrasing does not inflate forward closeness, and I apply the same time-decay weight,

$$w_{ij} = \frac{1}{1 + (y_j - t_i)}.$$

I then define the forward adoption counts using an analogous empirical thresholding rule, calibrated separately on the forward window of each filing year. Let $\mathcal{N}_i^F$ denote the 5,000 nearest successors of patent i . The global forward count is

$$\text{FSA}_{i,g}^{\text{raw}} = \sum_{j \in \mathcal{N}_i^F} w_{ij} \mathbb{1}\{s_{ij}^{\text{CSLS}} \geq \tau_g^F(t_i)\},$$

where $\tau_g^F(t_i)$ is set at the 96th percentile of the empirical distribution of future pairs. Within-class and cross-class analogues are constructed in the same way. Because I evaluate only nearest neighbors, these counts measure retrieved downstream adoption.¹⁵

To separate how much later patents drew on a patent’s ideas from how much that patent already echoed prior art at filing, I scale the forward signal by the patent’s own backward count. Let $\text{BSA}_{i,g}^{\text{raw}}$ be the global backward count defined above. The baseline influence signal is the log ratio

$$I_i = \log\left(1 + \frac{\text{FSA}_{i,g}^{\text{raw}}}{\text{BSA}_{i,g}^{\text{raw}} + 1}\right),$$

which captures how often a patent’s ideas are reused per unit of inherited language. The log compresses the scale so that very large ratios receive less weight before the robust z-score stage. The +1 in the denominator avoids division by zero and dampens cases with very small backward counts.

I then place I_i on a common scale within broad time–class cells. Let $d(i)$ denote the decade of t_i and let $c(i)$ be the patent’s primary technology class. Within each (d, c) cell I compute a robust z-score using the median and the median absolute deviation (MAD),

$$\text{Influence}_i = \frac{I_i - \text{median}_{d(i),c(i)}[I]}{1.4826 \cdot \text{MAD}_{d(i),c(i)}[I] + \varepsilon},$$

where 1.4826 rescales the MAD to the standard deviation under normal tails and ε is a small constant for numerical stability.¹⁶

¹⁴I avoid issues of right censoring by expanding the patent database on the level of patent titles through the 1930s. The post-1900 titles come from the Espacenet patent database, which draws on the same INPI filing records as the pre-1900 data, so the title format is consistent across the boundary.

¹⁵For descriptive and robustness analysis, I also compute normalized forward rates and within-year, within-class percentiles. The headline influence measure is built in a second step from the raw forward and backward counts.

¹⁶The robust standardization uses $\text{MAD} = \text{median}(|I - \text{median}(I)|)$. The factor 1.4826 puts the MAD on the same scale as a standard deviation under a Gaussian benchmark. Numerically, a value of +1 on the robust z-scale is about one robust standard deviation above the median, which corresponds to roughly 1.4826 MAD above the median. I set $\varepsilon = 10^{-12}$ as a numerical guard against a zero MAD. An audit of the realized decade-by-class denominators across 197 cells and 307,513 patent observations confirms that this guard rarely binds: the patent-weighted fifth percentile of cell MAD is 0.22, and only 0.02 percent of patent observations fall in cells with MAD below 0.01.

Interpretation is direct. Positive values indicate that the patent’s language was taken up more than is typical for patents from the same decade and primary class, conditional on how much it resembled the past when it arrived. Negative values indicate below-typical propagation. Because the statistic is standardized within decade and technology class, it is comparable across periods and technology classes without further adjustments. Within-class and cross-class variants replace the numerator with the corresponding forward count while holding fixed the same global backward baseline in the denominator.

3.2.3 Do the measures hold?

A credible influence measure should rank the inventions that historians already treat as landmarks above ordinary patents, for example Pasteur’s preservation method, the Jacquard programmable loom, or the Lumière cinématographe. I benchmark the influence score against forty-one notable nineteenth-century patents drawn from standard histories and encyclopedias of invention.¹⁷ The set spans chemistry, metallurgy, precision instruments, machinery, navigation, construction, lighting and heating, and road transport. None of these labels enter the construction of the measure; they serve only as an external yardstick for whether the language of influential inventions echoes forward into later titles. If the score captures technological reach, known landmarks should float to the top without being hard-coded into the measure.

I summarize performance with rank-based accuracy. The area under the ROC curve (AUC) is the probability that a randomly chosen famous patent receives a higher influence score than a randomly chosen non-famous patent.¹⁸ Intuitively, draw one famous and one ordinary patent at random; AUC is the chance the score puts the famous one higher. Across the full 1791 to 1900 sample, the influence score attains an AUC of 0.609. Focusing on the period where forward semantic adoption is most informative, 1850 to 1900, the pooled (micro) AUC rises to 0.673, and averaging by decade in that window yields a macro AUC of 0.693.¹⁹ A permutation check that shuffles the fame labels within the late window centers near 0.500 with a dispersion of about 0.056, which confirms that the observed separation is not a mechanical artifact of sampling. Permutation functions as swapping name tags on the same patents; if the score had no signal, its rank accuracy would match the shuffled baseline. I focus on 1850 to 1900 for headline results because the twenty-year forward window is well populated; results are similar when I restrict to patents with a complete forward window of twenty years.

The signal strengthens over time. By decade, late-century accuracies are highest, with an AUC of 0.872 in the 1880s, and solid performance earlier in the window, for example 0.702 in the 1850s and 0.660 in the 1860s. By technology class, the measure performs best where imitation and adaptation were active. Marine and navigation, mining and metallurgy, chemical arts, and precision instruments all show high rank accuracy. Where ideas were traded and reused more often, language leaves more footprints, and a semantic measure can see those traces. In classes with only a single famous case, such as agriculture or construction in this sample, AUCs are unstable and should be interpreted cautiously. Low AUC in a cell with one famous case is due to small-sample volatility, not a change in construct validity.

I next ask whether the influence score adds information beyond fixed effects and a control for how derivative the patent was at filing. I estimate a logistic model for fame, defined as membership in the same forty-one-patent benchmark used above, with decade and major-class indicators and log backward resemblance as baseline predictors. This asks whether the score helps once we already know when and where a patent was filed and how derivative its title was. In five-fold cross-validation over 1850 to 1900 the baseline attains an AUC of 0.629. Adding the standardized influence score raises the cross-validated AUC to 0.661, an improvement of 0.032. At this level of class imbalance, a three-point AUC gain is economically meaningful. Out-of-time tests tell the same story. Training the model on filings through 1879 and evaluating on subsequent decades yields an AUC of 0.615 with influence included versus 0.571 without it. Training through 1889 and testing

¹⁷Table C.1 in Appendix C lists the forty-one benchmark patents, their short titles, technology classes, and application codes.

¹⁸An AUC of 0.50 corresponds to random ranking. An AUC of 0.67 means the score correctly orders about two-thirds of random famous versus non-famous pairs. Under class imbalance, AUC remains a stable pairwise-ranking metric, which is why I foreground it and use lift and gains curves to summarize the tail.

¹⁹The pooled (micro) AUC treats every patent equally across decades. The macro AUC averages decade-specific AUCs, giving each decade equal weight.

after yields 0.579 with influence versus 0.514 without. Training on earlier decades and testing later asks whether a rule learned before widespread electrification and large-scale steel production still flags the right inventions afterward. A cluster bootstrap that resamples decade-by-class cells produces an average AUC of 0.612 with a 95 percent interval from 0.519 to 0.708, which indicates that the separation is not driven by a handful of clusters.²⁰ Resampling whole decade-by-class blocks preserves within-cluster correlation, so the interval captures genuine uncertainty rather than overconfident independence. The construction is stable across reasonable choices of the ridge parameter in the denominator and of the winsorization cutoffs. Tuning within 1850 to 1889 selects $\lambda = 1$ and delivers the same headline AUCs.

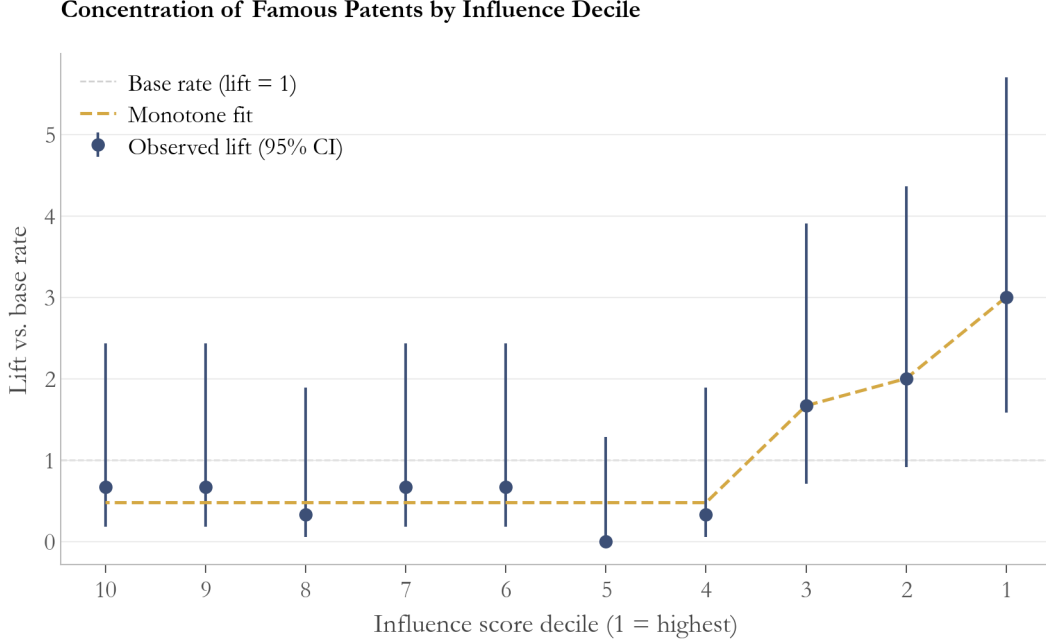


Figure 2: Concentration of Historically Famous Patents by Influence Decile

The tail behavior reinforces these rank statistics. In the late window, famous patents are several times more frequent in the upper tail of the score distribution than in the population as a whole. In pooled enrichment curves, the top one to two percent of patents by the influence score contain famous inventions at roughly six to seven times the base rate. A lift above one means famous patents are more common in that bin than in the population; for example, lift of three triples the base rate. Figure 2 shows the decile lift for 1850 to 1900, with a monotone decline across deciles, a top decile well above one, and Wilson confidence intervals that do not overlap the base rate.²¹ At extremely small K , precision at K is mechanically low because famous cases are rare. The absence of hits at K in per-decade 0.1 percent slices is therefore not informative by itself; the pooled enrichment and gains curves are the correct diagnostics under this class imbalance.²²

As a robustness check, I re-express the forward signal as a residual. Intuitively, this isolates the part of forward take-up that is not explained by how derivative the language was or by time and technology class. I regress log forward counts on log backward counts with decade and class fixed effects, take residuals, and standardize within decade by class. This alternative “excess influence” statistic achieves an AUC of 0.668 in the late window, which is nearly identical to the primary score, and confirms that the results are not specific to any one transformation.

²⁰I re-estimate the score using $\lambda \in \{0.5, 1, 2, 5\}$ in the ratio $\log(1 + \frac{\text{FSA}_{i,g}^{\text{raw}}}{\text{BSA}_{i,g}^{\text{raw}} + \lambda})$. For winsorization of the log-ratio, I compare percentiles (0.1, 99.9) and (1, 99.5). Headline AUCs for 1850–1900 are identical to three decimals across these settings, and the by-decade ordering is unchanged. Ridge λ controls variance when $\text{BSA}_{i,g}^{\text{raw}}$ is near zero; winsorization protects the robust z from extreme ratios. Both are standard regularization choices.

²¹The plotted intervals use Wilson binomial confidence intervals for per-decile hit rates, and the dashed curve is an isotonic smooth of lift across deciles.

²²Famous patents are extremely rare relative to the population. Under such imbalance, precision-recall summaries and pooled lift or gains curves are more informative than hit counts at very small K within tiny strata.

The influence score also correlates with a second external indicator: patent duration under the pre-1844 regime, where inventors chose five, ten, or fifteen-year terms at increasing cost. Among 9,027 patents filed between 1830 and 1844, mean influence increases monotonically with the term chosen ($p < 10^{-10}$).²³

The influence score behaves as intended. It assigns higher values to inventions that historians have judged to be important, it performs best in periods and technology classes where diffusion was active, it adds information beyond fixed effects and baseline dependence, and it remains stable in out-of-time checks. The signal is moderate rather than perfect, which is appropriate given the small benchmark set and the extreme rarity of famous cases. Within those limits, the measure tracks the forward reach of ideas as they appear in language and move through the subsequent record.

Novelty and influence capture different dimensions of a patent’s contribution, and the benchmark set confirms this. I classify each of the 41 patents as pioneering, those that opened new fields, or improving, those that refined existing technology. Pioneering patents, among them Gramme’s ring-armature dynamo (1BB87938), Pasteur’s wine preservation (1BB67006), and the Lumière cinématographe (1BB245032), score higher on novelty than improving patents such as Panhard’s gearbox (1BB273375), Michelin’s detachable tire (1BB214256), and Hennebique’s reinforced-concrete system (1BB223546): 0.575 versus 0.425 (Mann-Whitney $p = 0.048$). The influence score does not separate the two types ($p = 0.980$), because both pioneering and improving landmarks were widely adopted. I validate novelty on this pioneering-versus-improving margin rather than on the fame benchmark used for influence, because fame tracks impact rather than originality. The two measures track different things, and both behave as intended.

4 What collaboration changes: switching from solo to team

I begin by exploring what happens when inventors switch from working alone to working with others. In this study, working with others is measured as co-patenting. I focus on inventors who filed at least one solo patent before their first collaboration and track four outcomes: annual patent output, the novelty and influence of their work, and whether they move outside their home technology class. The sample runs from 1791 to 1900 and includes inventors with at least two patents, the minimum needed to define a career span and estimate within-inventor effects.²⁴

Throughout this section, I use event-study designs that align inventors by the year of their first collaborative patent, T_i . Event time $k = t - T_i$ measures years before or after this switch, with $k = 0$ marking the year of first collaboration. I estimate Sun–Abraham (SA) stacked specifications that compare early switchers to inventors who switch later or never collaborate, controlling for inventor fixed effects, home-class-by-year fixed effects, and stack (cohort-comparison) fixed effects. I define each inventor’s home class as the technology class in which they filed the most main patents before first collaboration. A main patent in this study, *brevet d’invention*, is the original patent grant, different from a certificate of addition, *certificat d’addition*, which is an incremental improvement filed against the existing main patent. Standard errors cluster at the inventor level because the same inventor’s outcomes across years are not independent.²⁵ The SA design accounts for staggered timing: different inventors start collaborating in different years, and a standard two-way fixed-effects estimator would implicitly use already-treated inventors as controls for newly-treated ones, biasing the estimates. Stacking cohort-specific comparisons avoids this by ensuring that each treated cohort is compared only to clean controls (Sun and Abraham, 2021). For productivity, novelty,

²³I focus on 1830 to 1844 because very few patents were filed before 1816, leaving earlier filings with a sparse backward comparison window that limits the semantic measure’s ability to rank quality. Including all pre-1844 filings ($N = 11,713$) weakens the association ($p < 10^{-8}$), but the monotonic ordering of means holds in both samples. The modest magnitude may partly arise because patent duration also captures inventor wealth and risk tolerance alongside patent quality.

²⁴Sample sizes differ across specifications because of data and design. All four outcomes (productivity, novelty, influence, and diversification) use main patents only. The productivity sample is largest ($N = 3,565,365$ inventor-years) because years with no main patents yield a valid zero for MainCount_{it} . Influence and novelty use smaller samples ($N = 1,057,062$ and $1,057,035$ respectively) because years with no main patents produce missing values, one cannot compute a mean influence or novelty score when there are no patents, which the regressions drop. Diversification is smallest ($N = 98,052$) because it excludes never-treated inventors and requires non-missing home-class assignment, which depends on having filed at least one patent at $k \leq -1$. These differences stem from outcome definitions and modeling choices, not selective data loss.

²⁵Clustering adjusts the standard errors so that statistical tests account for the correlation among repeated observations from the same inventor, rather than treating each inventor-year as if it were an independent draw.

and influence, controls include both never-treated inventors and not-yet-treated switchers; for diversification, where technology-class choice is the outcome itself, I restrict to not-yet-treated switchers only to avoid contamination from never-collaborators whose class choices may stem from permanent unobserved differences (see below).

4.1 No sustained productivity gain

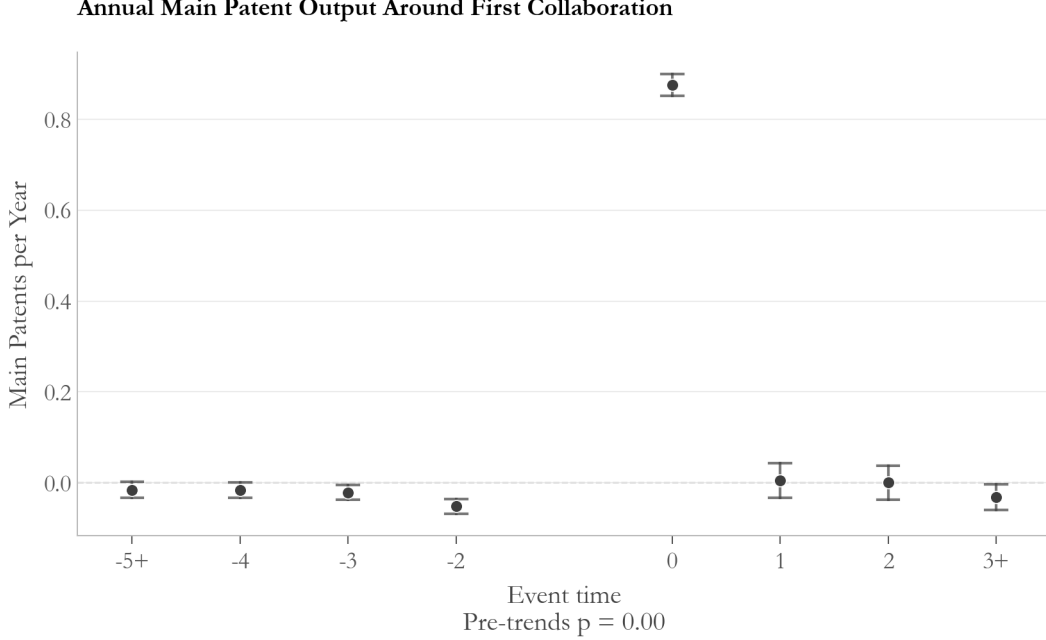


Figure 3: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is the number of main patents filed by inventor i in year t . Event time $k = 0$ marks the year of first collaboration; $k = -1$ is the omitted category. The sample includes inventors who filed at least one solo patent before collaborating (switchers) and inventors who never collaborate (never-treated controls), 1791–1900. The specification absorbs inventor fixed effects and home-class-by-year fixed effects. Event times $k \geq 3$ are binned into a single coefficient. Standard errors are clustered at the inventor level. The pre-event mean ($k \leq -2$) is 0.27 patents per year. Pre-trends are rejected ($F = 14.8$, $p < 0.001$). $N = 3,565,365$ inventor-year observations.

Figure 3 plots annual main-patent output around the first collaboration. The outcome is MainCount_{it} , the number of main patents inventor i files in year t . The estimating equation is:

$$y_{it} = \sum_{k \in \{-5, -4, -3, -2, 0, 1, 2\}} \beta_k D_{it}^k + \beta_{3+} D_{it}^{k \geq 3} + \alpha_i + \gamma_{c_i^{\text{pre}}, t} + \delta_s + \varepsilon_{it},$$

where D_{it}^k equals one if inventor i is k years from first collaboration and zero otherwise, α_i captures inventor fixed effects, $\gamma_{c_i^{\text{pre}}, t}$ captures home-class-by-year fixed effects, δ_s captures stack (cohort-comparison) fixed effects, and $k = -1$ is omitted. The notation c_i^{pre} denotes the inventor’s home class. Using the same fixed-effects structure across all four outcomes ensures that differences in results reflect differences in outcomes, not differences in specification. I bin event times at or beyond $k = 3$ into a single coefficient to preserve precision.

Before collaboration, switchers average 0.27 main patents per year. At $k = 0$, output spikes to $\hat{\beta}_0 = 0.88$ (s.e. 0.01). The first collaborative patent itself accounts for most of the jump. After $k = 0$, the effect disappears. At $k = 1$, the coefficient is $\hat{\beta}_1 = 0.00$ (s.e. 0.02); at $k = 2$, it is $\hat{\beta}_2 = 0.00$ (s.e. 0.02). For $k \geq 3$, the binned coefficient is $\hat{\beta}_{3+} = -0.03$ (s.e. 0.01), statistically significant at the 5% level, suggesting a modest decline at longer horizons.

The specification fails the parallel-trends test. The event-study design assumes that treated and control groups follow the same trajectory before the event, so that any post-event divergence reflects

the event itself rather than pre-existing differences. A joint test that pre-event coefficients equal zero rejects strongly ($F = 14.8$, $p < 0.001$). The pre-event coefficients are all negative relative to $k = -1$, with the largest departure at $k = -2$. Patent output was lower in all earlier years than in the year before first collaboration. Inventors enter collaboration when their output is rising, not falling. All results in this subsection therefore describe conditional correlations, not the effect of collaboration itself. Even aside from the pre-trend failure, excluding the first collaborative patent from the count removes the spike entirely (Appendix D.1, Figure D.1). Outside the mechanical increase at $k = 0$, I find no evidence that collaboration raises inventors' subsequent patenting rates.

4.2 No increase in novelty

If collaboration leads to more original ideas, we might expect patents filed during or after the first team project to be more novel. That is, more distinct from prior art. I test this using a patent-level novelty score that measures how much a patent's language and concepts differ from the existing corpus at the time of filing. I construct it as the inverse percentile rank of backward semantic adoption: patents that draw less heavily on prior language score higher. For each inventor-year, I average novelty across all main patents filed that year.

I estimate a Sun–Abraham stacked specification using switchers and never-treated controls from 1791 to 1900. The estimating equation matches the productivity specification above, with mean novelty replacing patent count as the outcome. Event times $k \geq 3$ are binned into a single coefficient, and standard errors are clustered by inventor.

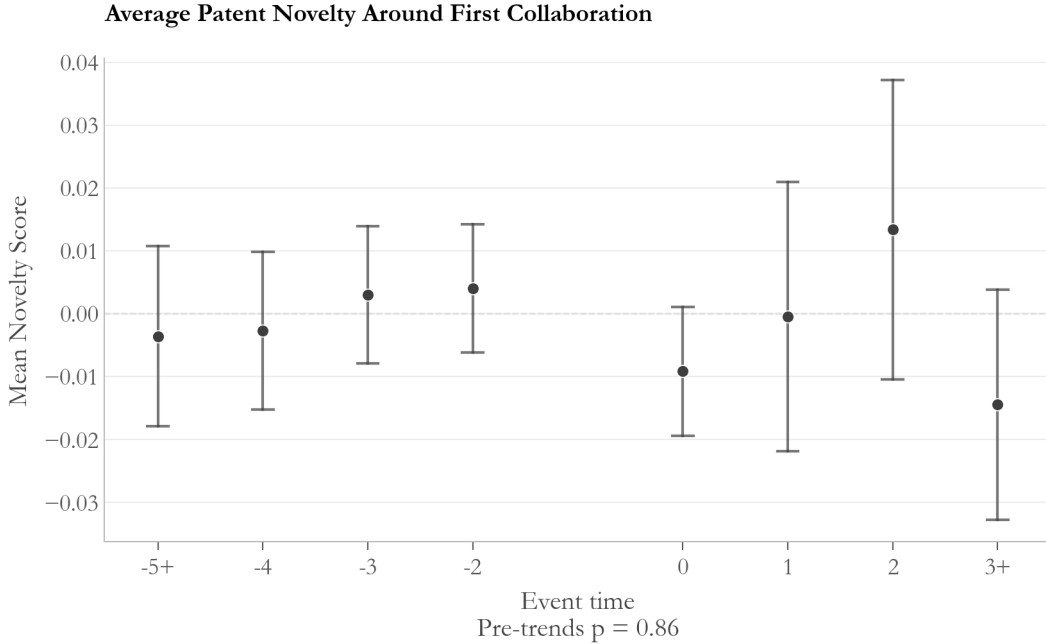


Figure 4: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is the mean novelty score across all main patents filed by inventor i in year t . Novelty is a patent-level percentile rank (0 to 1) measuring how distinct a patent's language and concepts are from prior art, constructed as the inverse of backward semantic adoption: higher values indicate greater originality. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The sample includes switchers and never-treated controls, 1791–1900. The specification absorbs inventor fixed effects and home-class-by-year fixed effects to avoid conditioning on post-treatment class choice. Event times $k \geq 3$ are binned into a single coefficient. Standard errors are clustered at the inventor level. Coefficients are small and statistically indistinguishable from zero at all event times, indicating no systematic change in patent originality following collaboration. Pre-trends are not rejected ($F = 0.32$, $p = 0.86$). $N = 1,057,035$ inventor-year observations.

Figure 4 shows essentially no effect. Coefficients at all event times are small and statistically indistinguishable from zero. At $k = 0$, the estimate is $\hat{\beta}_0 = -0.01$ (s.e. 0.01); at $k = 1$, it is

$\hat{\beta}_1 = 0.00$ (s.e. 0.01); at $k = 2$, it is $\hat{\beta}_2 = 0.01$ (s.e. 0.01); and for $k \geq 3$, it is $\hat{\beta}_{3+} = -0.01$ (s.e. 0.01). Pre-event coefficients are similarly small and mixed in sign; a joint test does not reject parallel trends ($F = 0.32$, $p = 0.86$). The pattern holds when I split novelty into within-class and cross-class components (Appendix D.2, Figure D.2 and Figure D.3).

This null is well-powered. A natural concern is that novelty scores might be stable within an inventor’s career, leaving the event study little variation to work with. I decompose the total variance of novelty into a between-inventor component and a within-inventor component. Only 19 percent of the variance separates one inventor from another; the remaining 81 percent comes from changes within the same inventor over time. The event study exploits precisely this within-inventor variation, and there is plenty of it. Given the standard errors, I can detect novelty shifts as small as 1.5 percentile points on a zero-to-one scale at 80 percent power. The 95 percent confidence interval at $k = 0$ ($[-0.019, 0.001]$) rules out practically meaningful positive effects. Collaboration does not make patents more original.²⁶

4.3 An immediate but temporary rise in influence

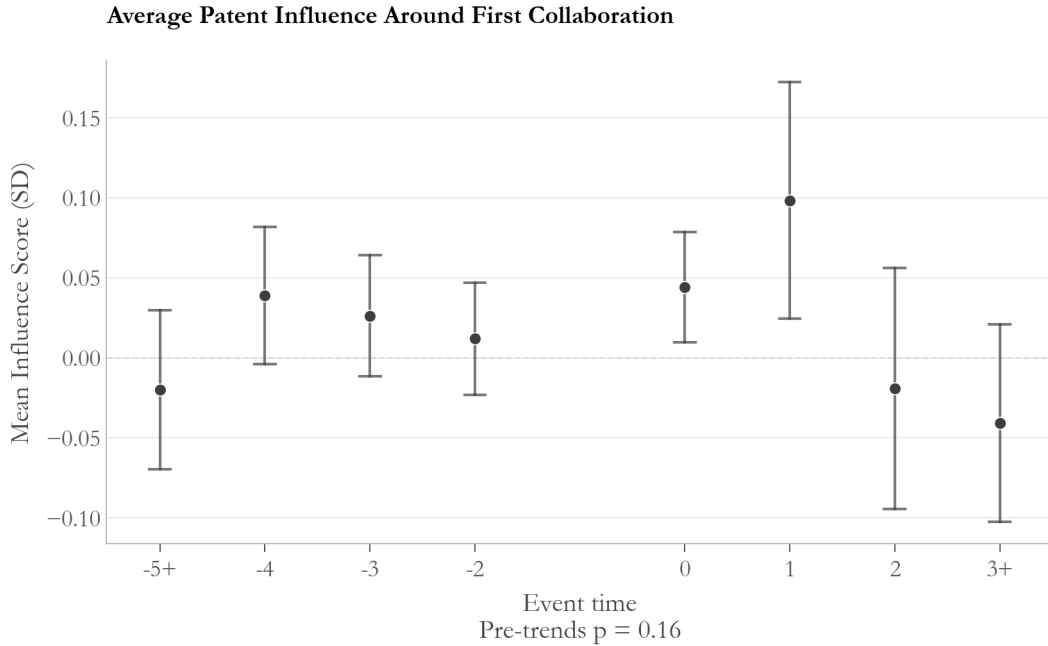


Figure 5: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is the mean influence score across all main patents filed by inventor i in year t . Influence is a patent-level z-score (standardized within decade and technology class) measuring how intensively future patents draw on a patent’s ideas, constructed as the log ratio of forward to backward semantic adoption. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The sample includes switchers and never-treated controls, 1791–1900. The specification absorbs inventor fixed effects and home-class-by-year fixed effects to avoid conditioning on post-treatment class choice. Event times $k \geq 3$ are binned. Standard errors are clustered at the inventor level. Pre-trends are not rejected ($F = 1.65$, $p = 0.16$). $N = 1,057,062$ inventor-year observations.

Influence measures how much future inventors draw on a patent’s ideas. I construct it as a z-score: for each patent, I compute the log ratio of forward semantic adoption (how much future patents borrow from this patent’s language) to backward semantic adoption (how much this patent borrows from prior patents’ language), then standardize within decade and technology class. Higher values mean greater downstream impact.

²⁶A model-free check confirms this null. I construct a lexical novelty score equal to the fraction of content words in a patent title that have never appeared in any prior patent in the same technology class, requiring no embeddings or language model. The stacked event study finds no change at first collaboration ($\hat{\beta}_0 = 0.002$, s.e. 0.003), with flat pre-trends ($F = 0.21$, $p = 0.93$). See Section D.3.

For each inventor-year, I average influence across all main patents filed that year. I then estimate a Sun–Abraham stacked event study:

$$y_{it} = \sum_{k \in \{-5, -4, -3, -2, 0, 1, 2\}} \beta_k D_{it}^k + \beta_{3+} D_{it}^{k \geq 3} + \alpha_i + \gamma_{c_i^{\text{pre}}, t} + \delta_s + \varepsilon_{it},$$

where all terms are defined as in the productivity equation above. As there, $\gamma_{c_i^{\text{pre}}, t}$ absorbs home-class-by-year fixed effects. I use the home class c_i^{pre} , rather than the inventor’s current class, to avoid conditioning on post-treatment class choice, which the diversification analysis below demonstrates is affected by collaboration. This controls for technology-specific time trends (e.g., surging patent activity in mechanical engineering during the 1860s) while avoiding “bad controls”: absorbing the inventor’s current class would condition on an outcome of collaboration itself. Event times $k \geq 3$ are binned, and standard errors are clustered by inventor.

Figure 5 shows the path. At $k = 0$, influence rises by $\hat{\beta}_0 = 0.04$ standard deviations (s.e. 0.02). It peaks at $k = 1$ with $\hat{\beta}_1 = 0.10$ (s.e. 0.04), then falls back: $\hat{\beta}_2 = -0.02$ (s.e. 0.04) is indistinguishable from zero, and estimates for $k \geq 3$ are small and imprecise. Pre-event coefficients are small and mixed in sign; a joint test does not reject parallel trends ($F = 1.65$, $p = 0.16$).

Influence rises on impact, peaking one year after first collaboration, and then returns to its pre-collaboration level. A balanced-window specification that requires each switcher to have at least one solo patent before $k = 0$ and one collaborative patent after yields similar magnitudes (Appendix D.4, Figure D.4).

In sum, collaboration brings an immediate, small, and temporary bump in influence, roughly 0.1 standard deviations at its peak, but no sustained step-change. This modest effect contrasts sharply with the large, persistent diversification across technology classes I document next.

4.4 Collaboration diversifies work across technology classes

To see where collaboration redirects effort, I examine whether inventors move outside their home technology class. For each inventor, I define c_i^{pre} as their most frequent broad technology class (one of twenty categories) computed from all main patents filed before first collaboration.²⁷ I then construct:

$$\text{OutsideClassShare}_{it} = \frac{\text{Number of main patents in year } t \text{ outside } c_i^{\text{pre}}}{\text{Total main patents in year } t}.$$

This share ranges from zero (all patent output in the home class) to one (all patent output elsewhere). Anchoring to c_i^{pre} , determined before collaboration begins, avoids conditioning on post-treatment choices and makes coefficients directly interpretable as percentage-point changes in diversification. Years in which an inventor files no main patents produce an undefined share and are dropped from the outcome.

I estimate a Sun–Abraham stacked event study using only not-yet-treated switchers as controls, since both groups ultimately collaborate and differ mainly in the timing of their switch, making them more comparable to treated inventors than never-collaborators, whose technology-class choices may arise from permanent unobserved factors such as specialization in classes ill-suited to teamwork. I absorb inventor fixed effects and home-class-by-year fixed effects, bin $k \geq 3$, and cluster standard errors by inventor.

Figure 6 shows the result. Before collaboration, solo-to-collab switchers are not entirely specialized: on average, 23% of their patents fall outside their home class. Collaboration more than doubles that share. At $k = 0$, the outside-class share rises by 34.5 percentage points ($\hat{\beta}_0 = 0.345$, s.e. 0.011). The effect drops to 27.6 percentage points at $k = 1$ and then stabilizes, standing at 26.9 percentage points at $k = 2$ and 26.1 percentage points at $k \geq 3$. The initial attenuation is modest,

²⁷The twenty INPI classes are broad: “Textiles” spans cotton spinning, silk weaving, dyeing, and finishing, activities that involve genuinely different knowledge. Because real knowledge boundaries are finer than the classification, some collaborations I classify as within-class actually cross substantive knowledge boundaries. This coarseness makes the cross-class diversification estimates conservative lower bounds on genuine boundary-crossing.

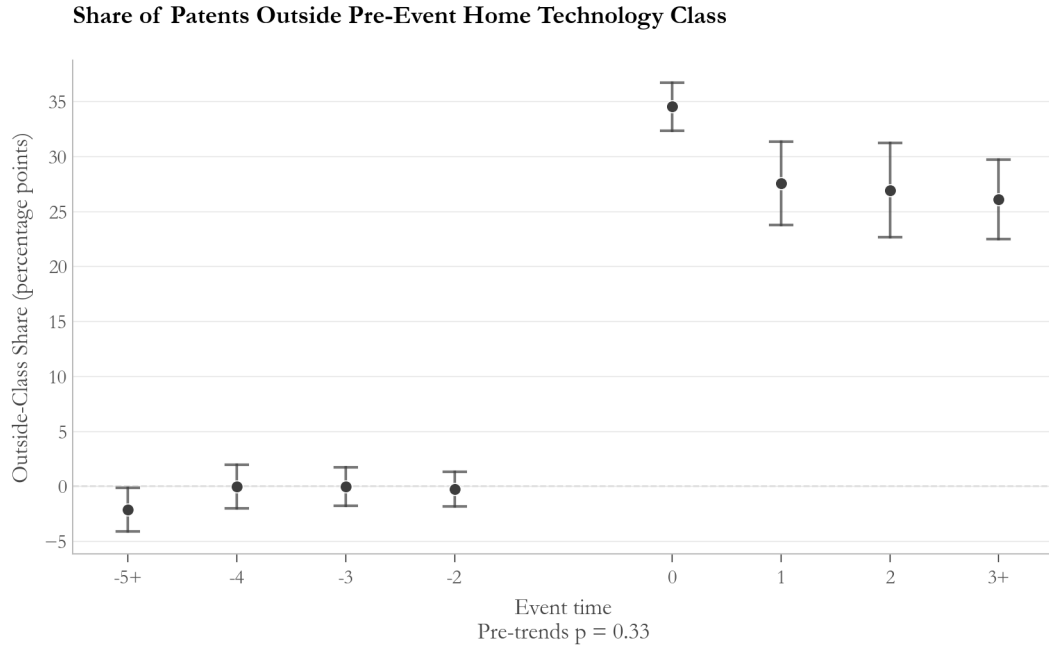


Figure 6: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is the share of inventor i ’s main patents in year t that fall outside their home technology class. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The sample includes switchers only, 1791–1900, with not-yet-treated switchers as the control group. The specification absorbs inventor fixed effects and home-class-by-year fixed effects. Event times $k \geq 3$ are binned. Standard errors are clustered at the inventor level. Pre-trends are not rejected ($F = 1.16$, $p = 0.326$). Before collaboration, switchers average 23% of patents outside their home class; at $k = 0$ this rises by 34.5 percentage points (s.e. 0.011). $N = 98,052$ inventor-year observations.

and outside-class shares remain elevated well above the pre-collaboration baseline, indicating a persistent shift rather than a brief spell of exploration. A joint test on pre-event coefficients finds no violation of parallel trends ($F = 1.16$, $p = 0.326$). The productivity pre-trend shows that inventors time collaboration to periods of rising output, but rising output does not mechanically predict which technology classes that output will land in. The selection on timing that explains the productivity pre-trend does not carry over to diversification: the diversification pre-trends remain flat.

The magnitude is large relative to the pre-collaboration baseline: a 34.5 percentage-point jump from 23% shifts the average switcher from filing roughly one-fifth of output outside their home class to filing roughly 58% there. One potential concern is mechanical. For inventors whose first collaborative patent falls outside their home class, the outside-class share may rise at $k = 0$ by construction rather than reflecting genuine reorientation of effort. To test this, I re-estimate the model excluding each inventor’s first collaborative patent from the outcome. If the result were purely mechanical, the estimate would collapse. It does not: the $k = 0$ estimate is 38.6 percentage points (s.e. 0.020), slightly larger than the headline 34.5 percentage points, and the entire post-collaboration path is virtually unchanged (Appendix D.5, Figure D.5, $F = 1.07$, $p = 0.371$). The diversification effect is real, not an artifact of how the first patent is counted.

The diversification is not merely compositional. Restricting the outcome to solo patents only, excluding all collaborative output, still shows a 26.9 percentage-point increase in outside-class patenting at $k = 0$ (s.e. 0.023), with effects persisting at 22.7 to 22.4 percentage points through $k \geq 3$ (Appendix D.5, Figure D.6). Pre-trends are not rejected ($F = 1.38$, $p = 0.240$). Collaboration redirects the inventor’s independent research direction, not just their team output. Adèle Amélie Longuemare’s career traces this redirection in detail. In 1885 she patented a method for making bellows nozzles from a single sheet of metal, clasped rather than soldered (1BB169949). Three years later she collaborated with Doria on a soldering lamp that vaporized mineral essence under pressure and projected the flame through a capillary nozzle (1BB193388). By 1891 she had filed solo patents generalizing the same vaporization principle into a heating apparatus (1BB211778), then into an autogenous soldering device (1BB237230), a petroleum burner for gas engine igniters (1BB240575), and by 1897 a carburetor for automobile engines (1BB264884). Before the collaboration she worked in sheet metal; after it, every patent involved pressurized fuel combustion.

A further concern is that some collaborations may be driven by cost-sharing rather than knowledge complementarity. Under the pre-1844 patent regime, upfront fees ranged from 300 to 1,500 francs, enough to exclude most artisans working alone (Section 2.3). Two inventors splitting the cost of a fifteen-year patent had a financial motive to co-patent that had nothing to do with exchanging knowledge. The 1844 reform replaced these lump sums with annual installments of 100 francs, making solo patenting broadly accessible and largely removing the financial case for collaboration. If the diversification association partly captures inventors who pooled resources rather than knowledge, the effect should weaken in the post-reform period when cost-driven collaborations became unnecessary. To test this, I interact the event-time indicators with a reform dummy equal to one if the inventor’s first collaboration occurred in 1845 or later ($N = 332$ pre-reform and 4,475 post-reform switchers).

Figure 7 shows the result. Pre-trends are flat for both the main path ($F = 0.69$, $p = 0.56$) and the interaction ($F = 0.78$, $p = 0.51$). The pre-reform diversification effect at $k = 0$ is 41.7 percentage points (s.e. 4.8). The post-reform increment is -7.5 percentage points ($p = 0.13$), yielding a post-reform effect of 34.2 percentage points. The difference is not statistically significant, though the small pre-reform sample limits the test’s ability to detect moderate changes. The pattern is the same when the first collaborative patent is excluded from the outcome (41.6 vs. 38.4 percentage points, interaction $p = 0.78$). The diversification link does not depend on the financial incentive structure for collaboration.

Collaboration does not raise productivity, novelty, or influence in any sustained way. What it does is redirect effort across technology classes. On average, switchers shift roughly 35 percentage points of annual output outside their historical home class the moment they begin collaborating, and this redirection sustains for years. The effect is not mechanical (it survives when the first collaborative patent is excluded), not compositional (solo-only output shifts too), and not driven by cost-sharing (it is indistinguishable before and after the 1844 reform). Diversification, not productivity or originality gains, is the primary channel through which collaboration changes inventive work. The next section examines where, specifically, this redirected effort lands, and what network structures

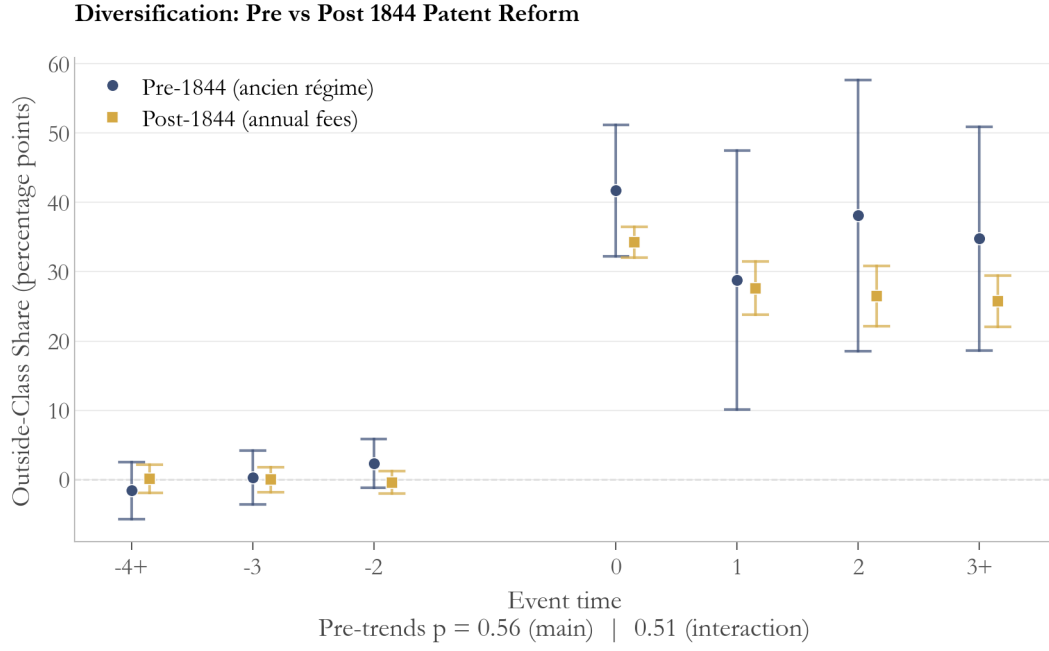


Figure 7: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design with reform interactions. The pre-reform path (solid circles) shows $\hat{\beta}_k$, the diversification coefficients for inventors whose first collaboration preceded the 1844 patent reform. The post-reform path (dashed squares) shows $\hat{\beta}_k + \hat{\gamma}_k$, the implied coefficients for inventors whose first collaboration occurred under the post-reform regime of annual fees. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The event window begins at $k = -4$ because very few pre-reform inventors are observed at longer horizons, making earlier coefficients unstable. Pre-trends are flat for both the main path ($F = 0.69$, $p = 0.56$) and the interaction ($F = 0.78$, $p = 0.51$). The reform increment at $k = 0$ is -7.5 percentage points ($t = -1.5$, $p = 0.13$). $N = 92,537$ inventor-year observations.

facilitate it.

4.5 Cross-class partnerships drive immediate diversification

To distinguish whether diversification reflects directed knowledge transfer from a cross-class partner or a more diffuse capacity built by collaboration of any kind, I classify each inventor’s first collaboration according to whether the first collaborative patent was filed outside the inventor’s home class. I then estimate the Sun–Abraham diversification specification separately for each group, using not-yet-treated switchers within the same group as controls. The outcome excludes the first collaborative patent from both numerator and denominator, so that the $k = 0$ estimate reflects the inventor’s remaining output in the year of first collaboration, not the collaborative patent itself.

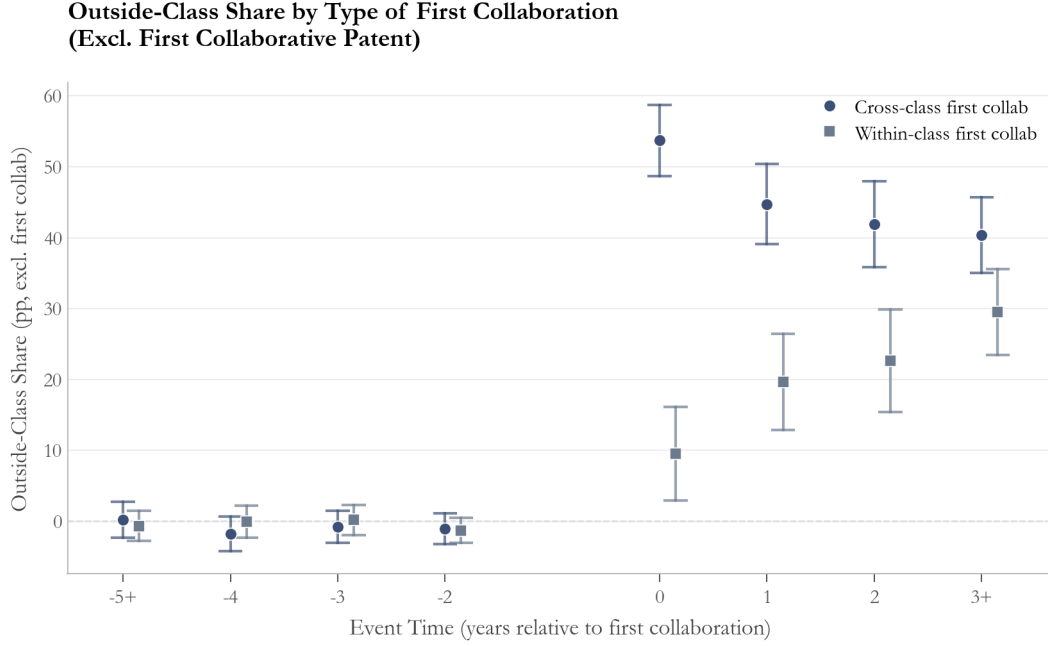


Figure 8: Point estimates and 95% confidence intervals from Sun–Abraham stacked event-study designs estimated separately for inventors whose first collaborative patent crossed a technology boundary (solid circles, $N = 2,880$) and inventors whose first collaboration was within their home class (dashed squares, $N = 1,927$). The outcome excludes the first collaborative patent from both numerator and denominator to remove the mechanical component. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. Controls are not-yet-treated switchers within each group. Standard errors cluster at the inventor level.

Figure 8 shows the split. Among the 2,880 inventors whose first collaboration crossed a technology boundary, the outside-class share rises sharply by 53.7 percentage points at $k = 0$ and remains elevated at about 40.3 percentage points by $k \geq 3$. Among the 1,927 inventors whose first collaboration stayed within their home class, the increase is smaller at impact, 9.5 percentage points at $k = 0$, but grows steadily to 29.5 percentage points by $k \geq 3$. Pre-trends are flat in both groups ($F = 1.10$, $p = 0.357$ for cross-class; $F = 0.66$, $p = 0.617$ for within-class).

The two trajectories point to different mechanisms. Cross-class first collaborations produce an immediate, fully realized shift. Exposure to the partner’s specific technology class appears to redirect the inventor’s own subsequent output in that same year, with little additional accumulation thereafter. Within-class first collaborations produce no comparable initial jump but generate a steadily growing capacity to diversify. Collaboration here likely builds general skills or network access that enables independent exploration into new technology classes over the following years.

4.6 No prior drift toward the partner’s technology class

A remaining concern is selection. Inventors who want to enter a new technology class may seek out a collaborator in that class. In that case, the estimate would not measure the effect of collaboration

on diversification, but the prior intent of inventors already moving toward the partner’s class. If selection drives the result, the inventor’s solo work should already be drifting semantically, that is, in the language and technical concepts of their patent titles, toward the partner’s technology class in the years before they meet. I test this using the same patent embeddings that underlie the novelty and influence measures (Section 3). For each of the 2,974 cross-class first collaborators, I compute the cosine similarity between the inventor’s solo patents in year t and the centroid of all patents in the partner’s technology class filed in $[t - 20, t - 1]$, minus the average cosine similarity to all other outside-class centroids.²⁸ This relative pull measure captures whether the inventor is moving specifically toward the partner’s technology class, net of general broadening.

I estimate the same Sun–Abraham stacked specification used throughout this section, with not-yet-treated cross-class switchers as controls. I also estimate a placebo in which the partner’s class is replaced by a randomly assigned outside class.

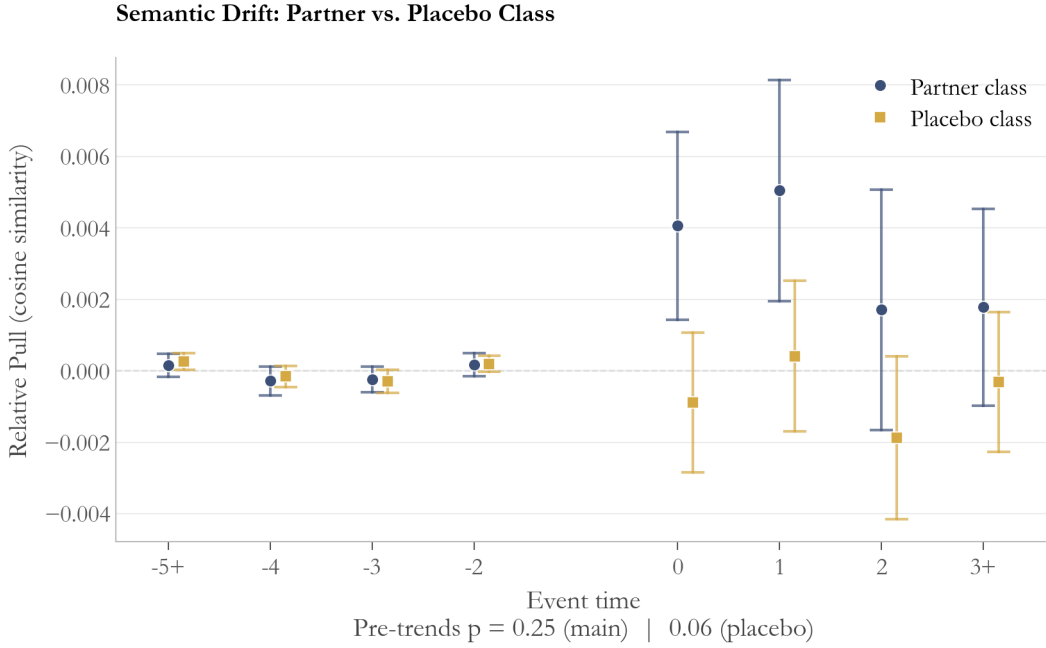


Figure 9: Point estimates and 95% confidence intervals from Sun–Abraham stacked event-study designs. The outcome is the relative semantic pull: cosine similarity between inventor i ’s solo patents in year t and the partner’s class centroid, minus the average cosine similarity to all other outside-class centroids. The solid series uses the actual partner’s class; the dashed gray series uses a randomly assigned outside class (placebo). Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The sample includes 2,974 cross-class first collaborators, 1791–1900, with not-yet-treated cross-class switchers as controls. The specification absorbs inventor fixed effects and home-class-by-year fixed effects. Event times $k \geq 3$ are binned. Standard errors are clustered at the inventor level. Pre-trends are not rejected for the main specification ($F = 1.36$, $p = 0.25$). $N = 23,535$ inventor-year observations.

Figure 9 shows the result. The pre-event coefficients are small and mixed in sign; a joint test does not reject parallel trends ($F = 1.36$, $p = 0.25$). A linear trend fitted to the pre-event path yields an annual slope of 0.000010 ($p = 0.95$), indistinguishable from zero. At $k = 0$, relative pull toward the partner’s class jumps by $\hat{\beta}_0 = 0.004$ cosine units (s.e. 0.001, $t = 3.02$, $p = 0.002$) and peaks at $k = 1$ ($\hat{\beta}_1 = 0.005$, s.e. 0.002). The placebo series is flat at all event times ($k = 0$: -0.001 , $p = 0.37$). The inventor’s solo work shows no directional movement toward the partner’s technology class for five years, then shifts at the moment of first collaboration, and this shift is specific to the partner’s class.

²⁸The centroid is the soft-label-weighted mean embedding: each patent’s contribution to a class centroid is weighted by its classification probability for that class, so a patent classified as 60% textiles and 30% machinery contributes to both centroids proportionally. I restrict the inventor side to solo patents only, because collaborative patents could mechanically reflect the partner’s technology class.

A split by pre-event proximity to the partner’s class sharpens the interpretation. Among inventors with below-median pre-event cosine similarity to the partner’s technology class, the $k = 0$ effect is positive but not significant ($\hat{\beta}_0 = 0.003$, $p = 0.28$; pre-trends: $F = 1.57$, $p = 0.18$), emerging at $k = 1$ ($\hat{\beta}_1 = 0.009$, 95% CI [0.002, 0.015]; Appendix D.5, Figure D.7). Even inventors with no prior semantic connection to the partner’s technology redirect their solo work after collaboration; the one-year delay matches what we would expect when new knowledge takes time to appear in independently filed patents. Among inventors with above-median pre-event proximity, the $k = 0$ effect is larger ($\hat{\beta}_0 = 0.007$, $p = 0.02$), though this subgroup shows a significant pre-trend ($F = 2.60$, $p = 0.03$), which indicates partial selection among inventors already somewhat close to the partner’s technology class. The main specification, pooling both groups, shows no such violation.

The diversification effect is not driven by inventors who were already heading toward the partner’s technology class.

5 The inventor network

Section 4 established that collaboration coincides with a large, persistent expansion of technological scope, driven by partnerships that cross technology-class boundaries. This section examines the network through which such partnerships form: its structure, who gains access to it, and how the composition of collaborative ties differs between men and women.

5.1 Node to node: structure of the network

I construct an inventor network where two inventors are linked if they appear together as applicants on the same patent. The structure of the inventor network reveals that collaboration in nineteenth-century France was rare, fragmented, and highly unequal. Of the 252,860 inventors in the sample, 174,309 worked entirely alone, corresponding to 68.9% of all inventors. They filed their patents solo and improved their designs solo. The median inventor had zero collaborators.

Collaboration was not only rare but fragmented. The average inventor had just 0.41 collaborators, a number pulled toward zero by the large isolate majority. Among those who did collaborate, most had only one or two co-inventors. Inventors split into 207,898 separate components, most of them pairs or tiny clusters: 13.0% were dyads, and only 0.04% contained more than ten inventors. The median non-isolate component contained just 2 inventors. The largest component held 77 inventors, and within it the average distance between any two was only 4.16 steps, short enough for knowledge to pass through indirect connections.

Where collaboration did occur, it was concentrated and self-reinforcing. When patents involved three or more inventors, those groups almost always formed closed triangles in which everyone collaborated with everyone else (transitivity 0.64, where 1 means every group is fully connected). Inventors with many collaborators tended to work with other well-connected inventors rather than with newcomers (assortativity 0.38, on a scale from -1 to 1). Access to the collaborative network depended on who you already knew.

This sparse, fragmented network makes collaboration a scarce resource.

5.2 Who collaborates? Gender, family, and cross-class composition

Women faced a sharp access barrier to collaboration. Of 4,295 female inventors, 82.3% patented only once, a rate similar to men’s 83.4%. The gap was not about persistence but about collaboration: 76.5% of women were isolates, compared to 68.8% of men, a gap of 7.7 percentage points. Only 1,010 women appear in the collaboration network. Women made up 1.7% of inventors and appeared on 1.8% of patents. The most prolific woman in the dataset never collaborated at all. Denise Ralu, born in Martinique in 1857 and working in Paris, filed 20 main patents between 1881 and 1885. She died at 29 (Merouani and Perrin, 2026a). Her inventions sat at the boundary between food processing and industrial chemistry: pineapple liqueurs (1BB143666), fruit preservation in alcohol (1BB144786), sugarcane distillation (1BB149254), and carbonated tar-water (1BB171806). She worked where the two classes met, yet without a collaborator to bridge them.

Conditional on entering the network, however, women had as many collaborators and belonged to

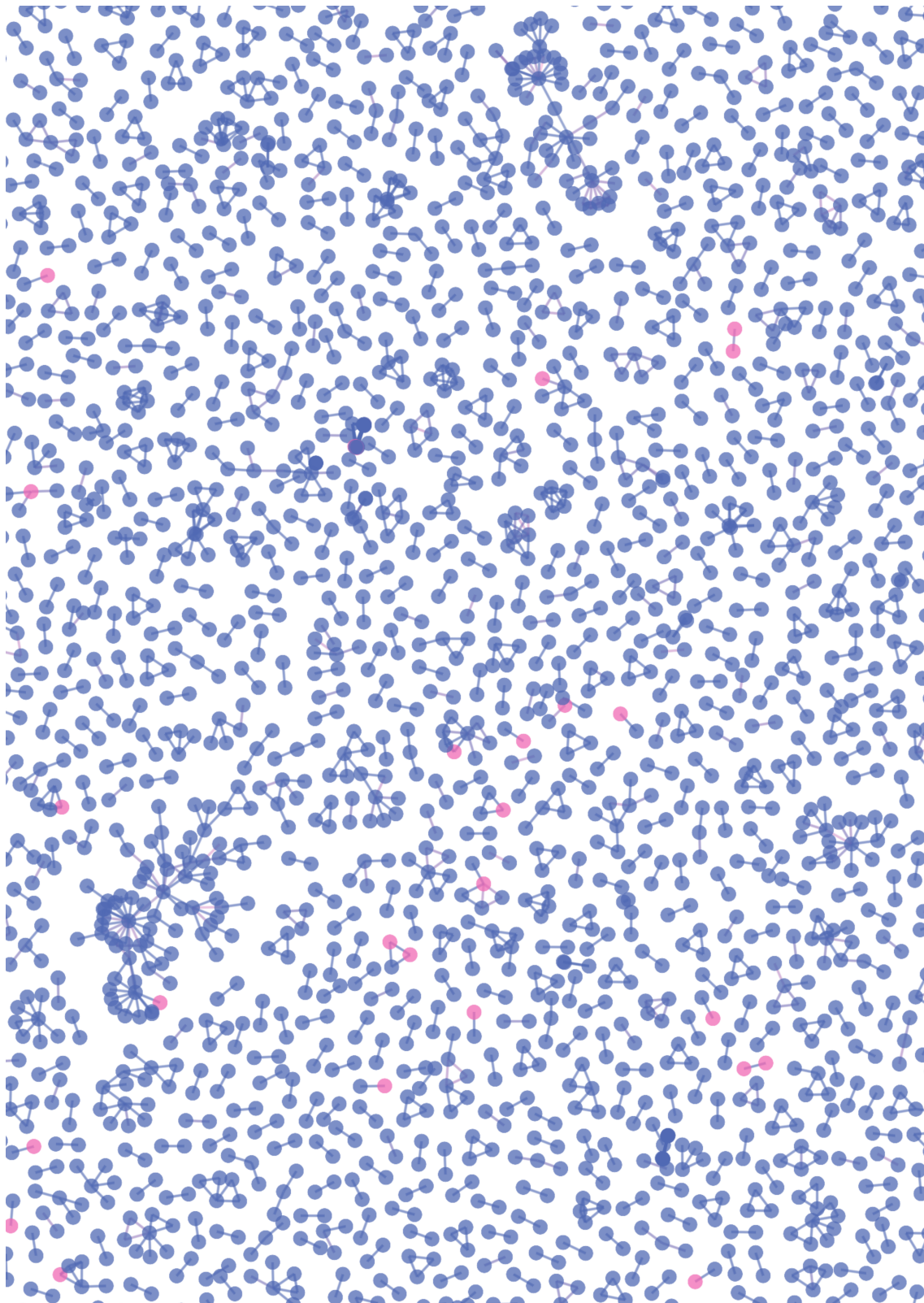


Figure 10: Visualization of the largest component (dense cluster, upper left) and some of the smaller components of the collaboration network. Red nodes are women, blue nodes are men.

groups of the same size as men. Among non-isolates, women averaged 1.39 collaborators versus 1.32 for men ($p = 0.052$), and average component sizes were statistically indistinguishable ($p = 0.20$). Betweenness centrality, which measures how often an inventor lies on the shortest path between two others, was near zero for both genders; the gender difference is statistically significant ($p = 0.0002$) but substantively uninformative, because in a network where the median component contains just 2 inventors, almost no one lies on a shortest path between others. The barrier operated at the boundary of the network, not within it.

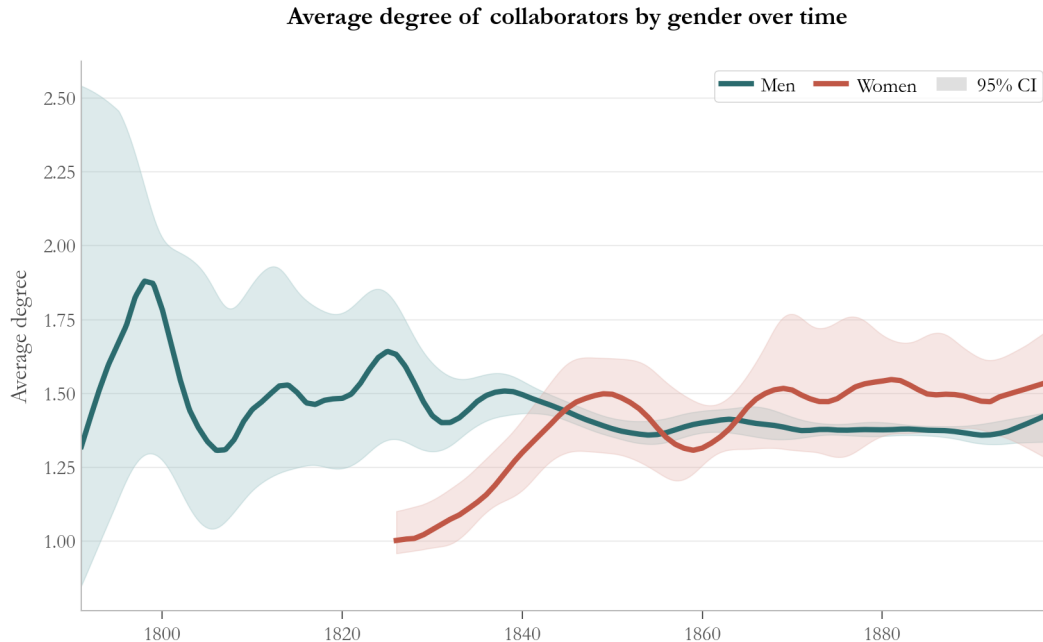


Figure 11: Average number of collaborators per inventor by gender (LOWESS smoothed). Shaded bands denote 95% confidence intervals. The female series begins in the 1820s because too few women patented before that decade for a reliable estimate.

Figure 11 confirms this pattern over time: after the 1840s, women’s average number of collaborators at times surpassed men’s, and after the 1860s the crossover became more regular. Women were scarce in all technology classes, but those who collaborated were not peripheral. Merouani and Perrin (2026b) show that female-linked patents were least likely in male-dominated classes such as marine and navigation, construction, and mining and metallurgy. Yet it is precisely in these classes that the cross-class composition gap documented below is largest (Figure 12).

Family ties provided the key entry point into inventive collaboration for many women. I mark ties as family ties if the connected inventors shared a surname, including maiden names. Just 9.4% of all ties were family links, but for women the share was 35.8%. For men it was only 8.8%. Among women with two or more collaborative ties, 66.2% of those whose first tie was a family link also had a family second tie. Only 33.8% transitioned to a non-family second tie, compared to 53.9% for men.²⁹ Family ties were persistent and within-class: tie by tie, family ties were 2.2 percentage points less likely to cross a technology-class boundary ($p < 0.0001$), controlling for source-class fixed effects.³⁰

Despite entering through within-class family ties, women’s ego networks, each inventor’s set of direct collaborators, were significantly more cross-class than men’s, and the gap holds within each technology class. Comparing non-isolate inventors within the same entry class, women’s share

²⁹The transition matrix covers 214 women and 17,339 men with at least two distinct collaborative ties, ranked by the year of each tie’s first co-patent.

³⁰The network framework treats each inventor as an individual node. For married couples and family firms, the decision about who patented, in which class, and with whom was likely shaped by household economic strategy rather than individual choice alone. The patent system itself imposed no constraint on household participation. The largest application in the database listed twelve individuals, with surnames indicating a family including in-laws. The analysis captures what happened after entry into collaboration, not the household-level decisions that determined who entered.

of cross-class ties exceeded men's by 5.6 percentage points ($t = 4.9$, $p < 0.0001$). The gap was positive in 16 of 20 technology classes. Figure 12 shows the class-by-class breakdown, ordered by the female-male gap. The figure also shows that this average difference is sectorally uneven. Road transport, precision instruments, and chemistry show the clearest positive gaps among the statistically marked class cells, while agriculture is the exception. None of the sixteen women entering from agriculture had a cross-class tie.

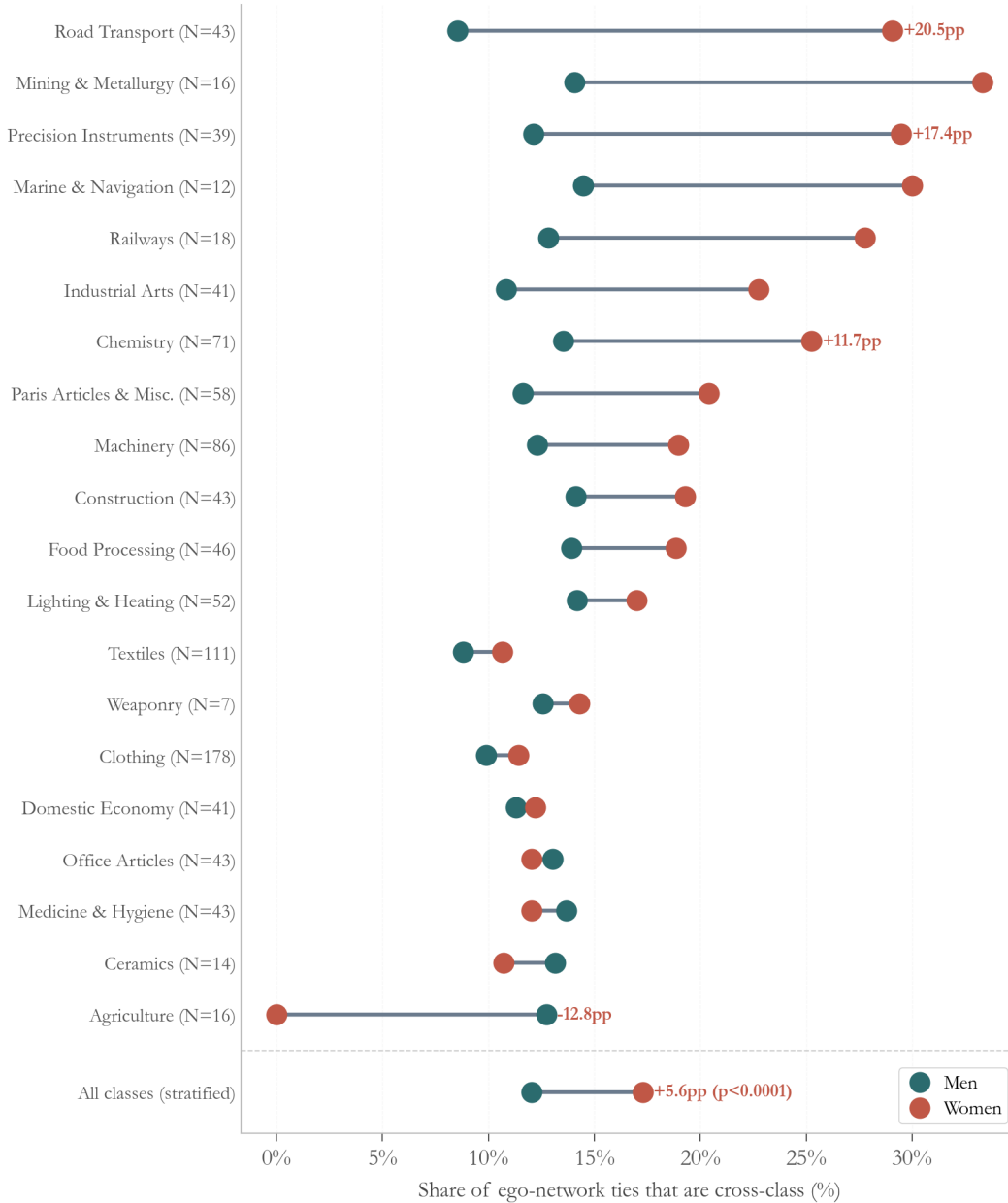


Figure 12: Cross-class share of collaborative ties by gender and technology class, among non-isolate inventors. Each dot shows the mean share of an inventor's collaborative ties that cross a technology-class boundary. Classes ordered by the female-male gap. Bold annotations mark statistically significant differences ($p < 0.05$). Female sample size per class in parentheses. Bottom row shows the overall within-class stratified difference.

The composition difference extends to the first act of collaboration itself. Among all non-isolate inventors, women's first co-patent was 4.4 percentage points more likely to cross a class boundary than men's ($p = 0.0002$, $N = 975$ women, 77,190 men). Adding entry-class fixed effects strengthens the gap to 4.8 percentage points ($p = 0.0001$), because women's concentration in technology classes with moderate cross-class rates was attenuating the raw difference. The gap is larger

among switchers and exclusively collaborative inventors, though smaller samples prevent sharp conclusions.³¹

Women’s cross-class advantage came from outside the family. 22.2% of women’s non-family ties crossed a class boundary, compared to 15.1% of their family ties. For men the corresponding rates were 14.3% and 12.3%. Women’s non-family ties crossed class boundaries often enough to more than offset the within-class concentration of their family ties.

Men who collaborated with women were themselves a selected, broadly connected group. The 921 men with at least one female collaborator had more patents (2.35 vs. 1.60), more technology classes (1.54 vs. 1.22), and more collaborators (1.87 vs. 1.32) than the 76,620 men who collaborated only with other men. Critically, these men’s own male-male ties were more cross-class (39.6%) than their ties to women (21.8%), a gap of 17.9 percentage points. Women matched with the most connected segment of the male inventor population, and these men brought cross-class breadth from their other partnerships.

This selection makes it misleading to compare individual ties between men and women directly, but it does not confound the inventor-level comparison: the within-class stratified gap of 5.6 percentage points compares all female non-isolates to all male non-isolates in the same entry class, regardless of partner identity. The cross-class distinctiveness of women’s networks increased over the century, growing by roughly 2.5 percentage points per decade in the stratified comparison.

The Rohart family illustrates how these patterns combined in a single career. François Ferdinand Rohart, a manufacturing chemist, filed 29 main patents over five decades, mostly in industrial chemistry but extending into agriculture through a long campaign against the phylloxera vine pest. He was exactly the kind of man who collaborated with women: prolific, cross-class, and broadly connected. His wife Marie Léontine entered the patent record in 1881 with a joint patent for transforming petroleum bitumen into vaseline (1BB144051). She then filed solo patents on ferric sulfate chemistry in 1882 (1BB151606) and 1884 (1BB159837), establishing an independent line of work. The cross-class expansion came through the collaborative tie. Between 1886 and 1888, the couple co-patented across four technology classes: degreasing wool fabrics (textiles, 1BB178894), reusing spent bleaching lime (construction, 1BB176874), destroying crop pests (agriculture, 1BB187817), and manufacturing artificial beeswax (chemistry, 1BB188226). Marie Léontine’s solo work stayed in chemistry; the boundary-crossing happened when she worked with her husband.

5.3 Knowledge flow through collaborative ties

Section 5.2 established that women who collaborate build more cross-class networks than men in the same technology class. Whether this compositional advantage translates into diversification depends on whether knowledge actually moves through cross-class ties. The diversification documented in Section 4 could reflect random exploration or directed movement toward technology classes encountered through collaboration. I test this by tracking whether solo patents filed after first collaboration land in the specific technology class of a prior collaborator.

Among post-collaboration solo patents that fall outside the inventor’s entry class, 22.0% land in the class of a first-year cross-class collaborator. Under random assignment to the 19 outside classes, the expected rate is 5.3%. The realized rate is 4.2 times chance, based on 8,339 patent-collaborator observations from 1,850 inventors.³² Diversification is not random exploration; it is directed movement toward the specific technology class the inventor encountered through a collaborative tie, not toward boundary-crossing in general.

The same pattern holds when the collaboration itself was not about crossing a class boundary. Among the 1,835 inventors whose first collaboration stayed within their home class, most partners were co-specialists. Only 194 had prior patents outside the shared home class. I track whether the focal inventor’s subsequent outside-class solo patents land in one of those partner-experienced technology classes. Because the collaboration was about the shared home class, there is no reason

³¹Among switchers, women’s first collaboration was 9.1 percentage points more likely to cross a class boundary ($p = 0.11$, $N = 62$ women, 4,706 men). Among exclusively collaborative inventors, the cross-class share gap is 3.7 percentage points ($p = 0.14$, $N = 193$ women, 11,005 men).

³²The female-specific sample is too small for a meaningful gender breakdown (32 patent-collaborator observations from 18 female inventors). The gender-universality claim rests on the regression interactions in Section 5.4 and the broader pattern across Section 5.2, not on the knowledge flow statistic alone.

the inventor would have chosen this partner for their outside-class knowledge. Yet among the 74 inventors for whom the partner brought outside-class experience and the focal inventor later filed at least one outside-class solo patent, 22.1% of outside-class entries land in a partner-experienced class, against a 10.8% random baseline ($2.0\times$, $t = 4.25$, $p < 0.001$). Restricting to partners whose outside-class patents predate the focal inventor’s entire career yields a ratio of $1.7\times$ ($p = 0.019$, $N = 45$). Knowledge flows through collaborative ties even when the collaboration itself does not cross a technology boundary.

The timing of class entry sharpens this interpretation. If inventors were diversifying under their own steam and happened to collaborate with people from technology classes they were already entering, the timing of entry into a collaborator’s technology class should look like entry into any other class. It does not. I track 541 entries by 444 inventors into a technology class where they have a collaborator from that class, and compare 4,255 entries into technology classes where they have no such collaborator. The median gap between first collaboration with someone from a given technology class and first solo patent in that class is 1 year. For entry into technology classes where the inventor has no collaborator, the median gap from first collaboration to first solo patent is 9 years (Mann-Whitney $p < 0.001$). Entries into a collaborator’s technology class happen fast; entries into unrelated technology classes are slow and spread out.

The patent titles make the mechanism visible. Louis-Antoine Possoz, a chemist with 38 patents and 8 solo patents in chemistry, collaborated with Valuet et compagnie, a food processing firm, in 1857 on methods for extracting glucose and alcohol from Jerusalem artichoke (1BB33008). One year later Possoz filed a solo patent for glucose fabrication from various sweet vegetables, classified under food processing (1BB38270). Alphonse Couvreur, a construction engineer with 10 solo patents in construction, collaborated with the machinery specialist Louis Combe in 1859 on a mobile crane (1BB40159). Two years later Couvreur filed a solo patent in machinery for a rotary elevator hoist (1BB51492). Marie Aimable Gouache, a lamp maker, collaborated with Louis Alphonse Hureau in 1875 on a petroleum burner (1BB108593). Seven years later she filed solo patents not only on gas burners in her home class of lighting (1BB147569) but also on a chemical apparatus for producing carburetted hydrogen gas from acid, metal, and essential oils (1BB148419), classified under industrial chemistry. A lamp maker had crossed from burning gas to generating it. In each case, the inventor’s solo work moved into the collaborator’s technology class. The knowledge that entered through the tie was specific enough to redirect subsequent independent work.

Only 0.4% of inventors who eventually entered a collaborator’s technology class had any patent in that class before they ever collaborated with anyone. The entry came after the collaboration, not before it.

5.4 Which network features drive diversification?

To ask which network features best predict this directed diversification, and whether the association differs by gender, I estimate inventor-year panel regressions. I restrict the outcome to solo patents because collaborative patents mechanically create the network ties that serve as regressors, and I measure the outcome over a forward window ($t + 1$ through $t + 5$) to avoid simultaneity between year- t network formation and year- t outcomes. The forward window addresses timing but not selection. Inventors who value boundary-crossing may accumulate cross-class ties and diversify for related reasons. The case that collaboration itself shifts diversification rests on the within-inventor event study and identification tests in Section 4. The regression here decomposes which features of the cumulative network predict subsequent diversification and tests whether the association differs by gender.³³

I define each inventor’s entry class, C_i^{entry} , as the technology class of their first patent, ordered by application year and, when multiple patents share the same year, by the sequential archival dossier number recorded at filing. This differs from the home class used in the event study, which is the most frequent technology class before first collaboration; for most inventors in this sample the two

³³Inventor fixed effects would absorb the gender indicator and eliminate the gender test. They would also offer little additional identification: the event study in Section 4 shows that diversification shifts once at first collaboration and persists, so within-inventor time variation in the forward outcome is small (a variance decomposition confirms that 75 percent of the total variance in forward solo diversification is between inventors, leaving only 25 percent for fixed effects to exploit). The regression complements the event study by decomposing which features correlate with diversification across inventors.

coincide, because the majority file few patents before first collaboration. The outcome is the share of solo patents filed outside that entry class in years $t + 1$ through $t + 5$:

$$Y_{i,t+1:t+5}^{\text{solo}} = \frac{\text{Solo patents of } i \text{ in } [t+1, t+5] \text{ outside } C_i^{\text{entry}}}{\text{Solo patents of } i \text{ in } [t+1, t+5]}.$$

I construct the collaboration network cumulatively. A tie between inventors i and n appears in the year of their first co-patent and persists forward. All regressors are measured through year t . I capture three channels, drawing on the structural holes and cohesion literatures reviewed in Section 2. *Range* is the share of an inventor’s collaborators whose entry class differs from the inventor’s own, measuring how far the inventor’s ties reach across technology-class boundaries (Burt, 2004). *Cohesion* distinguishes repeated ties by whether the pair has already crossed a class boundary together: repeated ties with prior boundary-crossing capture the established relationship that makes subsequent crossings easier (Fleming et al., 2007b; Reagans and McEvily, 2003), while repeated ties that have never left the home class serve as a comparison. *Peer exposure* is the average share of each collaborator’s own patents that fall outside the inventor’s entry class, testing whether observing collaborators’ boundary-crossing spills over independent of direct joint work.

The specification is:

$$\begin{aligned} Y_{i,t+1:t+5}^{\text{solo}} = & \alpha + \beta_1 \text{CrossClass}_{it} + \beta_2 \text{RepOutside}_{it} + \beta_3 \text{RepHome}_{it} + \beta_4 \text{NbrOutput}_{it} \\ & + \gamma F_i + \delta_1 (\text{CrossClass}_{it} \times F_i) + \delta_2 (\text{RepOutside}_{it} \times F_i) \\ & + \delta_3 (\text{NbrOutput}_{it} \times F_i) \\ & + \mathbf{X}_{it}' \lambda + \mu_{C_i} + \phi_t + \varepsilon_{it}, \end{aligned} \quad (1)$$

where F_i is a female indicator, $\mathbf{X}_{it}$ collects five controls: degree (number of collaborators), non-redundancy (share of non-overlapping ties), class reach (number of technology classes spanned by collaborators), family-tie share, and cumulative career patents, μ_{C_i} are entry-class fixed effects, and ϕ_t are year fixed effects. All continuous regressors are standardized. Standard errors are clustered by inventor. The sample contains 60,339 inventor-year observations from 7,076 inventors. I include additive rather than interacted entry-class $\times$ year fixed effects because the interacted specification (163 cells) absorbs gender variation in cells with zero women; the additive specification yields identical mechanism coefficients while retaining power for gender interactions.³⁴

Table 1 separates the contribution of each network feature. Range is the strongest predictor. A one-standard-deviation increase in cross-class partner share raises solo outside-class share by 0.068 (Model 3). Cohesion contributes independently: repeated ties that have already crossed a class boundary add 0.017, while repeated ties that have never left the home class are negatively associated (-0.021). Repetition by itself does not diversify; repetition after an initial crossing does. Peer exposure is modest (0.007) once range and cohesion are included. Neither the knowledge flow evidence nor the event study’s cross-class split could have separated these channels; the regression does.

The table also tests whether women received a different diversification benefit from these network features. The interaction between cross-class partner share and gender is -0.046 ($p = 0.33$), and the interactions on the other two channels are also not significant (repeated outside ties: 0.043, $p = 0.19$; neighbor output: -0.013, $p = 0.41$). The female main effect is -0.027 ($p = 0.61$). None of the gender interactions is individually significant. The point estimates do show the same directional pattern across specifications: the cross-class interaction is negative while the repeated-outside-tie interaction is positive, a pattern I examine further in Section 5.5 alongside the composition differences documented in Section 5.2.

5.5 Access, composition, and returns

Women collaborated at sharply lower rates than men: 7.7 percentage points more likely to be isolates, with only 1,010 in the collaboration network. The pattern of who got in points to a

³⁴With interacted entry-class $\times$ decade fixed effects (163 cells), all three gender interactions remain insignificant and the mechanism coefficients change by less than 0.002.

Table 1: Network Features and Solo Diversification (Forward Outcome, $t + 1$ to $t + 5$)

	(1) Range	(2) Exposure	(3) Combined
Cross-class partner share	0.069*** (0.005)		0.068*** (0.005)
× Female	-0.048 (0.047)		-0.046 (0.047)
Repeated outside-tie share	0.018*** (0.004)	0.028*** (0.004)	0.017*** (0.004)
× Female	0.041 (0.032)	0.028 (0.030)	0.043 (0.033)
Repeated home-only tie share	-0.021*** (0.005)	-0.027*** (0.005)	-0.021*** (0.005)
Neighbor outside output		0.012*** (0.002)	0.007** (0.002)
× Female		-0.018 (0.016)	-0.013 (0.015)
Degree	0.005 (0.007)	0.011 (0.007)	0.005 (0.007)
Non-redundancy	0.006 (0.006)	0.009 (0.006)	0.006 (0.006)
Class reach	-0.008 (0.006)	0.014* (0.006)	-0.010 (0.006)
Family tie share	0.007 (0.005)	0.006 (0.005)	0.007 (0.005)
Cum. career patents	0.007* (0.003)	0.009** (0.003)	0.007* (0.003)
Female	-0.027 (0.054)	-0.023 (0.054)	-0.027 (0.054)
Entry-class dummies	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
Inventor FE	No	No	No
N	60,339	60,339	60,339
Inventors	7,076	7,076	7,076
R^2	0.083	0.067	0.083

Notes: Outcome is solo outside-class patent share in years $t + 1$ to $t + 5$. All continuous regressors standardized. Additive entry-class and year fixed effects included. Standard errors clustered by inventor. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

barrier. Women who entered relied disproportionately on family ties and matched with unusually well-connected male partners (Section 5.2). Once inside, they built networks more cross-class than men’s within the same entry class, by 5.6 percentage points ($p < 0.0001$), with the gap positive in 16 of 20 technology classes. If the lower rate reflected indifference to collaboration, women who entered should have done so through the same professional channels as men, just less often. Instead, they entered through a fundamentally different channel, family ties that required having an inventor relative, and those who got in look like a selected group that needed strong pre-existing connections. The question is whether these more boundary-spanning networks delivered diversification, as they do for men.

They did. Cross-class partner share is the strongest predictor of diversification: a one-standard-deviation increase raises solo outside-class share by 0.068 (Table 1). Women held more of this cross-class exposure, and the advantage came from outside the family: 22.2% of women’s non-family ties crossed a class boundary, compared to 14.3% for men. If women received the same benefit per tie, this more cross-class exposure predicts at least as much diversification as men achieve. Women’s actual outcomes confirm the prediction. The raw gender gap in solo outside-class share is 2.7 percentage points ($N = 59,899$ male vs. 440 female inventor-years), and the female coefficient in the full regression is not significant (-0.027 , $p = 0.61$). As a separate check, I estimated the regression using only male inventors and used the resulting coefficients to predict female outcomes. The mean prediction error was not statistically distinguishable from zero ($p = 0.22$).³⁵ The network features that predict diversification among men predict it comparably when applied to women’s own, more boundary-spanning networks.

The regression also tests directly whether women received a different benefit from the same network features. The interaction between cross-class partner share and gender is -0.046 ($p = 0.33$), and the interactions on repeated outside ties (0.043 , $p = 0.19$) and neighbor output (-0.013 , $p = 0.41$) are also not individually significant. But the test is constrained by sample size: 86 female inventors contribute 440 inventor-year observations, and the smallest cross-class interaction the regression could reliably detect is 13.2 percentage points, roughly double the main effect. I can rule out that women received dramatically lower returns from cross-class ties; I cannot rule out moderate differences. The argument above does not depend on this test. It rests on the composition advantage, which is precisely estimated ($p < 0.0001$, $N = 978$ female inventors), and on the prediction test, which aggregates all coefficients simultaneously rather than testing one interaction at a time. Women’s more boundary-spanning networks explain why they diversified as much as men. Whether they received these benefits through the same channels as men is a separate question.

Although no single interaction is significant, the point estimates show a pattern that is jointly informative. The cross-class partner share interaction is negative: initial structural position was less predictive of diversification for women than for men. The repeated-outside-tie interaction is positive: trusted, repeated ties with partners who had already crossed a class boundary were more predictive for women. A permutation test that randomly reassigns gender labels across inventors rejects the null that both coefficients arose by chance ($p = 0.046$), and the sign pattern holds across all 7 alternative specifications I tested.³⁶ A separate prediction exercise confirms the same asymmetry: when I estimate the regression on men only and apply the resulting coefficients variable by variable, repeated outside ties predict women’s outcomes 2.7 times better relative to what they predict for men (bivariate $R^2 = 0.031$ vs. 0.011), while cross-class partner share predicts women’s outcomes less than one-third as well (0.012 vs. 0.038).³⁷ Women’s diversification was predicted more by repeated ties with partners who had already crossed a class boundary than by the share of partners from other classes.

This asymmetry is consistent with Burt (1998), who finds that women in a large American firm

³⁵I estimate the regression on 59,899 male inventor-year observations ($R^2 = 0.081$) and predict outcomes for 440 female observations. The prediction error for women (RMSE 0.454) is comparable to the male in-sample error (0.421).

³⁶I permute the gender indicator at the inventor level, preserving 86 female inventors and the cluster structure, and re-estimate the regression 5,000 times. The joint p -value is the fraction of permutations where the cross-class interaction is at least as negative and the repeated-outside-tie interaction is at least as positive as the observed values. The sign pattern also holds under entry-class-by-decade fixed effects, year fixed effects only, no fixed effects, and restrictions to collaborators only or multi-patenters only.

³⁷I regress solo outside-class share on each network variable alone, separately for men and women. Bivariate R^2 for repeated outside ties: 0.031 for women vs. 0.011 for men. For cross-class partner share: 0.012 for women vs. 0.038 for men.

had no less access to structural holes than men but could not use them directly; they benefited only through a well-connected partner who already bridged disconnected groups. Meng (2016) documents a parallel among university inventors. Industry collaboration significantly increases patenting for women but not for men, because men already access the same benefits through looser professional ties. In this setting, where women are roughly 1.7 percent of inventors, the same pattern appears. The repeated cross-class tie, not the initial structural position, is what predicts women’s boundary-crossing.

Two caveats qualify this interpretation.

I measure the outcome from solo patents, and only 25.8% of female switchers ever file a solo patent after collaboration, compared to 42.6% of males. If women continue to diversify through collaboration rather than solo work, the true benefit of their network ties is larger than what I capture here.

A simultaneous shock could in principle trigger both collaboration and diversification without knowledge transfer. The semantic drift test in Section 4 weighs against this. Inventors show no movement toward the partner’s specific technology class before they collaborate, and a placebo using a random outside class is flat at all event times. The within-class spillover above confirms that knowledge flows through ties even when the collaboration itself did not cross a technology boundary. I cannot fully rule out a shock that targets the partner’s exact class without appearing in the inventor’s solo work until the year of collaboration, but the cumulative evidence makes this an increasingly narrow account.³⁸

6 Discussion and conclusion

When inventors in nineteenth-century France moved from solo work to collaboration, they did not become more productive or create more original work.³⁹ Collaborative work was more influential, though the gain was modest and concentrated in the years immediately surrounding first collaboration. The more persistent change was in the direction of inventive effort. The share of output filed outside an inventor’s home class jumped by roughly 35 percentage points at first collaboration and remained elevated for years. Cross-class partnerships drove the diversification. Women entered collaboration at sharply lower rates than men, but those who did built networks with more ties across technology classes than men’s in the same technology class, despite relying on within-class family ties for entry. I cannot detect a gender difference in the overall diversification outcome. The path appears to differ: for men, having many partners from other classes predicted diversification; for women, the repeated relationship with a partner from another class mattered more.

Why did collaboration raise influence but not productivity or novelty? Team dominance of the highest-impact breakthroughs is a late-twentieth-century development (Wuchty et al., 2007), so the modest influence gain in nineteenth-century France is not surprising. Teams achieve higher impact than solo authors at virtually every combination of novel and conventional knowledge (Uzzi et al., 2013), and the gain here likely reflects joint work that drew on a wider base of expertise than either inventor commanded alone. Cross-class partnerships combined knowledge that was routine within each field but rarely connected, and joint work that bridged those domains proved more useful to subsequent inventors.

Novelty reflects a different process than influence. Disruptivity and influence need not move together, since the propensity to disrupt declines monotonically with each additional collaborator (Wu et al., 2019). Teams produce work that diffuses widely and shapes subsequent developments without departing from existing practice. The null results on novelty hold across system-wide, cross-class, and within-class measures (see Appendix D.2; also Appendix F). If anything, the only significant effect appears within the technology class, where collaboration is associated with lower novelty. The gain came from combining existing knowledge across fields, not from generating new ideas within one.

³⁸As supplementary evidence, a separate exercise exploiting the disappearance of patent agents from the record yields a consistent pattern: among 2,657 inventors whose patent agent disappeared from the record and who reappeared with a different agent, the change in intermediary sector exposure is associated with later sector entry ($p = 0.047$ under the strictest exit definition; $p = 0.060$ under the medium definition). Appendix G reports the full design, regression table, and robustness checks.

³⁹Pre-trends suggest that the productivity null stems from prior trajectory rather than collaboration itself.

The diversification results show knowledge moving through partners. When inventors entered a collaborator’s technology class, they usually did so quickly. The median gap was one year, compared to nine years for unrelated classes. The same pattern appears even when the first collaboration itself stayed within the inventor’s home class. Later solo patents still landed in the partner’s other classes at twice the random baseline rate. This is hard to explain as random exploration. The regressions tell the same story. What mattered was not having more collaborators. It was having the right collaborators: partners from other classes, and repeated partners with whom the inventor had already crossed a class boundary. The network literature calls these channels range and cohesion: a tie that reaches into a different pool of knowledge, and a repeated relationship through which that knowledge can be used (Fleming et al., 2007b; Reagans and McEvily, 2003). Co-patenting captures family ties and financial arrangements alongside genuine collaboration (Chanteux, 2023), yet co-inventor ties reliably predict knowledge flows even in other settings (Singh, 2005). The diversification effect survived the 1844 cost reform, which removed much of the financial incentive to share patent fees. A pattern driven primarily by cost sharing would be difficult to reconcile with that persistence.⁴⁰

Even in a network with no giant component, where most ties were dyadic and the vast majority of inventors worked alone, direct collaboration was one observed channel through which knowledge moved across technology boundaries. The French network was not unusual in its fragmentation (Andersson et al., 2019), but the mechanism the evidence reveals differs from the division of labor among deep specialists that drives modern teamwork (Jones, 2009). Classic network theory predicts that acquaintances bridge disconnected communities (Granovetter, 1973), but in this setting close collaborative ties appear to have performed part of that role, supplying access to dispersed knowledge associated with inventors’ subsequent solo work in new technology classes.

The barrier for women was getting into collaboration. Women were less numerous than men and more likely to work alone. 76.5% were isolates, compared to 68.8% of men. Those who entered collaboration did so disproportionately through family ties that tended to remain within their home technology class. Yet conditional on collaborating, women occupied comparable network positions and built networks with significantly more cross-class ties. These ties explain why women who collaborated diversified as much as men (Section 5.5). The mechanism was not identical across genders. For men, having many partners from other classes mattered more; for women, repeated work with a partner who had already crossed a class boundary mattered more. Women appear to have gained access to cross-class knowledge through a well-connected partner rather than by occupying brokerage positions themselves (Burt, 1998). The men who collaborated with women were themselves a selected group, with more patents, more collaborators, and broader technology portfolios than men who collaborated only with other men. It also parallels evidence that women benefit more from direct, high-trust collaboration than from looser professional ties (Meng, 2016). The small female network sample means I cannot rule out moderate gender differences in returns to specific network features. The narrower and better-supported claim is that the largest observed inequality was access to the collaborative ties through which diversification occurred.

These findings come with limitations. Patents record formal credit, not the full social process of invention, so co-patenting may mix genuine joint problem solving with financial partnerships, hierarchical relationships, or family arrangements. The network analysis is observational and cannot fully separate selection into certain kinds of ties from their effects. The gender interaction tests are further constrained by the small number of female inventors in the collaboration network and cannot detect moderate differences in how network features predict diversification for women versus men. The novelty and influence measures are text based, derived only from patent titles, and rely on the assumption that semantic distance maps to conceptual originality and impact of work, an assumption that large-scale comparisons of text-based and citation-based patent measures validate directly (Yang, 2025). Even where citations are available, they capture only one dimension of patent quality (Higham et al., 2021). Future work with full patent descriptions can sharpen both measures. Finally, the focus on inventors with at least two patents omits one-time patentees, who

⁴⁰Transaction records from the 1890s provide a direct check on whether diversified output had commercial value. Among inventors who had not yet collaborated, patents filed outside their home class were ceded or licensed at lower rates than home-class patents (4.1% vs. 5.3%, $p = 0.04$). After collaboration, the penalty disappeared: outside-class patents transacted at 5.2%, comparable to home-class patents at 4.6% ($p = 0.27$). The pattern is consistent with collaboration equipping inventors to produce commercially viable work in the new class, rather than merely redirecting effort into unfamiliar territory. The test covers 45,118 inventor-patent observations from 1890 to 1900.

may have different collaboration patterns and returns.

I also do not observe the geographic distance between collaborators. The addresses recorded on patents often reflect the patent agent's office rather than the inventor's domicile or workplace ([Merouani, 2026](#)), which makes reliable distance measures infeasible with current data. Whether proximity shaped who collaborated, and whether geographic constraints compounded the access barrier women faced, remains an important question for future work with better-geocoded sources.

Why, then, was collaboration so limited? The modern literature links collaboration to the burden of knowledge. As knowledge deepens, inventors need partners because no one can master enough alone ([Jones, 2009](#)). Nineteenth-century France faced a different version of the same problem. Knowledge was less codified and less institutionally organized, but it was spread across crafts, workshops, and technology classes. Collaboration paid when it connected an inventor to that outside knowledge. The difficulty was finding the right partner. The network was not organized around a few powerful brokers; it consisted mostly of isolates, dyads, and small components. A centralized, elite-dominated system would have produced a different shape. The evidence instead points to broad access to patenting but thin access to useful partners. Many inventors could file, but few found collaborators who gave them access to another field. Collaboration was rare because useful cross-class matches were hard to form. When they formed, they changed the direction of invention.

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A Record linking

A.1 Extracting keywords from patent titles

To distinguish if two patents fundamentally work with the same product or process despite having different technology classification, I extract keywords from the titles of patents in the French Patent Database, $T = \{t_1, t_2, \dots, t_N\}$. Each title is first reduced to its essential terms by lemmatizing words and removing stopwords, punctuation, and named entities, producing a refined version t'_i . I then apply a Term Frequency–Inverse Document Frequency (TF–IDF) weighting, which highlights the words that appear often in a given title but rarely across the rest of the corpus (Bafna et al., 2016). Formally,

$$\text{TF-IDF}(w, t'_i, T) = \text{TF}(w, t'_i) \times \text{IDF}(w, T),$$

where

$$\text{TF}(w, t'_i) = \frac{f_{w, t'_i}}{\sum_{w' \in t'_i} f_{w', t'_i}} \quad \text{and} \quad \text{IDF}(w, T) = \log\left(\frac{|T|}{|\{t \in T : w \in t\}|}\right).$$

From each title, I then select the highest-weighted words, $K_i = \{w_1, w_2, \dots, w_k\}$. These keywords summarize the central focus of the invention and provide a way to compare patents on the basis of the vocabulary they share. In my linking strategy, overlapping keywords serve as evidence that two observations may belong to the same underlying entity.

A.2 Blocking rules: generating candidate pairs

For each type of entity, whether male inventors, women inventors, family firms, or non-family firms, I construct blocking rules that generate candidate record pairs. These rules restrict attention to observations sharing plausibly identifying features, making the task computationally manageable while minimizing missed matches. Their design reflects the attributes most useful for distinguishing entities: names, occupations, and technological domains. All fields listed below are available as `field_l` and `field_r` for the left and right record respectively.

Field	Description
<code>patent_code</code>	The code of the patent family associated with the inventor record.
<code>patent_id</code>	The identifier of the patent application associated with the inventor record.
<code>dep_num</code>	The applicant order number associated with the observation.
<code>app_year</code>	The application year of the patent.
<code>surname</code>	The surname of the inventor; functions as firm name for firms.
<code>maiden_name</code>	The maiden name of the inventor.
<code>first_name</code>	The first name(s) of the inventor.
<code>gender</code>	The gender of the inventor ("F" or "M").
<code>mar_status</code>	The marital status of women inventors (married , widow , or unmarried).
<code>mother / father</code>	Binary flag equal to "1" if an individual shares names with their progeny; used to separate parent–child entities (mother for female records, father for male).
<code>married</code>	Binary flag equal to "1" unless a woman's marital status is unmarried .
<code>hisco</code>	The HISCO code for the inventor's occupation.
<code>keywords</code>	A list of technical keywords extracted from the patent title (see Section A.1).

Field	Description
tech_class	The major technology classification of the patent (20 categories).
tech_minor_class	The minor technology classification of the patent (97 categories).
place_id_list	A list of city-level place identifiers associated with the patent.

A.2.1 Firm inventor blocking rules

I use five blocking rules for firm inventors, whether family-owned or otherwise. All rules require surname and gender agreement and exclude pairs from the same patent application. Rules 1–3 additionally require distinct patent families and applications within 50 years. Rule 4 operates within the same patent family and imposes no time window. Rule 5 requires distinct patent families but no time window.

Table A.2: Firm inventor blocking rules

Rule	Description
1	Same name and occupation. Firms are compared if they share a HISCO occupation code.
2	Same name and shared technical keyword. Firms are compared if at least one keyword from the patent description overlaps.
3	Same name and technology class. Firms are compared if they share a major technology class.
4	Within patent-family differentiation. Firms sharing an applicant order (<code>dep_num</code>) within the same patent family are compared.
5	Female-linked firms only. Comparisons are allowed across distinct patents whenever both records are associated with female inventors.

A.2.2 Female inventor blocking rules

I use six blocking rules for female inventors. All rules exclude pairs from the same patent application and require the mother indicator to match, to avoid conflating mothers and daughters who share a surname. Rules 1–3 and 5–6 additionally require surname agreement; Rule 4 relaxes this to link records across name changes at marriage.

Table A.3: Female inventor blocking rules

Rule	Description
1	Same surname, marital status, and applicant order. Records are compared when marital status and applicant order (<code>dep_num</code>) both match.
2	Same surname and first name, same married flag. Records are compared when first name matches and the binary married flag is equal.
3	Same surname and maiden name, same married flag. Records are compared when maiden name matches and the married flag is equal.
4	Cross-surname link via maiden name. Records with different surnames and different married flags are compared if one record’s maiden name equals the other’s surname, capturing transitions between unmarried and married or widowed status.

Rule	Description
5	Same surname and marital status. Records are compared when marital status matches.
6	Same surname only (residual). The broadest rule: any two records sharing a surname are compared.

A.2.3 Male inventor blocking rules

I use six blocking rules for male inventors. All rules require surname agreement, applications within 30 years, a matching father indicator, and distinct patent applications. Rules 1–4 and 6 additionally require distinct patent families. To improve match quality, addresses corresponding to Paris (place identifier 641322) are excluded from location comparisons, since they occur very frequently.

Table A.4: Male inventor blocking rules

Rule	Description
1	Same surname and occupation. Records are compared if they share a HISCO occupation code.
2	Same surname and shared technical keyword. Records are compared if at least one keyword from the patent description overlaps.
3	Same surname and first name. Records are compared if the first name matches.
4	Same surname and minor technology class. Records are compared if the minor technology classification (97 categories) matches.
5	Within patent-family differentiation. Records sharing an applicant order (<code>dep_num</code>) within the same patent family are compared.
6	Same surname and overlapping place IDs. Records are compared if at least one city-level place identifier (excluding Paris) overlaps.

A.3 Comparison features and scoring

For each candidate pair generated by the blocking rules, I compute comparison features organized into five categories: names, occupations, technology, locations, and patent agents. These features serve as inputs to the matching pipeline described in Section 3.

Names. The Fellegi–Sunter model compares surnames at four levels: exact match, Damerau–Levenshtein distance of one or fewer, Jaro–Winkler similarity of 0.9 or above, and Jaro–Winkler similarity of 0.8 or above. I compare first names, which are missing for 33.5 percent of male inventor observations, at three levels of token overlap (exact, at least two shared tokens, at least one) and supplement the comparison with edit distance, Jaro–Winkler similarity, and cosine similarity between embeddings. I flag records containing initials, because initials can match many full names and inflate false positive rates.

Family-name disambiguation. French patent records frequently include kinship terms in the first-name field to distinguish same-surname relatives. I extract four pairs of complementary terms, listed in Table A.5. When complementary terms appear across the two records in a pair, I flag the pair as a definite non-match.

Table A.5: French kinship terms used for family-name disambiguation

Term pair	French	Meaning
younger / elder	<i>jeune / aîné</i>	Age distinction
father / son	<i>père / fils</i>	Parent–child

Term pair	French	Meaning
junior / senior	junior / senior	Generational
nephew / uncle	<i>neveu / oncle</i>	Extended family

Occupations and technology. I compare HISCO occupation codes hierarchically, from an exact match on the full code down to agreement on the first digit; 54.7 percent of observations lack a recorded occupation. Technology classification follows a similar hierarchy, combining extracted keywords from the patent description with the minor (97 categories) and major (20 categories) technology classes. Patent titles have no missing values; technology classifications are missing for 0.01 percent of observations and extracted keywords for 0.26 percent. Co-inventor overlap contributes both Jaccard similarity and embedding similarity, though 75 percent of observations lack co-inventor information because most patents name a single inventor. I compare application years by the size of the time gap between the two records.

Locations and patent agents. I compare city-level place identifiers for exact match or list overlap, excluding Paris because its frequency dominates. The presence of a patent agent (mandataire) interacts with location evidence, because the address on a patent filed through an agent may reflect the agent’s office rather than the inventor’s residence.

Scoring and calibration. Two record-level sentence embedding models (gtr-t5-large and all-MiniLM-L12-v2) each embed the full record as text and compute similarity as $1/(1 + d)$ where d is the Euclidean distance between the two records’ embeddings; I take the maximum as the combined score. A lighter model (all-MiniLM-L6-v2) separately computes cosine similarity for individual features: first names, occupations, co-inventor lists, and patent titles. The Fellegi–Sunter model yields a match probability from the log Bayes factor across all comparison levels, adjusted for term frequency. The RoBERTa classifier produces a probability from the sigmoid of its logits, which I calibrate using isotonic regression on a held-out development set because the uncalibrated scores are not well-calibrated near zero and one. I stratify the training sample by agreement between the embedding and Fellegi–Sunter scores to ensure coverage across the full difficulty spectrum.

B Extracting first names using machine learning

About 65 percent of male inventor-observations in the French Patent Database lack a first name. The original database records only surnames, but the underlying patent application documents, digitized scans provided by INPI, contain this information in handwritten form on the cover page (see Figure B.1 and Figure B.2 for examples). With more than 150,000 documents missing a first name, manual transcription is not feasible. I therefore build an automated pipeline that combines custom object detection with frontier vision-language models.

Each application cover page contains a section listing patentee information: names, addresses, and occupations. I train a YOLO11 object detection model to locate and crop this section automatically from the full page, because feeding an entire cover page to a vision-language model introduces noise. The model may read irrelevant printed or handwritten text elsewhere on the page, which increases both the error rate of the transcription and the burden of post-processing. Cropping to the relevant section focuses the transcription task on the handful of lines that contain the inventor’s name.

I pass each cropped section, together with the known surname, to two frontier vision-language models: Qwen2.5-VL and InternVL3, running on the Alvis GPU cluster (NAISS/C3SE). I do not rely on either model alone, because no single model is reliable enough at this scale. Instead, I retain a transcribed first name only when both models agree. This agreement-based approach filters out hallucinations and misreadings without requiring manual review of every output, and it proved more reliable in practice than fine-tuning a single model on curated training data.

Where the two models disagree, I resolve the case manually, prioritizing the most frequent surnames where errors compound most. I use OpenRefine’s clustering features to clean and standardize the extracted names. Throughout the process, I carry out extensive random spot-checks of the results. Where the two models agree, errors almost never go beyond a single letter. The full pipeline reduces first-name missingness from 65 to 33 percent for male inventor-observations.

50

Déposé le 18^{juin} 1890.
Parti le 18^{juin} 1890.

N° 1530 D'ENREGISTREMENT

081702

Gracmigo & August Whitehead
Mason jun (Jr) & Leigh (Evan Arthur)
représentés par la Société internationale des Inventeurs Modernes
à Paris, 30, boulevard des Capucines

BREVET D'INVENTION de 15 ans pour perfectionnement
des machines servant à tordre et blanchir les textiles
de toute autre manière, appliqués sur bobines ou mis sous
une autre forme compacte.

PIÈCES DÉPOSÉES SUIVANT PROCÈS-VERBAL		CESSIONS, TRANSMISSIONS, MUTATIONS, OPPOSITIONS, ETC.	
de	6 mai 1890	HEURE	MINUTES.
1°	requête		
2°	description		
3°	dessin		
4°	échantillon		
5°	bordereau		
6°	procuration		
1°	certificat d'addition pris le		
2°			
3°			
4°			
5°			
6°			
7°			
8°			
9°			
10°			
11°			
12°			
13°			
14°			
15°			
1°	annuité payée le		
2°	27 février 1891		
3°	27 février 1892		
4°	27 février 1893		
5°	27 février 1894		
6°	27 février 1895		
7°	27 février 1896		
8°	27 février 1897		
9°	27 février 1898		
10°	27 février 1899		
11°	27 février 1900		
12°			
13°			
14°			
15°			

Figure B.2: Example of a patent application document cover. This is patent application with archive code 1BB204189 from year 1890

C Notable patents from nineteenth-century France

I assembled this benchmark set of 41 patents by surveying the standard reference literature on nineteenth-century technology and French industrial history, identifying inventions routinely cited as landmark contributions, and matching each to a record in the INPI historical patent database (data.inpi.fr). No single source lists all 41, because no such catalog exists for the French patent system. I drew them instead from multiple overlapping references, spanning general surveys of technology and the specialized histories of particular sectors. Mokyr (1990) discusses over a fourth of the full set, including the Leblanc soda process, the Jacquard loom, the Fourneyron turbine, the Lenoir and Beau de Rochas internal-combustion engines, the Gramme dynamo, and the Gaulard-Gibbs transformer. Landes (1969) treats an overlapping but not identical set: Leblanc’s soda, Thimonnier’s sewing machine, Fourneyron’s turbine, Lenoir and Beau de Rochas on internal combustion, Gramme’s dynamo, the Hall-Hérout aluminium process, and Michelin’s tyres. Derry and Williams (1960), a comprehensive survey of technology from antiquity to 1900, names many of the remaining, including Sorel’s galvanization of iron, Giffard’s dirigible and steam injector, Monier’s and Hennebique’s reinforced concrete, Mege-Mouries’s margarine, the Lumiere cinematographe, and the Panhard-Levassor automobile layout.

The exercise is related in spirit to the validation strategy of Nuvolari et al. (2023), who validated their patent-duration quality indicators for 1791-1844 by matching patentee names against four independently compiled biographical databases of historically notable individuals and showing that patents filed by more famous inventors had higher quality.

Table C.1: Table of Notable French Patents (1791–1900)

Year	Inventor(s)	Title (short)	Reason	Sector	Application code
1791	Nicolas Leblanc	Large-scale soda-ash process	Birth of modern alkali industry	Chemistry	1BA12
1797	Firmin Didot	Stereotype printing process	Mass-edition printing	Industrial Arts	1BA2005
1800	Abraham-Louis Breguet	Tourbillon timekeeping regulator	Landmark in precision horology	Precision Instruments	1BA1792
1800	Joseph-Marie Jacquard	Programmable loom (punch-cards)	Automates patterned weaving	Textiles	1BA134
1806	Nicéphore & Claude Niépce	“Pyreolophore” heat-engine	Early internal-combustion engine	Machinery	1BA395
1820	Charles-Xavier Thomas de Colmar	Arithmometer calculator	First successful calculating machine	Precision Instruments	1BA1447
1830	Barthélemy Thimonnier (w/ A. Ferrand)	Chain-stitch sewing machine	First practical French sewing machine	Textiles	1BA3587
1832	Benoît Fourneyron	Water turbine	First efficient reaction turbine	Machinery	1BA3972
1839	Stanislas Sorel	Hot-dip galvanizing of iron	Corrosion protection by zinc coating	Mining & Metallurgy	1BA8634
1846	Adolphe Sax	Saxophone family	New woodwind instrument	Industrial Arts	1BB3226
1849	Charles-Xavier Thomas de Colmar	Industrial arithmometer	Mass-produced calculator	Precision Instruments	1BB8282
1851	Henri Giffard	Steam-powered airship (dirigible)	First powered dirigible concept	Marine & Navigation	1BB12226

Year	Inventor(s)	Title (short)	Reason	Sector	Application code
1854	André-Adolphe Disdéri	Carte-de-visite photography	Made portrait photography mass-market	Industrial Arts	1BB21502
1857	Édouard-Léon Scott de Martinville	Phonautograph (sound writing)	First device to record sound	Industrial Arts	1BB31470
1858	Henri Giffard	Self-acting steam injector	Simple boiler feed device	Machinery	1BB36512
1859	Prudent-René Dagron	“Microscope-jewel” stereoscope	Microphotography pioneer	Precision Instruments	1BB41361
1859	Édouard-Philippe Carré	Absorption refrigerator / ice-maker	Early artificial refrigeration	Lighting & Heating	1BB41958
1860	Étienne Lenoir	Gas-fired internal-combustion engine	First commercially used IC engine	Machinery	1BB43624
1862	Alphonse Beau de Rochas	Four-stroke engine cycle	Principle of modern Otto cycle	Lighting & Heating	1BB52593
1865	Louis Pasteur	Wine-preservation by heating	Pasteurization applied to wine	Food Processing	1BB67006
1866	Hippolyte Marinoni	Mechanical perfecting press	High-throughput newspaper printing	Industrial Arts	1BB71339
1867	Hippolyte Marinoni	Six-feeder cylinder press	Faster multi-feeder rotary press	Industrial Arts	1BB76392
1867	Joseph Monier	Iron-and-cement flowerpots	Origin of reinforced concrete	Agriculture	1BB77165
1868	Louis Ducos du Hauron	Color photography methods	First practical color processes	Industrial Arts	1BB83061
1869	Hippolyte Mège-Mouriès	Oleomargarine process	Invention of margarine	Chemistry	1BB86480
1871	Zénobe Gramme	Ring-armature dynamo	Breakthrough DC generator	Precision Instruments	1BB87938
1874	Émile Baudot	Rapid telegraph system	Baudot code & multiplexing	Precision Instruments	1BB103898
1876	Paul Jablochhoff	Arc “candle” lamp	City-scale electric lighting	Precision Instruments	1BB112024
1878	Auguste de Méritens	Dynamo-electric induction machine	Widely used arc-lighting dynamo	Precision Instruments	1BB123766
1879	Léon & Henri Serpollet; Claudius Richard	Instantaneous (flash) steam generator	Foundation of steam cars/boats	Machinery	1BB133362
1881	Auguste de Méritens	Electric autogenous welding	Early arc-welding process	Mining & Metallurgy	1BB146010
1882	Lucien Gaulard & John-Dixon Gibbs	AC distribution by transformers	Step-down/step-up power networks	Precision Instruments	1BB151458
1886	Paul Héroult	Electrolytic aluminum smelting	Hall-Héroult breakthrough	Mining & Metallurgy	1BB175711
1890	Clément Ader	Fixed-wing “avion”	Early airplane concept	Marine & Navigation	1BB205155

Year	Inventor(s)	Title (short)	Reason	Sector	Application code
1891	Michelin & Cie (André & Édouard Michelin)	Detachable pneumatic tyre	Quick-change tire for cycles/cars	Road Transport	1BB214256
1892	François Hennebique	Reinforced-concrete system	Modern RC frames & beams	Construction	1BB223546
1895	Auguste & Louis Lumière	Cinématographe camera-projector	Birth of motion-picture projection	Industrial Arts	1BB245032
1895	Panhard & Levassor	Motor-car improvements	Standardized “Système Panhard”	Road Transport	1BB245276
1895	Louis-Michel Bullier	Practical acetylene for lighting/heating	Makes acetylene usable at scale	Lighting & Heating	1BB246000
1896	Georges Claude & Albert Hess	Acetylene storage system	Safe storage (toward dissolved gas)	Chemistry	1BB257679
1897	Panhard & Levassor	Gearbox with change-speed	Sliding-gear transmission	Machinery	1BB273375

D Event study robustness

D.1 Productivity

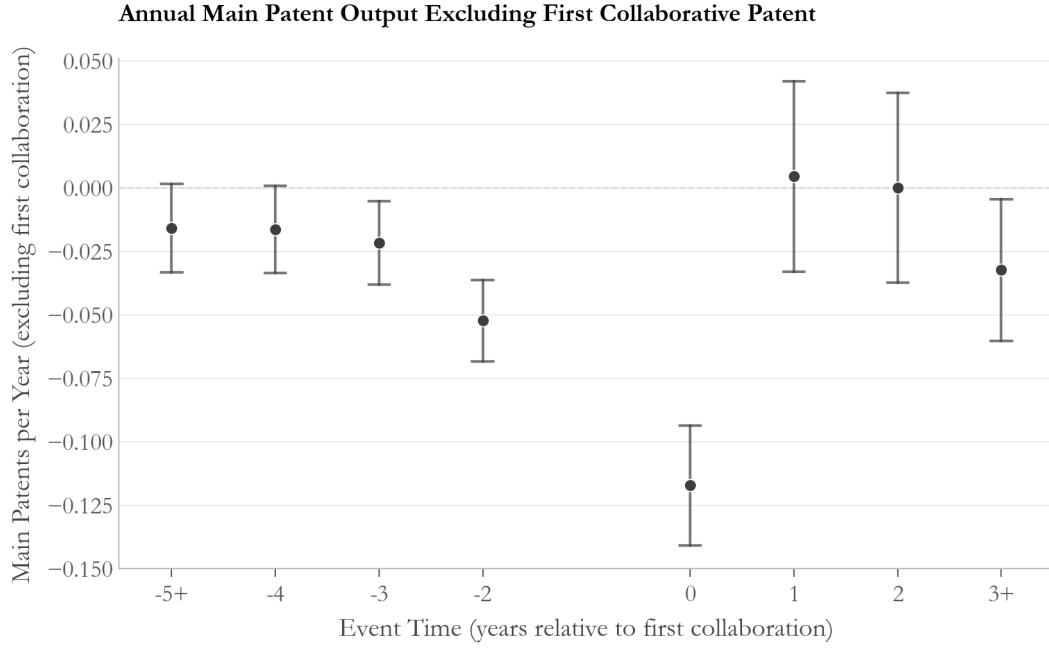


Figure D.1: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is `MainCount_excl_first_it`: identical to Figure 3 except that the count at $k = 0$ excludes the first collaborative patent itself. This robustness check isolates whether the $k = 0$ spike reflects the mechanical inclusion of the collaborative patent or a genuine productivity increase. The $k = 0$ coefficient (-0.12 , s.e. 0.01) is negative, confirming the spike in Figure 3 is mechanical: once the first collaborative patent is removed, output at $k = 0$ is actually below the reference year. Sample, fixed effects, and clustering as in Figure 3. $N = 3,565,365$ inventor-year observations.

D.2 Novelty

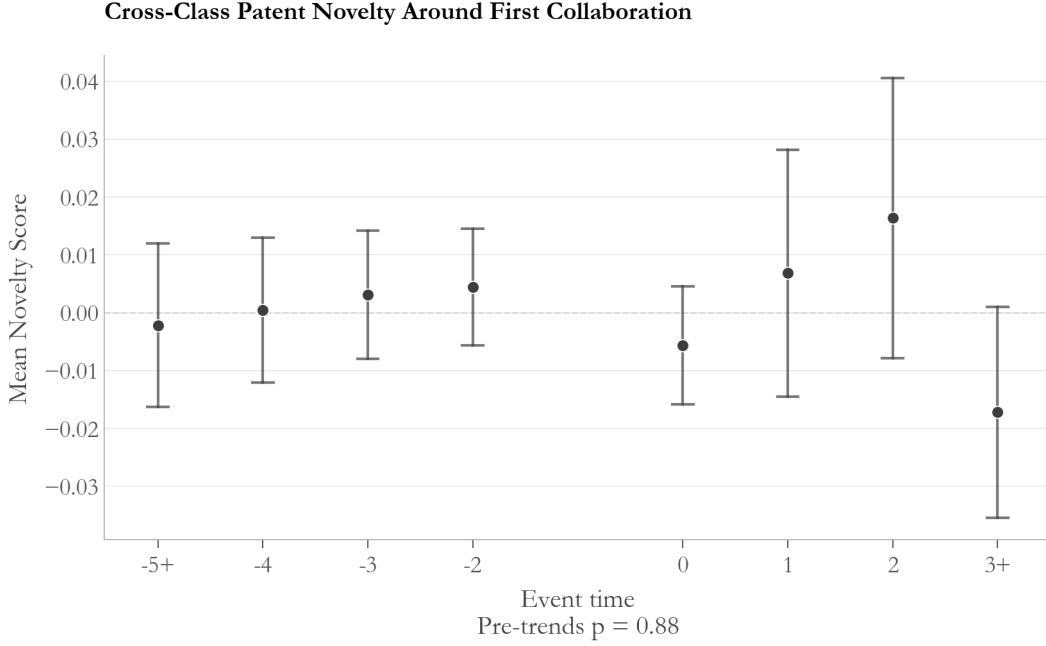


Figure D.2: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is mean novelty across main patents filed in year t , restricted to patents in technology classes different from the inventor’s home class (cross-class novelty). Novelty is measured as the inverse percentile rank of backward semantic adoption: higher values indicate greater distinctiveness from prior art. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The sample includes switchers and never-treated controls, 1791–1900. The specification absorbs inventor fixed effects and home-class-by-year fixed effects. Event times $k \geq 3$ are binned. Standard errors are clustered at the inventor level. Coefficients are small and statistically indistinguishable from zero, indicating no systematic change in cross-class patent originality following collaboration. $N = 1,057,035$ inventor-year observations.

D.3 Lexical novelty: a model-free robustness check

BSA novelty counts how often a patent’s title lands semantically close to prior titles, using a domain-adapted language model to represent meaning and CSLS to correct for generic phrasing. The concern is that the null result in Section 4 reflects the embedding geometry rather than the historical record. A completely model-free measure provides a clean check.

For each patent, I define lexical novelty as the fraction of content words in the title that have never appeared in any prior patent in the same technology class. I identify content words by stripping accents, converting to lowercase, removing common French stop words, and discarding tokens shorter than three characters. A score of zero means every word in the title had already appeared in that technology class; a score above zero means at least one word is new to the class vocabulary. The measure is entirely transparent: it counts, it does not model. Across the full sample, mean lexical novelty is 8 percent and the median is zero, reflecting that most titles recombine familiar technical terms rather than introduce vocabulary new to the class.

I aggregate to the inventor-year level and estimate the same Sun–Abraham stacked specification, with inventor fixed effects and home-class-by-year fixed effects. At $k = 0$, the estimate is $\hat{\beta}_0 = 0.002$ (s.e. 0.003), statistically indistinguishable from zero. Pre-event coefficients are flat; a joint test does not reject parallel trends ($F = 0.21$, $p = 0.93$). The stacked sample contains 1,058,496 inventor-year observations. First collaboration does not change the fraction of new vocabulary in an inventor’s patent titles. The null holds regardless of how novelty is measured.

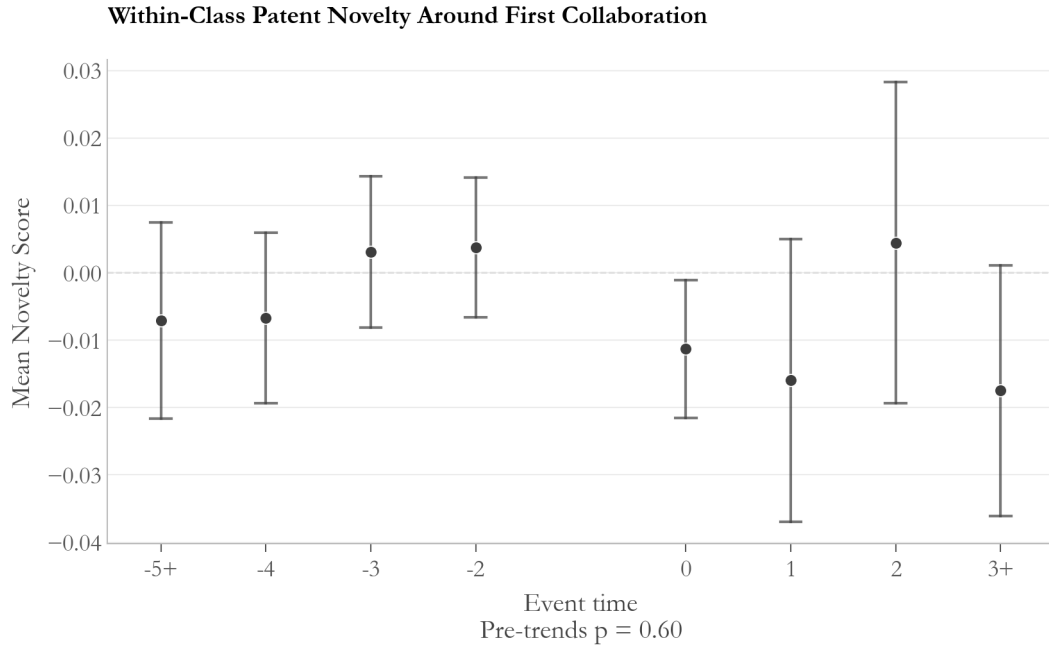


Figure D.3: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is mean novelty across main patents filed in year t , restricted to patents in the same technology class as the inventor’s home class (within-class novelty). Novelty is measured as the inverse percentile rank of backward semantic adoption. Event time $k = 0$ marks first collaboration; $k = -1$ is omitted. The sample includes switchers and never-treated controls, 1791–1900. The specification absorbs inventor fixed effects and home-class-by-year fixed effects. Event times $k \geq 3$ are binned. Standard errors are clustered at the inventor level. The $k = 0$ coefficient is small but statistically significant (-0.01 , s.e. 0.01 , $p < 0.05$) and negative, indicating that within-class novelty, if anything, declines slightly at first collaboration rather than improving. All other coefficients are statistically indistinguishable from zero. $N = 1,057,035$ inventor-year observations.

D.4 Influence

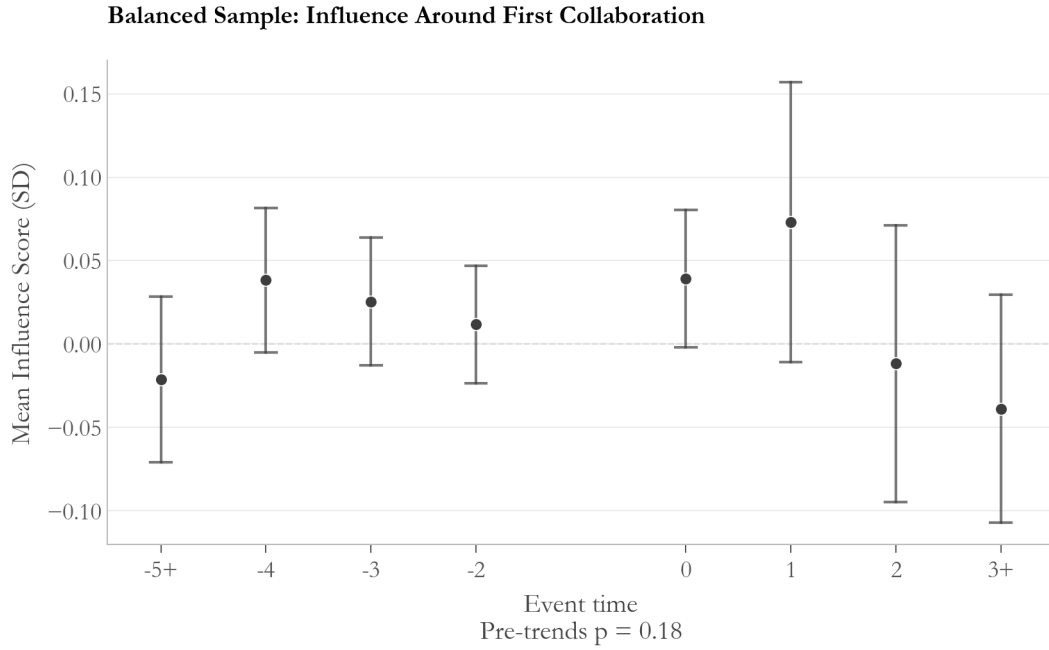


Figure D.4: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design on a balanced sample. The sample includes only switchers who have at least one solo main patent in $k \in [-5, -1]$ and at least one collaborative main patent in $k \in [0, 5]$, ensuring observable activity on both sides of first collaboration. The outcome and specification are identical to Figure 5. Point estimates are similar to the headline ($k = 0$ effect of 0.04 SD, s.e. 0.02; peak at $k = 1$ of 0.07 SD, s.e. 0.04), but neither is individually significant in this smaller sample. Pre-trends are not rejected ($F = 1.6$, $p = 0.18$). $N = 1,024,035$ inventor-year observations.

D.5 Cross-class diversification

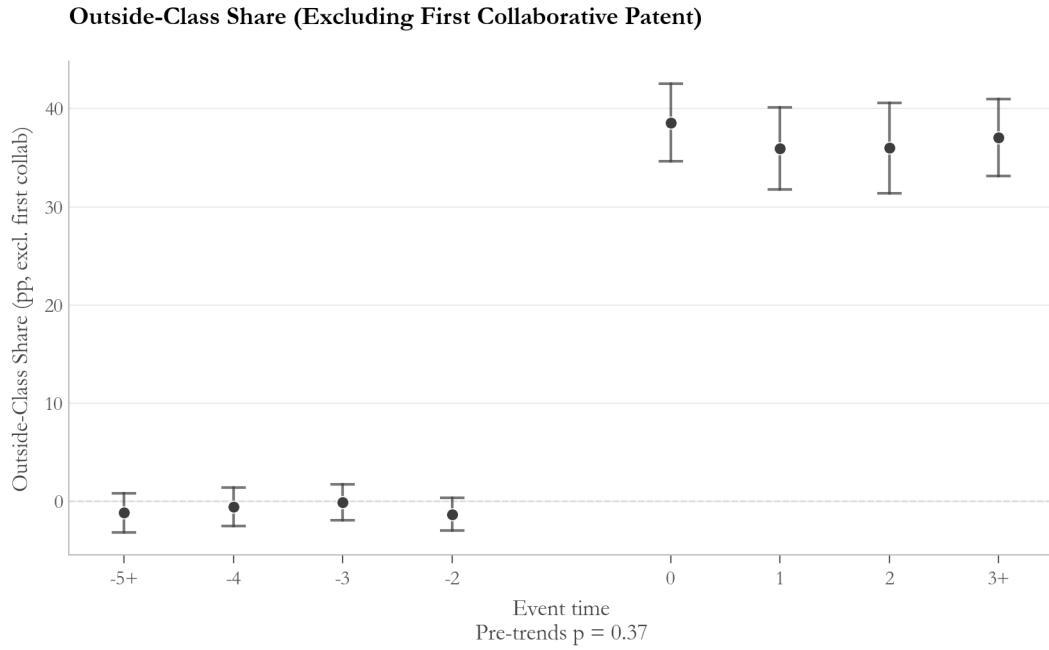


Figure D.5: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design, replicating the headline specification (Figure 6) but excluding each inventor’s first collaborative patent from the computation of the outside-class share. This tests whether the headline result is mechanical: if the first collaborative patent simply appears in a new technology class by construction, the $k = 0$ estimate would collapse once that patent is removed. Instead, the estimate is 38.6 pp (s.e. 0.020), slightly larger than the headline 34.5 pp, and the path at $k = 1, 2, 3$ is nearly identical to the headline (35.9, 36, 37 pp). All other design choices follow the headline: not-yet-treated switchers as controls, inventor fixed effects, home-class-by-year fixed effects, event times $k \geq 3$ binned, standard errors clustered by inventor. Pre-trends are not rejected ($F = 1.07$, $p = 0.371$). $N = 94,370$ inventor-year observations.

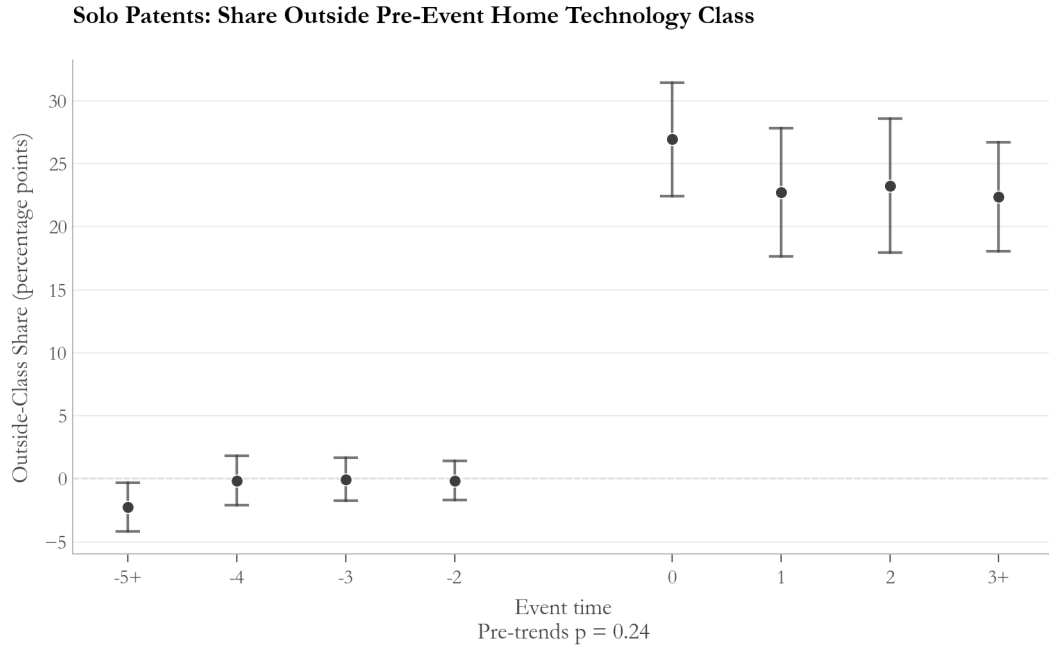


Figure D.6: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design, replicating the headline specification (Figure 6) but restricting the outcome to solo patents only. For each inventor-year, the outside-class share is computed using only patents with a single inventor, excluding all collaborative output. This tests whether the diversification effect is behavioral (solo work shifts to new technology classes) or compositional (only collaborative patents are cross-class). The solo-only effect at $k = 0$ is 26.9 pp (s.e. 0.023), with effects persisting at 22.7 to 22.4 pp through $k \geq 3$. All other design choices follow the headline: not-yet-treated switchers as controls, inventor fixed effects, home-class-by-year fixed effects, event times $k \geq 3$ binned, standard errors clustered by inventor. Pre-trends are not rejected ($F = 1.38$, $p = 0.240$). $N = 93,148$ inventor-year observations.

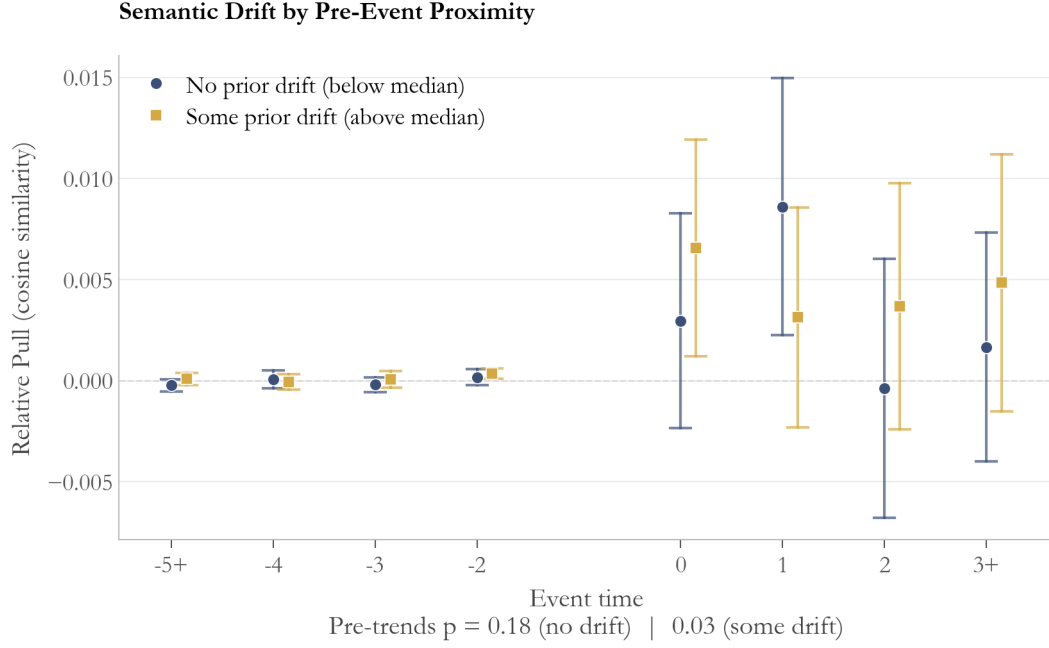


Figure D.7: Point estimates and 95% confidence intervals from Sun–Abraham stacked event-study designs, replicating the semantic drift specification (Figure 9) but splitting the cross-class flipper sample at the median of pre-event ($k \leq -2$) cosine similarity to the partner’s class. This tests whether the semantic shift at collaboration is driven by inventors who were already somewhat close to the partner’s technology class. The solid series includes inventors with below-median pre-event proximity (“no prior drift”); the dashed series includes those with above-median proximity (“some prior drift”). For the no-drift group, the $k = 0$ effect is positive but not significant ($\hat{\beta}_0 = 0.003$, $p = 0.28$), emerging at $k = 1$ ($\hat{\beta}_1 = 0.009$); pre-trends are not rejected ($F = 1.57$, $p = 0.181$). For the some-drift group, the $k = 0$ effect is larger and significant ($\hat{\beta}_0 = 0.007$, $p = 0.02$), though a marginal pre-trend violation appears ($F = 2.60$, $p = 0.035$), driven by a small positive $k = -2$ coefficient (0.0003). All other design choices follow the main specification: not-yet-treated cross-class switchers as controls within each group, inventor fixed effects, home-class-by-year fixed effects, event times $k \geq 3$ binned, standard errors clustered by inventor.

E Robustness to common-surname exclusion

Common surnames raise the risk of false matches in the record linking: two unrelated individuals named Martin who patent in the same department could be linked as one inventor, inflating their apparent patent count and biasing the event-study estimates. I test the sensitivity of the four headline event studies to this risk by dropping all inventors whose surname is Martin or Smith, the two most common surnames in the database.⁴¹

I re-estimate each headline specification on the restricted sample. All design choices are identical to the main text: Sun–Abraham stacked cohort differences-in-differences, the same fixed effects, the same event window ($k \in [-5, 5]$, $k \geq 3$ binned), and standard errors clustered at the inventor level. Figure E.1 through Figure E.4 overlay the restricted-sample estimates (grey squares) on the full-sample baseline (black circles). In all four cases, the restricted-sample coefficients are nearly indistinguishable from the baseline. The productivity effect at $k = 0$ moves from 0.875 to 0.874 (s.e. 0.012); novelty remains null (-0.0080 , s.e. 0.0052); influence is stable (0.0502, s.e. 0.0177); and diversification shifts from 34.5 to 34.5 percentage points (s.e. 0.011). Pre-trend tests are unchanged. Record-linking errors concentrated in the most common surnames are not driving the results.

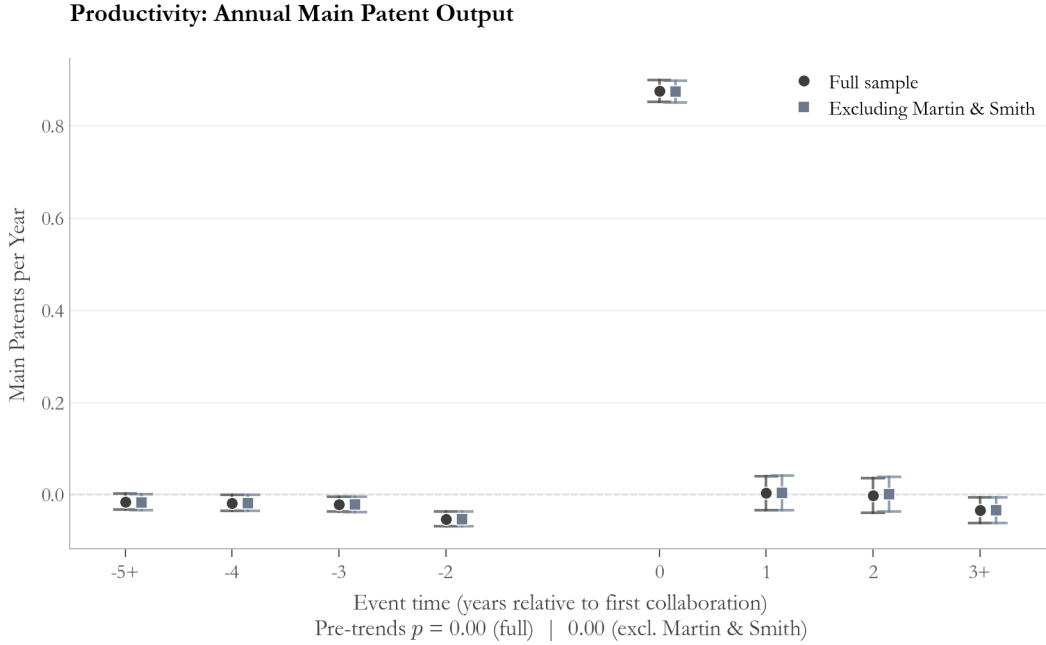


Figure E.1: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is the annual count of main patents. Black circles show the full-sample headline specification (Figure 3); grey squares show the same specification after dropping all inventors with surnames Martin or Smith (287 inventors, 49 flippers). Fixed effects: inventor, calendar year, and stack. Standard errors clustered at the inventor level. Event times $k \geq 3$ are binned; $k = -1$ is the omitted reference period.

⁴¹Martin is the most common surname in France; Smith appears frequently because the database includes patents filed by foreign nationals. Together these two surnames account for 957 inventor records. Of these, 287 appear in the balanced panel of inventors with at least two patents, and 49 are classified as flippers (the treated group in the event-study design).

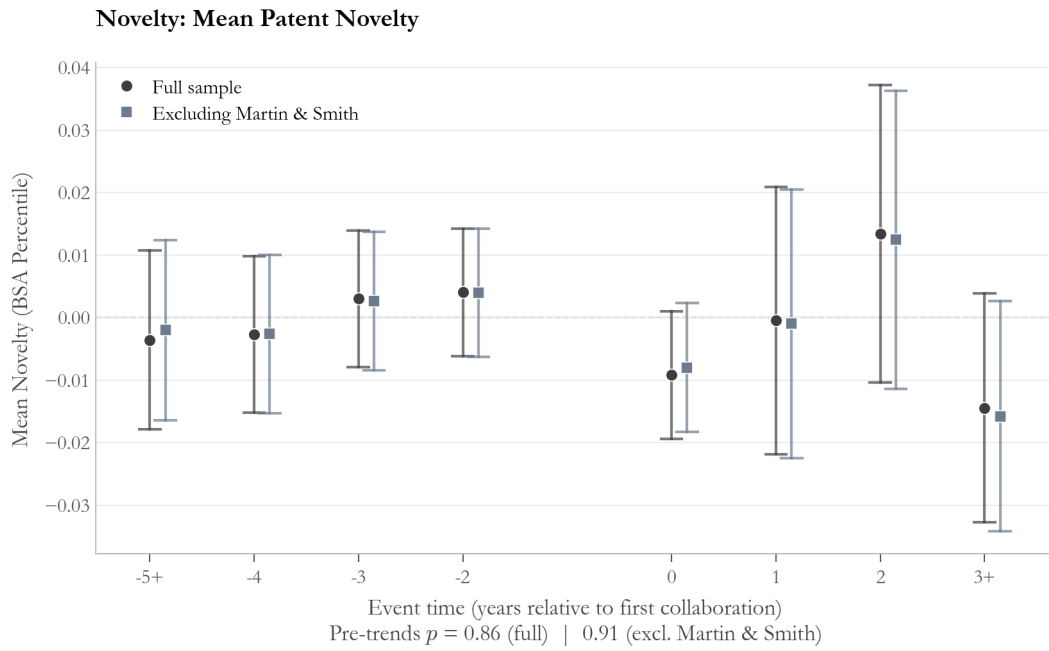


Figure E.2: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is mean patent novelty (BSA inverse percentile rank). Black circles show the full-sample headline specification; grey squares show the restricted sample excluding Martin and Smith inventors. Fixed effects: inventor, home-class-by-year, and stack. Pre-trends are not rejected ($p = 0.91$). Standard errors clustered at the inventor level.

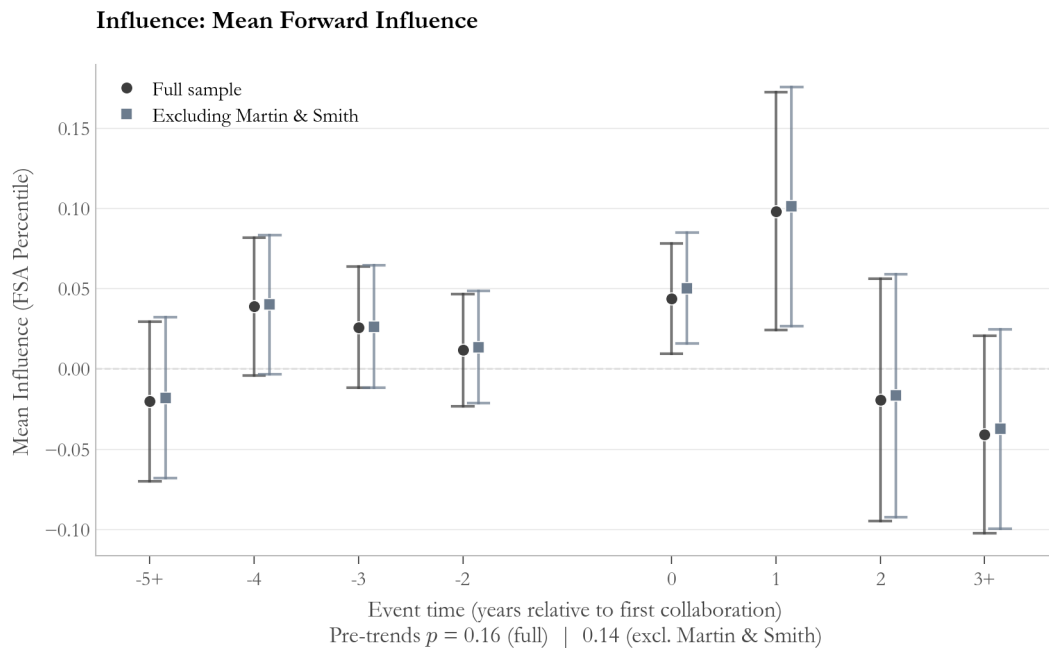


Figure E.3: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is mean forward influence (FSA inverse percentile rank). Black circles show the full-sample headline specification; grey squares show the restricted sample excluding Martin and Smith inventors. Fixed effects: inventor, home-class-by-year, and stack. Pre-trends are not rejected ($p = 0.14$). Standard errors clustered at the inventor level.

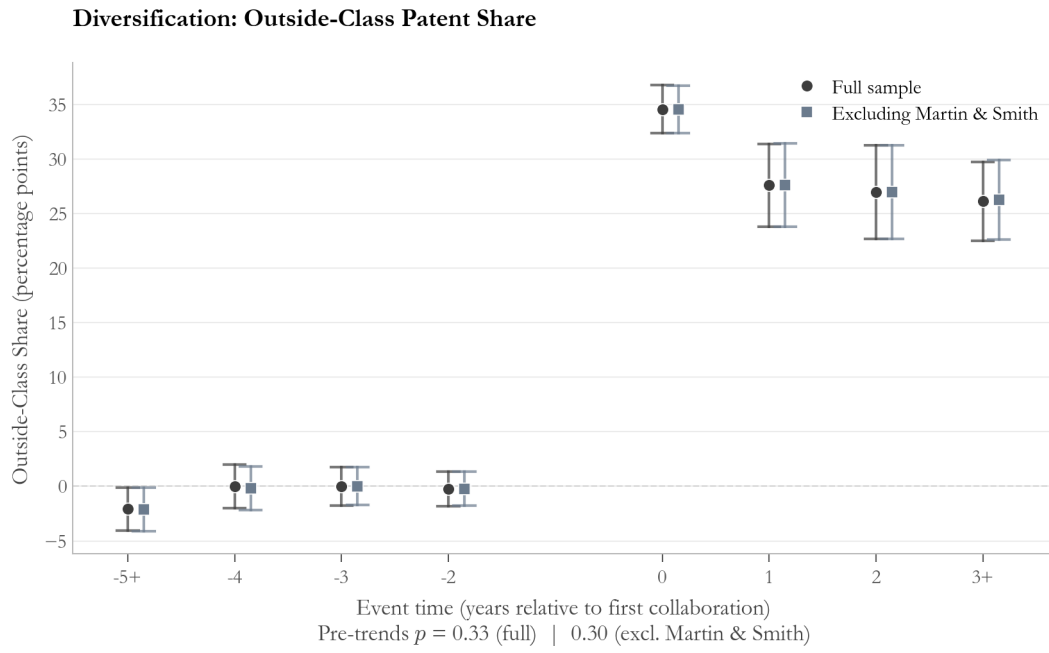


Figure E.4: Point estimates and 95% confidence intervals from a Sun–Abraham stacked event-study design. The outcome is the share of main patents filed outside the inventor’s pre-event home technology class. Black circles show the full-sample headline specification (Figure 6); grey squares show the restricted sample excluding Martin and Smith inventors. Fixed effects: inventor, home-class-by-year, and stack. Controls: not-yet-treated switchers. Pre-trends are not rejected ($p = 0.30$). Standard errors clustered at the inventor level.

F Patent level robustness: how different are collaboration patents?

Table F.1: Probit marginal effects

	<i>Outcome: Collaborative Patent</i>				
	Pre-reform	Post-reform	Full period	Full period	Full period
	(1)	(2)	(3)	(4)	(5)
15-years patent	-0.006 (0.008)	0.037*** (0.006)			
No. of Additions	-0.006* (0.004)	-0.012*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)	-0.011*** (0.001)
High Tech	0.029*** (0.007)	0.016*** (0.001)	0.017*** (0.001)	0.018*** (0.001)	0.018*** (0.001)
Post 1844-reform			-0.023*** (0.003)	-0.019*** (0.003)	-0.018*** (0.003)
Novelty (%)			-0.030*** (0.002)	-0.035*** (0.002)	-0.035*** (0.002)
Influence				0.004*** (0.001)	0.004*** (0.001)
Female-linked			0.047*** (0.004)	0.045*** (0.004)	0.045*** (0.004)
Firm				-0.084*** (0.003)	-0.084*** (0.003)
Foreign Import					0.012*** (0.003)
Observations	12573	295800	307463	306997	306997
Pseudo R^2	0.002	0.001	0.003	0.007	0.008

Note:

*p<0.1; **p<0.05; ***p<0.01

Inventors in France could choose patent terms of 5, 10, or 15 years. Before 1844, the choice came with steep upfront costs: 300 francs for five years, 800 for ten, and 1,500 for fifteen. Over half the patents were of the shortest duration, and a fifth were of the longest duration (Table F.2). One might expect that longer patents, requiring greater investment, would be associated with joint ventures. But the probit model, Table F.1, shows that a 15-year patent was no more likely to be collaborative than a shorter one.

The 1844 reform changed the system. Upfront fees were replaced by annual renewals of 100 francs. Suddenly, applying for 15 years became costless, an option that inventors could later abandon by simply ceasing payment. In this new regime, the data show a positive association between long patents and collaboration. But the effect is mechanical. After the reform, the 15-year patent became the default, so the variable no longer reflects inventors' choices.⁴²

Collaborative patents are less likely to generate the smaller improvement, or additions patents, (certificat de perfectionnement). Before the reform, the association is negative but does not reach the 5 percent significance level (-0.62 percentage points, $p = 0.098$); after the reform, the effect is larger and highly significant (-1.17 percentage points). The effect is persistent across different sample splits and with additional controls. Collaboration appears to deliver a perhaps more complete main invention, while solo inventors are more prone to pursue incremental extensions. Perhaps it was also harder to reassemble the team for smaller improvements.

High-tech⁴³ patents are far more likely to be collaborative. The effect is large: 2.85 percentage

⁴²The better measure is the patent's lapse date, which would give its actual duration. That information is not in my extracted data.

⁴³In technology sectors characterized by rapid rates of technical progress, such as textiles, machinery, mining, and marine and navigation.

points in the pre-reform period, 1.59 points after. This suggests that collaboration mattered most in sectors with rapid technical progress.

Novelty cuts one way and influence the other. In the full-period probit with baseline controls (Table F.1, columns 3–5), moving a patent one percentile up the novelty distribution makes collaboration less likely by -0.034 to -0.030 percentage points. By contrast, patents whose language is taken up more by subsequent inventors (influence) are slightly more likely to be collaborative: a one-unit rise raises the probability of joint invention by 0.37 to 0.39 percentage points (using Models 4 and 5). Put plainly, highly original ideas tend to be pursued solo, while ideas that later diffuse and become templates for others are a bit more likely to originate in teams.

Finally, collaborative patents are more likely to be female-linked, less likely to be linked to a firm, and more likely when the patent involves a foreigner.

Table F.2: Descriptive table for the probit marginal effects

	Missing	Overall	Pre-reform	Post-reform	P-Value
n		308373	12573	295800	
Collaborative Patent, n (%), n (%)	0	265534 (86.1)	10519 (83.7)	255015 (86.2)	<0.001
	1	42839 (13.9)	2054 (16.3)	40785 (13.8)	
15-years patent, n (%), n (%)	0	14469 (4.7)	9960 (79.2)	4509 (1.5)	<0.001
	1	293904 (95.3)	2613 (20.8)	291291 (98.5)	
10-years patent, n (%), n (%)	0	302038 (97.9)	9155 (72.8)	292883 (99.0)	<0.001
	1	6335 (2.1)	3418 (27.2)	2917 (1.0)	
5-years patent, n (%), n (%)	0	300239 (97.4)	6031 (48.0)	294208 (99.5)	<0.001
	1	8134 (2.6)	6542 (52.0)	1592 (0.5)	
High Tech, n (%), n (%)	0	218821 (71.0)	8442 (67.1)	210379 (71.1)	<0.001
	1	89552 (29.0)	4131 (32.9)	85421 (28.9)	
Female-linked, n (%), n (%)	0	302958 (98.2)	12380 (98.5)	290578 (98.2)	0.059
	1	5415 (1.8)	193 (1.5)	5222 (1.8)	
Firm-linked, n (%), n (%)	0	281388 (91.2)	12119 (96.4)	269269 (91.0)	<0.001
	1	26985 (8.8)	454 (3.6)	26531 (9.0)	
No. of Additions, mean (SD), mean (SD)	0	0.2 (0.7)	0.4 (1.0)	0.2 (0.7)	<0.001
Novelty (%), mean (SD), mean (SD)	910	0.5 (0.3)	0.5 (0.3)	0.5 (0.3)	<0.001
Influence, mean (SD), mean (SD)	1370	0.1 (0.9)	0.2 (1.0)	0.1 (0.9)	<0.001
Foreign Import, mean (SD), mean (SD)	0	0.0 (0.2)	0.1 (0.4)	0.0 (0.2)	<0.001

G Patent-agent switching as auxiliary evidence on intermediary exposure

Patent agents mediated the relationship between inventors and the patent office throughout the nineteenth century. Because agents served multiple clients across technology classes, they connected inventors indirectly to a network organized by technology sectors. This appendix tests whether that intermediary channel directed subsequent inventive effort using sustained disappearance of agents from the observed patent record as a proxy for intermediary disruption.

G.1 Design

I proxy agent exit by the last year in which the agent is observed filing on behalf of any client, requiring a gap of at least five years before the sample end to avoid right-censoring. Under the medium definition (at least 10 prior patents, at least 5 prior clients, last observed year no later than 1894), 310 agents satisfy these criteria. Among the approximately 15,700 inventors who used one of these agents within two years of exit, 3,395 continued patenting and 2,657 (78%) subsequently appeared with a different identifiable agent.

For each switcher i moving from old agent A to new agent B , I compute the sector distribution of each agent’s client base using a leave-out construction: agent A ’s pre-exit portfolio uses only patents filed by A ’s other clients before the exit year, excluding inventor i ; agent B ’s pre-switch portfolio uses only patents filed by B ’s other clients before inventor i ’s switch year, excluding inventor i . The treatment variable is the exposure shift $\Delta_{is} = \text{NewShare}_{is} - \text{OldShare}_{is}$, the change in intermediary sector exposure for each inventor–sector pair.

The regression takes the form

$$\text{entered_post}_{is} = \alpha_{A(i)} + \delta_s + \beta \cdot \Delta_{is} + \gamma \cdot \text{entered_pre}_{is} + \varepsilon_{is}$$

where $\alpha_{A(i)}$ absorbs old-agent fixed effects, comparing inventors who lost the same agent but switched to different new agents; δ_s absorbs sector fixed effects; and entered_pre_{is} controls for whether inventor i had already patented in sector s before the exit. Standard errors are clustered at the old-agent level. $\beta > 0$ means inventors entered sectors where their new agent had relatively more clients than their old one. The estimation sample is an inventor-by-sector panel covering all sectors, with pre-switch sector presence controlled directly rather than by sample restriction.

G.2 Results

Table G.1: Delta-exposure alignment regressions across exit-definition tiers. The dependent variable is an indicator for whether inventor i entered sector s after the agent switch. All specifications include old-agent and sector fixed effects and control for pre-switch sector presence. Standard errors clustered at the old-agent level in parentheses.

	Loose (1)	Medium (2)	Strict (3)
Δ exposure	0.044 (0.028)	0.067* (0.035)	0.086** (0.043)
Old-agent share (placebo)	0.043 (0.033)	0.048 (0.039)	0.068 (0.048)
Permutation p -value		0.000	
Observations	52,801	50,407	41,895
Old-agent clusters	317	215	135
<i>Note:</i> * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$			

The delta-exposure coefficient is positive across all three tiers and significant at the strict level ($p = 0.047$). Interpreted substantively, the strict estimate implies that a 10-percentage-point increase in the exposure shift corresponds to about 0.9 percentage points more entry; the medium estimate

implies about 0.7 percentage points. The old-agent placebo, which replaces the exposure shift with the old agent’s own sector share, is uniformly null, confirming that the result does not reflect pre-existing alignment with the old agent’s portfolio. A permutation test that randomly reshuffles inventor-to-new-agent assignments within old-agent groups yields $p < 0.002$ (500 permutations).

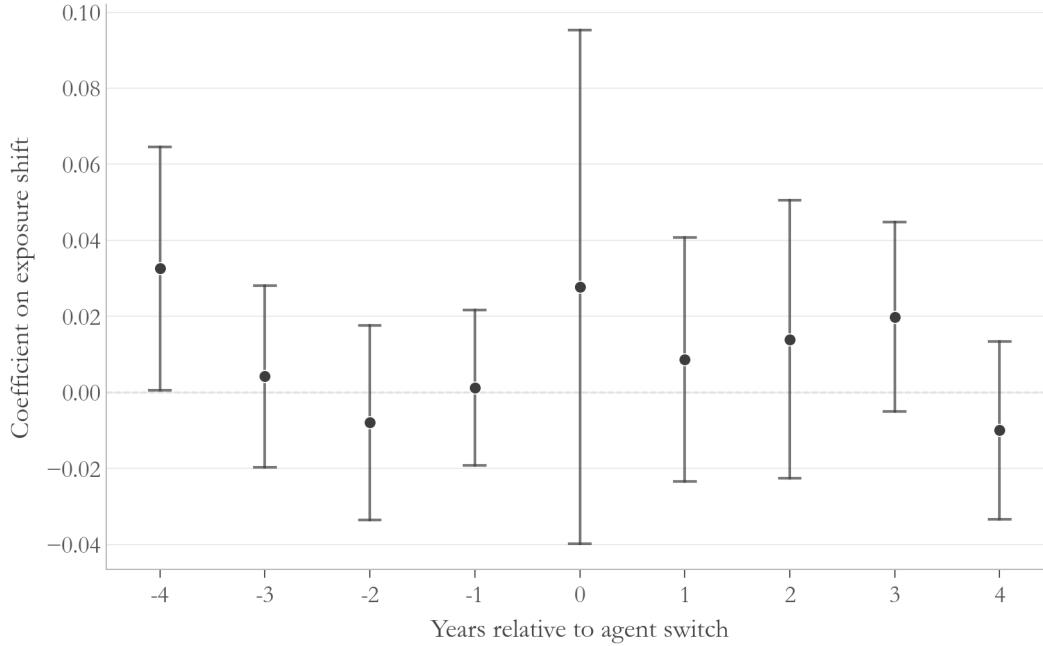


Figure G.1: Point estimates and 95% confidence intervals for the delta-exposure coefficient at each year relative to the agent switch ($k = -4$ to $+4$, estimated in separate cross-sectional regressions). The coefficient at $k = -4$ is significant ($p = 0.048$); the three nearest pre-period bins ($k = -3$ to -1) average -0.001 and are uniformly insignificant. The vertical dashed line marks the switch year.

Figure G.1 plots the coefficient on $\Delta_{i,s}$ at each year relative to the switch. The three pre-period bins nearest the switch ($k = -3$ to -1) are uniformly flat; $k = -4$ is significant at the 5% level, a pattern I note in the limitations below. The post-period coefficients do not display a sharp rise, suggesting diffuse adjustment over several years rather than a discontinuity at the switch.

G.3 Limitations

The pre-period placebo for the raw new-agent share (not the shift) is significant ($p = 0.002$), indicating that inventors were already moving toward their future agent’s sectors before the switch. The delta specification does not exhibit the same pre-period placebo failure, but four weaknesses remain. The coefficient at $k = -4$ in the event-time plot is significant ($p = 0.048$), suggesting that some selection toward the new agent’s sectors may have begun earlier than the switch year. The effect does not display a dose-response gradient: splitting switchers by the magnitude of their exposure shift does not produce monotonically larger coefficients. The effect is concentrated in pre-1875 exit cohorts; later cohorts show null or negative estimates. A forced-disruption subsample restricted to the cleanest cases (long-active agents, large gap to sample end, quick switches) produces a directionally positive coefficient ($\hat{\beta} = 0.098$) that is not significant ($p = 0.341$) because only 38 old-agent clusters remain; the delta pre-period placebo in that subsample is flat ($p = 0.608$). I treat the exercise as suggestive evidence that knowledge flows through intermediary networks, not as independent identification.